The Pollution–Productivity Curve: Non-Linear Effects and Adaptation in High-Pollution Environments*

Matthew S. Brooks[†] Faraz Usmani[‡]

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Abstract

This paper studies whether workers adapt to air pollution using ball-by-ball data from Indian cricket, where match schedules and training locations generate variation in exposure. A 10 $\mu g\,m^{-3}$ increase in match-day $PM_{2.5}$ reduces bowlers' performance relative to batters' by about 1 percent. This effect is statistically significant only in the top pollution quintile (over 5 times the U.S. annual average), although we cannot reject a linear dose–response. Higher exposure in the prior month attenuates this effect, consistent with adaptation, but lowers performance by more than attenuation offsets. Results suggest lowering baseline pollution, not merely averting extremes, yields productivity gains.

Keywords: Air pollution, labor productivity, adaptation
JEL codes: Q53, Q56, J24

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[†]Department of Agricultural and Resource Economics; University of California, Davis; One Shields Avenue, Davis, CA 95616; msbrooks@ucdavis.edu.

[‡]Global Energy Alliance for People and Planet, 2445 M Street NW, Suite 700, Washington DC 20037; faraz.usmani@energyalliance.org.

1 Introduction

The harmful effect of fine particulate matter ($PM_{2.5}$) air pollution on labor productivity is well documented (Archsmith et al., 2018; Adhvaryu et al., 2022; Chang et al., 2019, 2016; Graff Zivin and Neidell, 2012; He et al., 2019; Kahn and Li, 2020; Mullins, 2018). Yet this evidence comes overwhelmingly from settings where pollution is low by global standards. We know far less about its effects on the roughly 2.8 billion people—36% of the global population—for whom hazardous $PM_{2.5}$ is not an occasional shock but a chronic condition of daily life (Health Effects Institute, 2025).[1] A central open question in such settings is whether workers adapt: whether accumulated past exposure moderates the productivity effect of an acute pollution shock. This is, at its core, a question about the difference between short- and long-run responses to a changing environment—a distinction fundamental to economics and central to a large literature on how output responds to temperature (Burke and Emerick, 2016; Dell et al., 2014; Deschênes and Greenstone, 2007), but one that remains largely unexplored for pollution. Answering it requires individual-level data of a kind that is scarce precisely where it matters most: high-frequency performance measures that separately identify acute and chronic exposure, and variation in workers' pollution histories uncorrelated with their ability.

We study this question using ball-by-ball performance data from 773 matches in the Indian Premier League (IPL), India's premier professional cricket competition, paired with a machine learning-based dataset providing daily $PM_{2.5}$ estimates for each stadium (Wang et al., 2024).[2] This setting offers three features that together support credible identification. First, predetermined match schedules make contemporaneous $PM_{2.5}$ exposure orthogonal to player ability and performance. Second, the league's travel schedule, combined with an auction-based salary cap that distributes player quality roughly evenly across teams, generates variation in accumulated exposure that is plausibly unrelated to ability, which ball-by-ball data let us exploit through player-specific pollution

[1] Following Rentschler and Leonova (2023), we classify annual average exposure to $PM_{2.5}$ above 35 $\mu g\,m^{-3}$—the least stringent World Health Organization (2021) interim target—as "hazardous." For context, national annual average concentrations in the United States have generally remained below 10 $\mu g\,m^{-3}$ over the past decade.

[2] A "ball" is analogous to a pitch in baseball.

histories.[3] Third, because each delivery is a direct contest between a bowler and a batter, we can identify how pollution's effect differs across tasks with distinct physical demands. This contest also defines our outcome: whether a delivery results in a run measures the bowler's performance relative to the batter's, so our estimates are net effects—pollution that harms both players registers only to the extent that it harms the bowler more—and understate the harm to either player alone.

We find that workers chronically exposed to high levels of air pollution partially adapt, becoming less sensitive to acute shocks. However, this adaptation comes at a cost: accumulated exposure itself degrades baseline performance, and the protective effect never outweighs cumulative harm relative to a low-exposure benchmark. We report four sets of results.

First, pollution significantly impairs worker performance: a 10 $\mu g\,m^{-3}$ increase in $PM_{2.5}$—half a standard deviation—raises the run-scoring probability by 0.42 percentage points (0.7% relative to the baseline mean), a deterioration in the bowler's performance relative to the batter's. To gauge this magnitude, reducing match-day exposure from the typical IPL level (sample median, 36.7 $\mu g\,m^{-3}$) to the World Health Organization (WHO) daily safe limit (15 $\mu g\,m^{-3}$) would raise productive output by 2.3% for bowlers vis-à-vis batters. This asymmetry is consistent with the higher sustained physical exertion and respiration rates that bowling demands. Analyzing bowler-independent measures of performance, available only for a subset of matches, corroborates this interpretation, revealing a deterioration in bowler-specific performance as pollution rises.

Second, these effects are statistically detectable only in the highest pollution quintile—above 53 $\mu g\,m^{-3}$, over 5 times higher than U.S. annual averages.[4] Such high pollution levels are common in developing countries but rarely studied. We find no evidence of a threshold below which pollution is harmless, however: run-scoring probability rises steadily with pollution, and the dose–response is not distinguishable from linear.

[3] Our identifying assumption for the adaptation effect requires that long-term pollution exposure is not systematically correlated with player ability or other factors that may affect the response to contemporaneous pollution. The IPL's salary cap—which limits how much each team can spend on player compensation—provides support to the claim that player quality is roughly equal on average across teams. Empirically, we find no evidence that team or player performance is correlated with accumulated exposure (Figures H6 and H5).

[4] U.S. annual average $PM_{2.5}$ concentrations were consistently below 10 $\mu g\,m^{-3}$ in the past decade (Health Effects Institute, 2025).

Third, workers chronically exposed to high pollution become less sensitive to acute shocks, consistent with adaptation over both short and long time horizons. Below 50 $\mu g\,m^{-3}$, a 10 $\mu g\,m^{-3}$ increase in match-day $PM_{2.5}$ raises the run-scoring probability by 0.8 percentage points for bowlers at the 10th percentile of past 30-day exposure but has no detectable effect for bowlers at the 90th percentile; the difference, 1.1 percentage points, is statistically significant ($p < 0.05$). Career-average exposure attenuates the acute response similarly: the marginal effect of match-day $PM_{2.5}$ is 62% smaller for bowlers at the 90th percentile of career exposure than for those at the 10th ($p < 0.10$). This attenuation is concentrated along the lower part of the dose–response curve: above 50 $\mu g\,m^{-3}$, the acute effect is small and does not vary with exposure history.

Fourth, adaptation does not offset cumulative harm. Past exposure raises run-scoring probability directly, and although this cumulative harm declines as match-day pollution rises toward 50 $\mu g\,m^{-3}$, the total effect of exposure relative to a low-exposure benchmark is harmful at every observed combination of match-day and past exposure. For virtually all policy-relevant scenarios, both acute and chronic pollution unambiguously impair productivity.

Adaptation is biologically plausible: the environmental toxicology literature has established that repeated exposure to air pollution triggers adaptive responses in lung cells within days to weeks (Dimeo et al., 1981; Hackney et al., 1977; Hamade and Tankersley, 2009; West et al., 2003). Adaptation could also operate through behavioral channels (e.g., wearing masks or using inhalers) or psychological channels (e.g., coping strategies), though the IPL setting limits behavioral avoidance by requiring all players to attend every match in outdoor stadiums. We ground our empirical strategy in this toxicology literature, using it to inform the timescale over which we expect adaptation to emerge.

Our findings contribute to a literature on pollution and productivity that remains concentrated in low-pollution contexts (Aguilar-Gomez et al., 2022): key studies on pollution's physical productivity effects overwhelmingly examine the United States, Europe, or China (Archsmith et al., 2018; Chang et al., 2016; Graff Zivin and Neidell, 2012; Mullins, 2018; Chang et al., 2019; He et al., 2019; Guo and Fu, 2019; Kahn and Li, 2020), with only Adhvaryu et al. (2022) focusing on India. This

geographic concentration leaves the productivity effects of the high exposure levels typical of South Asia and Sub-Saharan Africa largely unmeasured. Moreover, prior work has largely left adaptation to chronic pollution unexplored: among these studies, only He et al. (2019) examine exposure beyond a week, and their distributed lag framework captures the level effect of past exposure rather than the interaction between past and contemporaneous exposure.

Two studies are closest in spirit to ours. Archsmith et al. (2018) study the effect of $PM_{2.5}$ on decision-making quality among baseball umpires in the United States, finding linear effects and testing for—but not detecting—non-linearities in a setting where mean $PM_{2.5}$ is approximately 10 $\mu g\,m^{-3}$. While we cannot formally reject a linear dose–response, we find that marginal effects are only statistically detectable at concentrations above 53 $\mu g\,m^{-3}$, well above their range. Mullins (2018) identifies adaptation to ozone among U.S. collegiate track and field athletes, finding that athletes accustomed to higher ozone concentrations in the seven days preceding the performance event experience smaller performance decrements when ozone is elevated.[5] We extend this finding to $PM_{2.5}$, a solid particulate rather than a gas and the pollutant responsible for the largest share of the global burden of disease from air pollution (Health Effects Institute, 2025).[6]

We make three contributions. First, we extend the study of environmental adaptation to a new margin. The climate economics literature has shown that long-run temperature averages moderate the short-run response to temperature (Burke and Emerick, 2016; Dell et al., 2014; Heutel et al., 2021); we demonstrate a parallel dynamic for air pollution, showing that the effect of an acute $PM_{2.5}$ shock depends on a worker's accumulated exposure. This implies that the productivity costs of short-run pollution shocks may differ substantially across populations with different long-run exposure, a heterogeneity that pooled estimates obscure.

Second, we develop a new person-based measure of pollution exposure to uncover this result. Unlike the climate economics literature which focuses almost exclusively on adaptation within

[5] In a related study, Qin et al. (2022) document that the performance of professional footballers in China is sensitive to the gap between match-day $PM_{2.5}$ and prior levels of exposure to $PM_{2.5}$, but leave open the question of how the magnitude of a same-day shock differs by past exposure levels.

[6] Mullins (2018) also tests for adaptation effects for $PM_{2.5}$ but does not find them. The divergent findings in our sample could be explained by the fact that levels of $PM_{2.5}$ in our sample are much higher than those in the U.S.

spatial units (such as agricultural fields, Census tracts, or nations), we shift the unit of analysis to the individual. We draw on the environmental toxicology literature to anchor the horizon over which adaptation should emerge in the physiological mechanisms identified in that field. Guided by this literature, we estimate the lag profile of recent exposure directly rather than imposing a window, and use it to fix the horizon over which recent exposure is measured for the adaptation specifications.

Third, we bring together two literatures that have studied the effects of air pollution largely in isolation. On one side, a quasi-experimental literature in environmental economics identifies the causal effect of contemporaneous exposure on health and economic outcomes (e.g., Deryugina et al., 2019; Graff Zivin and Neidell, 2012; Schlenker and Walker, 2015); on the other, a body of work in epidemiology relates long-run exposure to mortality and morbidity outcomes using prospective cohort designs that rely on selection-on-observables rather than quasi-experimental variation (e.g., Dockery et al., 1993; Lepeule et al., 2012; Pope III et al., 2002). Because these literatures study different outcomes with different research designs, the harm from contemporaneous exposure and that from accumulated exposure are rarely measured on a common scale. We develop a framework that places both effects—together with the protective effect of adaptation—on the same yardstick of performance, estimated within a single design. This allows us to compare their magnitudes directly: to ask not only whether contemporaneous and chronic exposure each impair performance, but how the per-unit harm of an acute shock compares to that of accumulated exposure, and whether adaptation is ever large enough to offset cumulative damage.

Our results have implications for environmental policy in emerging economies, where large swaths of the population work in outdoor, physical labor. Because we cannot reject a linear dose–response, the productivity gains from pollution control are not confined to extreme episodes: they accrue across the range of exposure that workers in this setting face, almost all of it above the WHO guideline. At the same time, while workers in chronically polluted environments develop partial resilience to acute shocks, cumulative performance losses from chronic exposure far outweigh this protective effect. Adaptation is not a solution to pollution—it is evidence of ongoing harm.

The remainder of this paper is organized as follows. Section 2 describes our setting and the

physiological basis for adaptation. Section 3 develops a conceptual framework. Sections 4 and 5 present our data and empirical strategy. Section 6 reports our results, and Section 7 concludes.

2 Background

2.1 Institutional details of cricket

Cricket is a bat-and-ball sport in which two teams alternate between batting and bowling, with the objective of scoring more runs than the opposition. The fundamental unit of play is the interaction between a bowler (analogous to a pitcher in baseball) and batter (analogous to a baseball batter), with each delivery of a ball representing a discrete episode of measurable performance. Bowlers are responsible for delivering the ball toward the batter's wicket, employing various techniques to make run-scoring difficult, while batters attempt to score runs through offensive batting.[7] Due to more intense physical activity (each delivery requires a running approach, for fast bowlers) resulting in an increased respiration rate, bowlers are more exposed to air pollution than batters.[8] We therefore expect bowler performance to deteriorate more than batter performance when air pollution rises.

Our empirical setting is the Indian Premier League (IPL), the world's premier Twenty20 cricket league.[9] The IPL operates as a franchise system with ten teams based in different cities that span India's diverse geography and differ substantially in average air quality owing to differences in vehicle traffic, industrial activity, and natural features (Figure 2). Each franchise is based at a home stadium in its host city, where it plays its home matches and conducts the majority of its training. The league fixes the full match schedule in advance of each season before air quality on any given match day is realized or can be reliably predicted. Players are allocated to teams through an annual auction conducted under a salary cap. Each franchise bids from the same fixed amount, so no team can outspend another in assembling its roster.

[7] Fielders, the third type of player, support the bowling side by catching batted balls and preventing runs.

[8] See Appendix A for details including physiological metrics.

[9] Twenty20, indicating two innings with 20 overs each, is one style of cricket that is designed to be played over an approximately three-hour period, in contrast to the traditional style in which matches can take days.

2.2 Physiological basis for adaptation

While the notion of adaptation to changing environmental conditions is not new to the economics literature (Dell et al., 2014; Burke and Emerick, 2016; Moore, 2017; Burke et al., 2024), it is typically conceived as a behavioral response to long-run changes, consistent with an optimization framework. The adaptation under study here is similar in that it involves a response to a changing environment; however, the optimizing agent we focus on is not an individual person, but rather biochemical processes in cells.

The environmental toxicology literature has established that mice become resistant to the acute damages of air pollution with repeated exposure (West et al., 2003; Kültz et al., 2015; Lee et al., 2018). The key adaptation mechanism that the environmental toxicology literature has identified is that, when exposed to air pollution repeatedly, cells augment the rate at which they can produce glutathione in response to incoming air pollution.[10] Studies have shown that this adaptation response occurs as quickly as after a week of repeated exposure to air pollution (West et al., 2003). However, little is known about how this response may amplify—or fade—over longer time horizons, motivating our empirical exploration of varying exposure windows (described in detail in Section 4.3.1).

3 Conceptual framework

Our primary outcome is worker performance. While productivity is a measure of output per input, we conceptualize performance as a measure of how well an outcome aligns with a desired goal. In our empirical setting, performance for a bowler is defined as the fraction of deliveries in which they do not concede a run. Generally, performance is a determinant of productivity: for a given level of input, worker performance determines the level of output.

Our conceptual framework (Appendix C) highlights three channels through which air pollution may affect worker performance: a non-linear dose–response relationship, adaptation to chronic exposure through behavioral and physiological pathways, and heterogeneous impacts across task

[10] See Appendix B for a detailed discussion of this mechanism.

types with different physical demands. We define a production function describing a cricket bowler's performance as $F(P, a)$ where P is pollution and a are adaptation measures that can occur in the long-run (either positive or negative). Equation 1 decomposes the total effect of pollution on performance into its constituent parts:

$$\underbrace{\frac{\mathrm{d}F}{\mathrm{d}p}}_{\text{Long-run effect}} = \underbrace{\frac{\partial F}{\partial p}}_{\text{Short-run effect}} + \underbrace{\frac{\partial F}{\partial a}\frac{\mathrm{d}a}{\mathrm{d}p}}_{\text{Adaptation effect}} \tag{1}$$

Standard reduced-form estimates capture only the short-run effect $\frac{\partial F}{\partial p}$. By interacting current pollution with a longer-term average exposure in our empirical specifications, we can disentangle the short- and long-run effects.

4 Data

We combine three data sources: ball-by-ball performance data from 773 IPL matches in India (2008–2022), high-resolution daily $PM_{2.5}$ estimates from both ground monitors and satellite-based sources, and weather controls.[11] This section describes each data source and discusses features relevant for our empirical analysis.

4.1 Cricket performance

We obtain ball-by-ball cricket performance data for the IPL from Cricsheet, an open-source repository of cricket match data (Cricsheet, 2024). Our analytic sample contains all matches in the IPL from 2008–2022 that took place in India, comprising over 180,000 deliveries (analogous to pitches in baseball) across 773 matches.[12] For each delivery, we observe detailed information including the

[11] Several seasons (and parts of seasons) of the IPL were played outside India due to security and public health concerns. We exclude these matches from our analysis to focus on productivity impacts within the high-pollution setting of India.

[12] 177 IPL matches took place outside India during these years due to security concerns; we exclude these from our sample.

identities of both the bowler and batter, the outcome of the delivery (such as the number of runs scored or wickets taken), and the precise match situation (such as the innings and over number). We combine these performance metrics with characteristics of each bowler and batter scraped from ESPN Cricinfo, which includes the national cricket team with which the player is associated, their age, and—for bowlers—their bowling style.[13]

Table 1 (panels B–E) describes summary statistics from these matches. We leverage the data's ball-by-ball granularity by defining our main outcome as a binary indicator for whether any runs were conceded on a given delivery.[14] The mean of the binary outcome—the run-scoring probability—is 59.9%.

4.2 $PM_{2.5}$ exposure

Our primary source of air pollution data comes from Wang et al. (2024), who provide daily estimates of ground-level fine particulate matter ($PM_{2.5}$) concentrations at a 10 km × 10 km resolution across India for the period 1980–2022. These estimates are derived from a machine learning model that combines satellite data, meteorological information, and ground monitor readings. The high spatial resolution of this dataset allows us to precisely match daily air quality conditions to each of the 20 stadiums where IPL matches were played during our study period, including both the ten permanent home stadiums and ten additional stadiums that periodically hosted matches (Figure 2).[15] We validate the Wang et al. (2024) data against the U.S. AirNow network and show that it better matches ground monitor $PM_{2.5}$ readings than aerosol optical depth (AOD) from the Moderate Resolution

[13] Bowling styles can be categorized into "pace" (fast and medium) and "spin." Pace bowlers optimize the speed of the delivery, requiring a long running approach, whereas spin bowlers focus on spinning the ball, requiring a substantially shorter running approach.

[14] This approach has three advantages over aggregate performance measures (such as total runs scored). First, it preserves the rich variation in the data, allowing us to control for both time-invariant characteristics of players and the evolving match situation. Second, it provides a player-level measure of performance based on specific bowler–batter interactions, rather than team-level outcomes that depend on many additional factors. Third, by focusing on the binary outcome of conceding any runs rather than the number of runs conceded, we avoid potential complications from cricket's non-linear scoring system where certain outcomes (e.g., boundaries worth four or six runs) occur discretely based on specific field events rather than incremental performance differences. As a robustness test, we conduct analyses accounting for the full range of run outcomes (0, 1, 2, 3, 4, 5, or 6) using an ordered logit model.

[15] Specifically, we matched the latitude and longitude coordinates of each stadium to the corresponding 10 km × 10 km cell in the Wang et al. (2024) gridded data.

Table 1: Summary statistics: IPL matches in India 2008-2022

Variable	Unit	N	Mean	SD	Min	Median	Max
Panel A: Treatment variables							
Match-day $PM_{2.5}$ ($\mu g\,m^{-3}$)	Match	773	42.29	20.05	14.23	36.65	159.91
Past 30-day $PM_{2.5}$ ($\mu g\,m^{-3}$)	Match-team	1,544	45.82	12.21	22.56	44.01	94.36
Career $PM_{2.5}$ ($\mu g\,m^{-3}$)	Bowler	445	42.62	9.66	17.83	41.67	103.16
Panel B: Cricket performance outcomes							
Runs scored	Delivery	183,572	1.255	1.630	0.000	1.000	6.000
$\mathbb{1}$ (At least one run scored)	Delivery	183,572	0.599	0.490	0.000	1.000	1.000
Panel C: Bowler-independent outcomes							
No-ball (=1)	Delivery	183,572	0.004	0.063	0.000	0.000	1.000
Delivery length (m)	Delivery	116,414	6.36	2.70	-2.00	6.32	20.39
Delivery speed (km/h)	Delivery	113,975	116.44	21.28	79.87	124.45	154.40
Panel D: Bowler characteristics							
South Asian (=1)	Bowler	434	0.620	0.486	0.000	1.000	1.000
Pace bowler (=1)	Bowler	445	0.616	0.487	0.000	1.000	1.000
Age (years)	Bowler	434	25.74	4.51	16.43	25.45	41.58
Seasons in IPL	Bowler	445	3.2	2.7	1.0	2.0	12.0
Panel E: Cricket statistics							
Home bowler (=1)	Delivery	183,572	0.317	0.465	0.000	0.000	1.000
Home batter (=1)	Delivery	183,572	0.319	0.466	0.000	0.000	1.000
Innings	Delivery	183,572	1.5	0.5	1.0	1.0	4.0
Over	Delivery	183,572	10.2	5.7	1.0	10.0	20.0
Panel F: Weather controls							
Temperature (°C)	Match	773	29.49	2.52	21.68	29.44	37.26
Humidity (%)	Match	773	52.17	19.90	10.02	54.25	90.41
Precipitation (m)	Match	773	0.0007	0.0024	0.0000	0.0000	0.0254
Pressure (1,000 Pa)	Match	773	97.41	3.52	84.60	98.07	101.23
Radiation (MJ/m^2)	Match	773	24.48	2.83	6.68	25.00	29.93
Wind speed (m/s)	Match	773	2.30	1.02	0.06	2.28	5.48

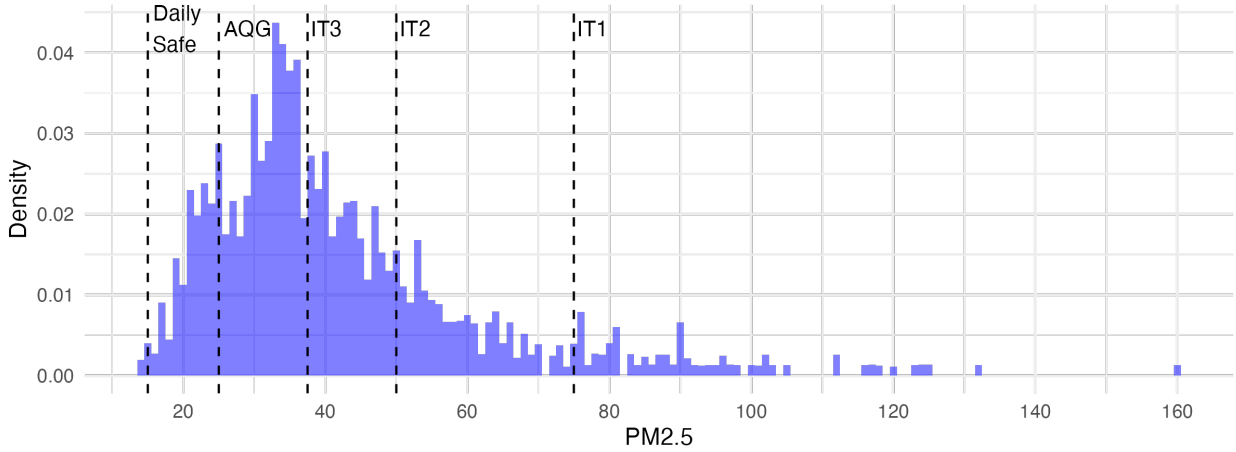
Notes: 183,572 deliveries in 773 IPL matches played in India, 2008-2022 (13 seasons, 20 stadiums, 445 bowlers, 575 batters). Delivery length and speed are observed only on the Hawk-Eye subset (512 matches). Past 30-day $PM_{2.5}$ is the bowling team's mean over the 30 days before the match; career $PM_{2.5}$ is the bowler's mean across all matches played. No-ball (=1) indicates a delivery ruled illegal on account of a bowler fault. Length is the distance in meters from the batter's stumps to the point where the delivery bounces, so a negative coefficient means fuller deliveries. Speed is the measured speed of the delivery in kilometers per hour, bottom-coded at the within-bowler-type (pace/spin) 5th percentile. Age is at IPL debut. Sources: cricsheet (Cricsheet, 2024); $PM_{2.5}$ from Wang et al. (2024); weather from ERA5-Land (Muñoz Sabater, 2019).

Imaging Spectroradiometer (MODIS).[16]

These data provide complete coverage of all IPL matches during our study period. As shown in

[16]See Appendix D for details.

Figure 1: IPL match $PM_{2.5}$ distribution with WHO thresholds indicated



Notes. This figure shows the distribution of $PM_{2.5}$ at IPL matches in the period 2008–2022 with WHO thresholds for daily $PM_{2.5}$ exposure indicated. WHO Interim Threshold 1 (IT1) is 75 $\mu g\,m^{-3}$, WHO Interim Threshold 2 (IT2) is 50 $\mu g\,m^{-3}$, WHO Interim Threshold 3 (IT3) is 37.5 $\mu g\,m^{-3}$, WHO Air Quality Guideline (AQG) is 25 $\mu g\,m^{-3}$, and The WHO Daily Safe exposure threshold (Daily Safe) is 15 $\mu g\,m^{-3}$. These thresholds are based on the levels of $PM_{2.5}$ associated with higher short-term mortality risk (World Health Organization, 2021).

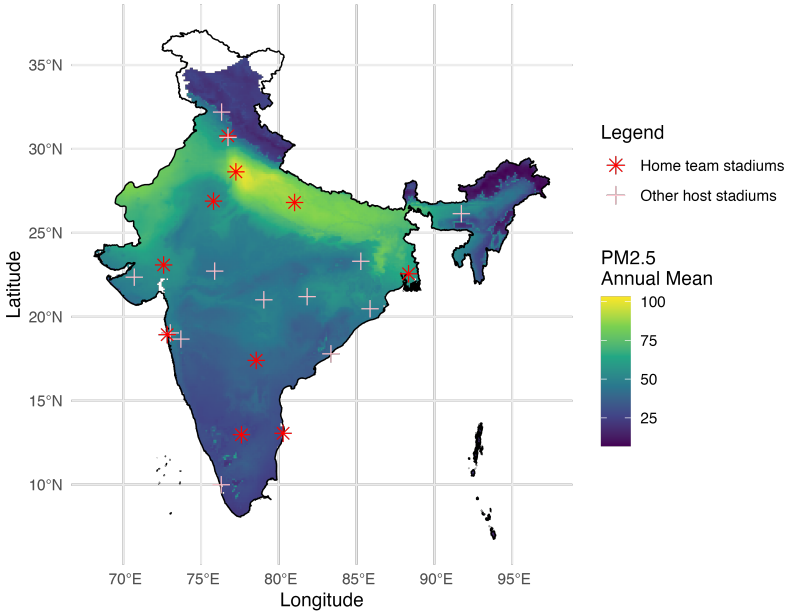
Figure 1, $PM_{2.5}$ concentrations at IPL matches are far higher than in high-income country settings, with a mean of 42.3 $\mu g\,m^{-3}$ on match days (median 36.7), a maximum of 159.9 $\mu g\,m^{-3}$, and over 85% of matches exceeding the WHO daily safe limit of 15 $\mu g\,m^{-3}$ (World Health Organization, 2021). In addition, there is a wide dispersion of pollution across space, as shown in Figure 2.

4.3 Defining past exposure to $PM_{2.5}$

To identify how the effect of contemporaneous exposure varies by past exposure to $PM_{2.5}$, we need to develop a sound measure of past $PM_{2.5}$. The challenge with doing so is that players in the IPL move both within a season and across seasons. Therefore, choosing one location on which to base a player's exposure to pollution is problematic. Instead, we design our measure of past exposure to $PM_{2.5}$ to track where a player is located based on their IPL travel schedule.[17]

[17] See Appendix E for an extended discussion of the justification for this approach.

Figure 2: Geographic distribution of cricket stadiums and $PM_{2.5}$ in the IPL



Notes. This figure shows annual mean $PM_{2.5}$ across India in 2019 as estimated by Wang et al. (2024) and the locations of cricket stadiums used in the Indian Premier League (IPL) from 2008–2022. Red asterisks indicate home stadiums of the ten IPL franchise teams: Chennai Super Kings (Chennai), Delhi Capitals (New Delhi), Gujarat Titans (Ahmedabad), Kolkata Knight Riders (Kolkata), Lucknow Super Giants (Lucknow), Mumbai Indians (Mumbai), Punjab Kings (Mullanpur), Rajasthan Royals (Jaipur), Royal Challengers Bengaluru (Bengaluru), and Sunrisers Hyderabad (Hyderabad). Pink crosses indicate other stadiums that have hosted IPL matches during this period but are not permanent home stadiums for any current IPL team. White space indicates no $PM_{2.5}$ data available from Wang et al. (2024) for a given area.

4.3.1 Short-term past exposure

We ground our definition of short-term pollution exposure in the environmental toxicology literature which finds that mice exhibit tolerance to air pollution when exposed to it repeatedly for days to weeks (West et al., 2003). In Section 5.3 we estimate the horizon over which recent exposure affects performance in our context. We consider a maximum window of 30 days since players begin training at their home stadiums approximately 30 days before the IPL season starts.

Our primary measure of past exposure to $PM_{2.5}$ is a bowler j's mean $PM_{2.5}$ over the past T days, where P indicates $PM_{2.5}$:[18] $\overline{P}_{J(j)d} \equiv \frac{1}{T}\sum_{\tau=1}^{T} P_{J(j)\tau}$. Our main specifications use $T = 30$ days, a choice we motivate in Section 5.3.[19] Since this measure extends into the past by at most one month

[18] Appendix F defines two alternative measures of past exposure (counts of days above a threshold and a degree-day analog) and explains why we rely on the mean.

[19] Results for T 1-30 are reported in Figure H12.

and there is a nine-month gap between seasons of the IPL, this past-exposure measure is assigned at the team-level, so that all bowlers on the same team receive the same level of exposure.[20] The measure of exposure is calculated over a rolling lookback window relative to a specific day, d.

Rules for assigning $PM_{2.5}$ exposure to teams To construct short-term past exposure, we assume the following. If they play a match on the next day: we assume they travel the day prior. So we assign them the value of $PM_{2.5}$ in the next match's stadium. If they do not play a match on the next day, but did play a match on the previous day: we assume the team is still in the location of the previous match. So we assign them the value of $PM_{2.5}$ in the previous match's stadium. If neither, we assign them the $PM_{2.5}$ for their home stadium.

In Figure 3, we compare the $PM_{2.5}$ estimates assigned to teams as opposed to the $PM_{2.5}$ at their home stadium for the 2018 season.[21] The red line shows $PM_{2.5}$ exposure at the team's home stadium; the blue line shows $PM_{2.5}$ exposure for each team given their assumed location. When the two lines overlap—as they do most of the time, since teams spend most of their time in their home stadium—the line is purple. Teams based in a high pollution location that travel to matches in lower pollution settings (e.g., Delhi Capitals), or vice versa (e.g., Chennai Super Kings) tend to have many days on which these two measures differ, reinforcing the need to more precisely estimate $PM_{2.5}$ exposure for each team.

4.3.2 Long-term past exposure

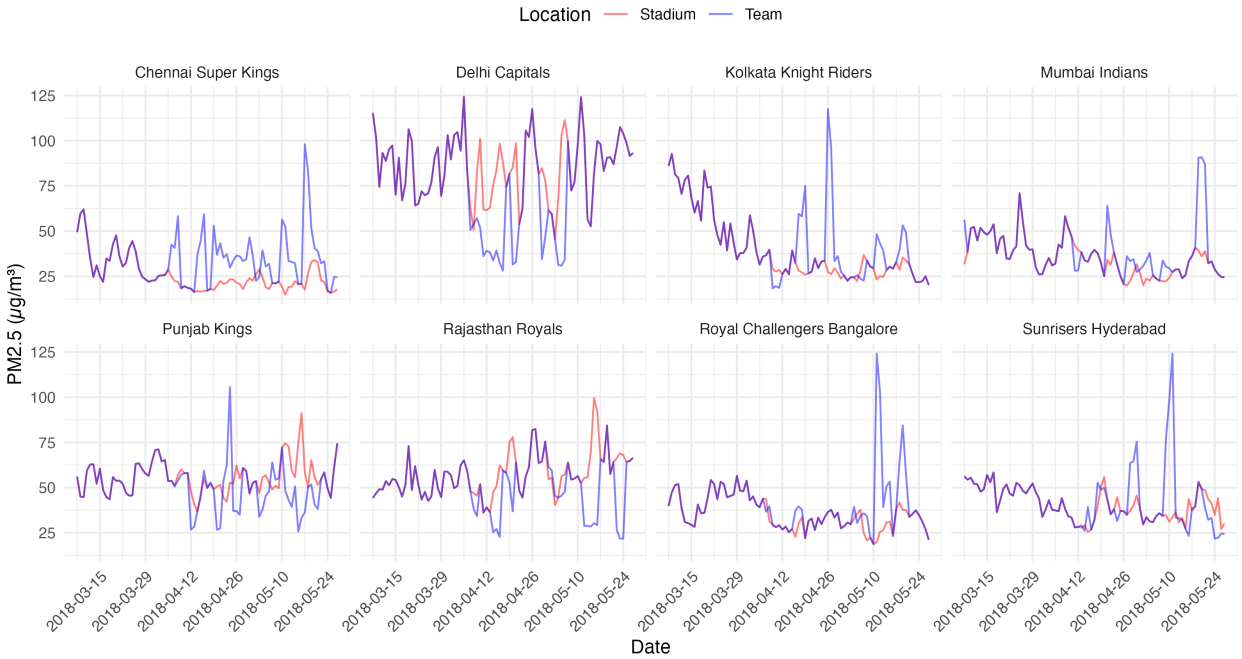
In addition to the measures of past exposure in Section 4.3.1 which vary for the same team (and individual) over time, we also test the robustness of our results for a measure that is fixed for each individual: the average of their exposure across all the matches they play in the IPL.[22] For the average player in our sample, this is measured over a time span of 3.2 seasons (range: 1 to 12 seasons). The measure consists not only of a player's home-stadium's average, but also includes

[20] The team-level of assignment is reflected in the subscript $J(j)$, where J is a function that maps bowler j to team J.

[21] Figure H3 shows the distribution of $PM_{2.5}$ exposure histories across all seasons for each of the eight teams with the highest numbers of matches in the IPL as of 2022. We combine the $PM_{2.5}$ exposure histories for teams that changed names across seasons but retained the same home city and stadium; the graph shows the most recent name of each team.

[22] The distribution of this measure is displayed in Figure H9.

Figure 3: Team $PM_{2.5}$ exposure vs. home stadium $PM_{2.5}$ exposure in 2018



Notes. This figure compares the $PM_{2.5}$ estimates assigned to teams (based on their travel itinerary, as in Section 4.3.1) relative to the $PM_{2.5}$ at their home stadium. The red line shows the level of $PM_{2.5}$ exposure at the team's home stadium whereas the blue line shows the $PM_{2.5}$ exposure for each team taking into account their away matches and travel schedule.

exposures from matches played in away-stadiums, though it is disproportionately weighted to the pollution levels at their home stadium (since teams play roughly half their matches at home). It only tracks their pollution exposure during IPL matches, not the rest of the year, and so is a measure of "on-the-job" exposure. If pollution exposure during cricket matches themselves contributes to a bowler's ability to tolerate pollution while playing cricket, this measure appropriately captures that type of exposure.[23]

4.4 Weather

$PM_{2.5}$ concentrations are influenced by many meteorological factors, which themselves may influence the performance of cricket players. We therefore include variables for temperature, humidity,[24]

[23] In Appendix G.3, we explore alternative definitions of long-term $PM_{2.5}$ exposure.

[24] ERA5-Land contains dewpoint temperature rather than relative humidity. We calculate relative humidity from temperature and dewpoint temperature using the meteorological approximation outlined by Lawrence (2005).

and wind in our regression specifications. We obtain these weather variables from ERA5-Land, a state-of-the-art reanalysis dataset developed by the European Centre for Medium-Range Weather Forecasts, accessed through Google Earth Engine (Muñoz Sabater, 2019). The ERA5-Land data provide global coverage at approximately 11 km $\times$ 11 km spatial resolution, enabling precise matching to stadium locations.

5 Empirical approach

We examine how air pollution affects cricket performance, with particular attention to how players' typical exposure levels moderate these effects. Our main analysis centers on ball-by-ball outcomes, using as our primary outcome variable a binary indicator for whether a bowler concedes a run on each delivery. The choice of a binary outcome is motivated by two considerations. First, conditional on scoring, the distribution of runs is heavily concentrated at one run (over 60% of scoring deliveries resulting in a single run, as shown in Figure H2), and higher run values (fours and sixes) are achieved through qualitatively different actions—hitting the ball to the boundary—rather than incremental performance differences, making a continuous measure potentially misleading.[25] Second, the binary outcome provides a clear interpretation: the marginal effect of pollution is the change in probability of a bowler conceding (or batter scoring) a run, where conceding a run indicates poor bowler performance.[26] We conceptualize run-concession as an error rate on the part of the bowler, analogous to the "correct-call" error rate for umpires used in Archsmith et al. (2018).

[25] In Appendix G.1 we specify and estimate ordered logit models that incorporate variation in the number of runs scored. We also test whether results are robust to an OLS model where the outcome is the number of runs scored per delivery.

[26] There is a slight wrinkle for bowlers in that they also seek to get the batter out, which may justify allowing a run in rare cases. We consider this a minor consideration and abstract away from it in our analysis.

5.1 Effects of contemporaneous air pollution on performance

We examine the relationship between match-day $PM_{2.5}$ levels and the run-scoring probability with the following linear probability model:

$$R_{ij\ell t} = \beta P_{\ell d} + \mathbf{X}'_{\ell d}\gamma + \psi_j + \phi_i + \delta_{\ell y} + \mu_m + \theta_{n(t)} + \eta_{o(t)} + \Lambda_i + \Delta_j + \varepsilon_{ij\ell t} \tag{2}$$

where $R_{ij\ell t}$ is an indicator for whether at least one run is scored on the tth delivery of the match, bowled by bowler j to batter i at location (stadium) ℓ. The unit of analysis is a single delivery (i.e., a single ball thrown from a bowler to a batter). The regressor of interest, P, is the daily mean value of $PM_{2.5}$ (measured in 10 $\mu g\, m^{-3}$) estimated by Wang et al. (2024) in the 10 km grid cell containing the latitude and longitude of the stadium at which the match is played.[27] Our coefficient of interest, β, thus captures the effect of match-day $PM_{2.5}$ levels on the run-scoring probability. A positive β indicates that the run-scoring probability rises with daily mean $PM_{2.5}$, consistent with poorer performance of the bowler relative to the batter.

Our preferred specification includes bowler fixed effects (ψ_j), batter fixed effects (ϕ_i), stadium-by-year fixed effects ($\delta_{\ell y}$), month fixed effects (μ_m), innings fixed effects ($\theta_{n(t)}$) and over-phase fixed effects ($\eta_{o(t)}$).[28] It also includes dummy variables for whether the stadium is the home stadium for the batter (Λ_i) or bowler (Δ_j).[29] Bowler and batter fixed effects control for any time-invariant player-specific factors (such as innate ability), stadium-by-year fixed effects account for stadium-specific temporal trends that might correlate with both pollution levels and cricket performance as well as fixed characteristics of stadiums (e.g., elevation) that may influence performance, and innings

[27] The standard measure of $PM_{2.5}$ is 1 $\mu g\, m^{-3}$. When measured in these units, the effects we observe are very small in magnitude. Given the distribution of $PM_{2.5}$ in our sample (mean 42.3 $\mu g\, m^{-3}$ and standard deviation 20.0 $\mu g\, m^{-3}$; for full distribution see Figure 1), measuring $PM_{2.5}$ in 10 $\mu g\, m^{-3}$ is appropriate. The effect sizes we observe are similar to those of other studies that measure effects in 10 $\mu g\, m^{-3}$ (Adhvaryu et al., 2022).

[28] The subscripts $n(t)$ and $o(t)$ indicate the innings, during which delivery t (synonymous to ball t) occurs. There are typically two innings per match. There are 20 overs per innings. The over-phase fixed effects coarsen the 20 overs into three phases: "powerplay overs" (1–6), where there are restrictions on fielders; middle overs (7–15); and "death overs" (16–20), the final overs in an innings.

[29] Note that some matches are played at stadiums that are a home stadium for neither team, which prevents these two terms from being collinear. Bowlers and batters change teams across seasons which means these terms are not absorbed by the bowler and batter fixed effects.

and over-phase fixed effects control for stage-of-match effects that could influence playing styles and strategies. Month fixed effects absorb within-season variation in $PM_{2.5}$, along with changes in playing strategy over the course of a season that may be correlated with that variation. The dummy variables for the home stadium of the bowler and batter account for home field advantage. A vector of weather controls for the day d and location ℓ of the match ($\mathbf{X}_{\ell d}$) includes a linear and quadratic term for temperature, and linear terms for relative humidity, atmospheric pressure, precipitation, solar radiation, and wind speed.[30]

Intuitively, Equation 2 exploits day-to-day variation in air pollution at the same stadium within a season, while controlling for differences in performance arising from bowler and batter average abilities and differing strategies throughout each match. We therefore interpret the coefficient of interest, β, as the marginal effect of a 10 $\mu g\,m^{-3}$ rise in daily mean $PM_{2.5}$ on the run-scoring probability, independent of any other factors that may also affect it, $PM_{2.5}$, or both.

Identification in Equation 2 comes from the fact that cricket matches are scheduled well in advance (before reliable air pollution forecasts are available) and cannot be rescheduled due to pollution levels, making match-day $PM_{2.5}$ exposure orthogonal to player characteristics and performance potential.[31]

We acknowledge two potential threats to this identification approach. First, it is possible that higher profile matches—where playing strategies may systematically differ—may be disproportionately scheduled to occur on weekends so as to reach wider audiences. This would be a problem for identification if $PM_{2.5}$ is lower on weekends (due to, say, reduced vehicle traffic or industrial activity), or if crowd levels differed systematically and crowds could influence playing strategies. We address this concern by estimating Equation 2 with day-of-week fixed effects included so that identification comes from variation in pollution among matches on the same day of the week.[32]

[30] See Appendix G.2 for a more flexible approach to controlling for weather variables and interactions between weather variables.

[31] The Board of Control for Cricket in India (BCCI) lists its criteria for selecting the venues for matches and does not include air pollution considerations among them (BCCI, 2026). The full match schedule for the season is released weeks before the season starts, far before any reliable air pollution forecast might be available, making match-day $PM_{2.5}$ unrelated to the choice of when and where to hold a match.

[32] We also note that the difference between average $PM_{2.5}$ levels during weekdays (457 matches with mean $PM_{2.5}$ of 42.83 $\mu g\,m^{-3}$) and weekends (316 matches with mean $PM_{2.5}$ of 41.50 $\mu g\,m^{-3}$) is negligible in magnitude and not

Second, players may adjust strategies once match-day $PM_{2.5}$ is realized. Even if $PM_{2.5}$ is orthogonal to player performance potential, it is plausible that players could adjust strategies once $PM_{2.5}$ is observed so that changes in performance cannot be interpreted as degradations per se. To address this concern, we empirically test whether there are systematic differences in strategies associated with $PM_{2.5}$. Specifically, we test whether $PM_{2.5}$ is associated with more experienced bowlers (defined as those with at least three prior IPL seasons) being chosen to bowl and whether $PM_{2.5}$ is associated with the order in which a bowler appears in the bowling sequence. We also estimate Equation 2 with bowler–batter pair fixed effects instead of individual bowler and individual batter fixed effects. Intuitively, this specification allows each bowler to have a different average ability depending on the batter they face, and vice versa. The specification only uses variation in $PM_{2.5}$ that occurs when the same bowler and batter face each other, alleviating the concern that certain bowler–batter pairs might be associated with $PM_{2.5}$ levels, perhaps as a result of playing strategies. In addition to these empirical tests, we emphasize that our outcome—run-scoring probability—captures a cricket metric that players are optimizing, regardless of strategies. If strategies were changing, we would view this as a mechanism by which this outcome of interest changes.

The validity of our estimates of the causal effect of air pollution on performance rests on the assumption that, conditional on our included fixed effects and controls, match-day pollution levels are uncorrelated with unobserved determinants of cricket performance.[33]

All specifications cluster standard errors two-way by match and bowler to account for potential serial correlation in performance for bowlers and the fact that pollution exposure is measured for each match.

statistically significant.

[33]One threat to this identifying assumption could arise if low (or high) quality players were chosen to play when pollution is high. Our individual batter and bowler fixed effects alleviate this concern by comparing performance within the same batter and bowler over matches with varying pollution concentrations. Another threat to identification could arise if coaches choose to field players who exhibit tolerance to pollution levels when pollution is high. While we cannot rule this out, such a pattern would make our estimates understate the effect of pollution on performance.

5.2 Non-linearity in the dose–response

To examine potential non-linearities in the pollution-performance relationship, we estimate the following specification replacing the continuous $PM_{2.5}$ measure with indicators for $PM_{2.5}$ quantiles:

$$R_{ij\ell t} = \sum_{k=2}^{5} \beta_k Q_k\left(\mathrm{P}_{\ell d}\right) + \mathbf{X}'_{\ell d}\gamma + \psi_j + \phi_i + \delta_{\ell y} + \mu_m + \theta_{n(t)} + \eta_{o(t)} + \Lambda_i + \Delta_j + \varepsilon_{ij\ell t} \tag{3}$$

where $Q_k(\mathrm{P}_{\ell d})$ represents indicators for the second through fifth quintiles of match-day $PM_{2.5}$, with the lowest quintile serving as the reference category.

We implement spline regression with varying numbers and placements of knots to identify non-linearities in the response of performance to pollution. We estimate

$$\begin{aligned} R_{ij\ell t} = &\sum_{m=0}^{p} \beta_m(\mathrm{P}_{\ell d}^m) + \sum_{n=1}^{N} \beta_{p+n}(\mathrm{P}_{\ell d} - \tau_n)^p \mathbf{1}\{\mathrm{P}_{\ell d} \geq \tau_n\} \\ &+ \mathbf{X}'_{\ell d}\gamma + \psi_j + \phi_i + \delta_{\ell y} + \mu_m + \theta_{n(t)} + \eta_{o(t)} + \Lambda_i + \Delta_j + \varepsilon_{ij\ell t} \end{aligned} \tag{4}$$

where we vary the number of knots N, the locations of the knots along the distribution of $PM_{2.5}$, and the degree of the polynomial p. We focus on two linear spline specifications. The first places knots at the four WHO thresholds for air pollution: 25, 37.5, 50, and 75 $\mu g\, m^{-3}$.[34] The second places a single knot at the 50 $\mu g\, m^{-3}$ threshold.

5.3 Effects of short-run past exposure on performance

Following Schwartz (2000); He et al. (2019), we specify a polynomial distributed lag model to estimate the effect of recent days of past $PM_{2.5}$ exposure on match-day performance:

$$\begin{aligned} R_{ij\ell t} = &\beta_0 \mathrm{P}_{\ell d} + \sum_{s=1}^{S} \beta_s \mathrm{P}_{J(j),d-s} \\ &+ \mathbf{X}'_{\ell d}\gamma + \psi_j + \phi_i + \delta_{\ell y} + \mu_m + \theta_{n(t)} + \eta_{o(t)} + \Lambda_i + \Delta_j + \varepsilon_{ij\ell t} \end{aligned} \tag{5}$$

where β_s is constrained to satisfy

[34] See Figure 1 for where these thresholds fall relative to the distribution of $PM_{2.5}$ in our sample.

$$\beta_s = \sum_{q=0}^{Q} \alpha_q \, g_q(s), \qquad s = 1, \ldots, S. \tag{6}$$

In Equation 6, Q is the degree of the polynomial (e.g., $Q = 2$ represents a polynomial of degree 2), and s indexes lags (i.e., $s = 1$ indicates $PM_{2.5}$ one day before the match, $s = 2$ indicates $PM_{2.5}$ two days before the match, and so on). S is the full window length (e.g., $S = 30$ when we consider the 30 days preceding the match). The parameter α_q is the coefficient on each orthonormal polynomial component, $g_q(s)$ of each β_s.

Our estimand of interest is the cumulative effect through a given lag window k (where $k \leq S$): $\sum_{s=1}^{k} \hat{\beta}_s$. This represents the effect of raising $PM_{2.5}$ by 10 $\mu g\, m^{-3}$ on each of the k days preceding the match.

5.4 Heterogeneity in the effect of contemporaneous air pollution by past exposure

The specifications above estimate the average and non-linear effects of contemporaneous pollution on performance, treating all workers symmetrically regardless of their exposure histories. We now turn to our central question: does a worker's accumulated past exposure to pollution moderate the contemporaneous effect? Our approach is to interact match-day $PM_{2.5}$ with measures of the player's past exposure, using the two categories defined in Section 4.3: short-term measures (days to weeks, across matches for a given player on a given team) and a long-term measure (career-average exposure, fixed within player). Since short-term measures are assigned at the team level—all players on a team share the same travel itinerary—while the long-term measure is player-specific,[35] we tailor our specifications accordingly.

To claim that past exposure is causally affecting the response to contemporaneous exposure, we would have to assume that a player's past exposure to pollution is uncorrelated with any characteristic that affects their response to contemporaneous pollution. While we cannot completely rule out the

[35] Half of the bowlers in the sample play on two or more teams across the span of their career in the IPL.

existence of any such feature, we provide two arguments in support of this claim. The first is our knowledge of the assignment mechanism for long-term exposure. The IPL auction, with its salary cap for each team, supports the idea that player ability—the feature that most plausibly may modulate the response to pollution—is roughly evenly distributed across teams, and thus past exposure levels. The second is empirical. We provide a number of descriptive plots that fail to find differences along observable characteristics across both contemporaneous and past exposure levels.[36]

5.4.1 Heterogeneity by short-term past exposure

To detect heterogeneity in the effect of contemporaneous air pollution by short-term past exposure, we estimate

$$\begin{aligned} R_{ij\ell t} = \beta_1 \mathrm{P}_{\ell d} + \beta_2 \mathrm{P}_{\ell d} \times \overline{\mathrm{P}}_{J(j)d} + \beta_3 \overline{\mathrm{P}}_{J(j)d} \\ + \mathbf{X}'_{\ell d}\gamma + \psi_j + \phi_i + \delta_{\ell y} + \mu_m + \theta_{n(t)} + \eta_{o(t)} + \Lambda_i + \Delta_j + \varepsilon_{ij\ell t} \end{aligned} \tag{7}$$

where $\overline{\mathrm{P}}_{J(j)d}$ is the past T-day mean $PM_{2.5}$ exposure defined in Section 4.3.1 for bowler j on team J. Other variables are as defined in Equation 2. We include the full interaction of present $PM_{2.5}$ and past $PM_{2.5}$ to disambiguate the level-effect of past $PM_{2.5}$ itself from the modulating effect that past exposure may have on the harms of present exposure. Importantly, because this measure of past exposure changes over time for an individual, it is not collinear with bowler fixed effects.

Identification in Equation 7 comes from changes in $PM_{2.5}$ levels that a team is exposed to within a season as a result of their travel itinerary and realizations of $PM_{2.5}$ where they are located.

Non-linearity in the heterogeneous effect By specifying a linear interaction term between two continuous variables, the specification in Equation 7 imposes the restriction that the magnitude of the adaptation effect is constant across current (match-day) pollution levels. To explore whether past exposure moderates the effect of present exposure differently at different levels of present exposure,

[36] Figure H6 shows that bowler performance (as indicated by the run-concession rate) is uncorrelated with career $PM_{2.5}$ exposure and Figure H5 which shows that team performance (as indicated by the win rate) is uncorrelated with home stadium $PM_{2.5}$.

we interact $\overline{\mathrm{P}}_{J(j)d}$ with the linear spline specification in Equation 4.

5.4.2 Heterogeneity by long-term past exposure

The short-term measures above vary within player across matches, capturing adaptation that develops over days to weeks. We complement these with a measure of career-average exposure that captures adaptation accumulated over years. For the long-term measure defined in Section 4.3.2—a bowler's mean $\mathrm{PM}_{2.5}$ across all IPL matches—we estimate:

$$\begin{aligned} R_{ij\ell t} = {} & \beta_1 \mathrm{P}_{\ell d} + \beta_2 \mathrm{P}_{\ell d} \times \overline{\mathrm{P}}_{j0} \\ & + \mathbf{X}'_{\ell d}\gamma + \psi_j + \phi_i + \delta_{\ell y} + \mu_m + \theta_{n(t)} + \eta_{o(t)} + \Lambda_i + \Delta_j + \varepsilon_{ij\ell t} \end{aligned} \tag{8}$$

where $\overline{\mathrm{P}}_{j0}$ is long-term exposure to pollution for a given bowler, and other variables are as defined in Equation 2. Note that every bowler in the sample is assigned a measure of $\overline{\mathrm{P}}_{j0}$; no bowlers are excluded from the estimation sample for these long-run adaptation effects. Identification in Equation 8 comes from the fact that average pollution levels at teams' home stadiums are widely divergent, generating variation in players' typical exposure levels that is uncorrelated with player ability.[37] Because career exposure is fixed within bowler, its level effect is absorbed by bowler fixed effects. We therefore omit the main effect of $\overline{\mathrm{P}}_{j0}$ and identify the adaptation effect solely through the interaction term β_2. As with the short-term past exposure measure, we interact $\overline{\mathrm{P}}_{j0}$ with the linear spline specification in Equation 4 to allow for non-linearity in the heterogeneous effect.

It is important to emphasize that while the short- and long-term measures of past exposure capture a similar concept, they rely on fundamentally different sources of identifying variation. The short-term measure relies on idiosyncratic variation in $\mathrm{PM}_{2.5}$ levels within a season, based on a team's travel itinerary. We argue that this recent short-term $\mathrm{PM}_{2.5}$ is not correlated with player performance. By contrast, the long-term measure of exposure uses variation in exposure across a career. A major concern in this approach is that there could be endogenous exit from the IPL:

[37] This assumption is supported by the IPL's auction and salary cap system, which ensures that each team has the same amount of money to buy players, resulting in teams having players of roughly equal ability on average.

specifically, the players who are most sensitive to $PM_{2.5}$ might leave. This could bias the estimate of β_2 in Equation 8 downward, overstating the adaptation effect.

We address this concern in two ways. First, we build a model to predict bowler exit. We then reweight the sample by an inverse-probability-weighting correction for attrition, so that those who exit early are upweighted. We then compare the estimate of the contemporaneous effect of $PM_{2.5}$ in the reweighted sample to the unweighted sample, to see if the effect of $PM_{2.5}$ is different. We also re-estimate the linear version of 8 and the spline version of 8 with one knot at 50 $\mu g\, m^{-3}$ using these weights and compare these results to the same regressions without weights. If results are similar, this would suggest that attrition correlated with sensitivity to $PM_{2.5}$ is not artificially manufacturing an adaptation effect.

Second, we estimate Equation 8 for a series of subsamples. Ideally, we would estimate the equation on a balanced panel (i.e., only consisting of bowlers who appear in all 13 IPL seasons in our sample) but this is not possible—the maximum number of seasons a bowler appears in is 12 (Table 1). We therefore explore a series of short-career subsamples: bowlers whose entire IPL career spans at most 1 season, at most 2 (the median), at most 3 (the mean), at most 5, and at most 8 seasons. While we cannot completely rule out concerns of endogenous exit, if results are consistent across these subsamples, it would provide strong evidence that our main estimate is not driven by endogenous exit.

6 Results

Our empirical analysis proceeds in four steps. We first establish that air pollution significantly impairs productivity, then show that these effects are statistically detectable only at extreme pollution levels. We next present evidence of adaptation to chronic exposure at both short and long time horizons, before showing that this adaptation does not offset cumulative harm from sustained exposure.

6.1 Effects of contemporaneous air pollution on performance

Table 2 presents estimates of how match-day $PM_{2.5}$ exposure affects the run-scoring probability. Our preferred specification (column 3) includes fixed effects for bowlers, batters, stadium-by-year, month, innings, and over-phase, along with home stadium indicators and weather controls.[38] A 10 $\mu g\,m^{-3}$ increase in $PM_{2.5}$—half a standard deviation—raises the run-scoring probability by 0.42 percentage points ($p < 0.05$).[39] Given a mean run-scoring probability of 59.9%, this represents a 0.7% relative increase in the run-scoring probability.[40]

To translate this estimate into policy-relevant terms, consider that median match-day $PM_{2.5}$ in our sample is 36.7 $\mu g\,m^{-3}$—more than twice the WHO daily safe limit of 15 $\mu g\,m^{-3}$. Under the linear specification, moving from the median-level exposure in the IPL sample to the WHO safe limit would result in at least a 2.3% increase in productive output.[41] Section 6.2 examines whether this per-unit effect varies with the level of pollution.

Several robustness checks support our main result. First, restricting the sample to the 27.4% of observations with a ground-monitor $PM_{2.5}$ reading from the U.S. AirNow network yields a positive but statistically insignificant linear estimate and imprecise quintile estimates (Table I9), reflecting the smaller sample. Second, applying the multiple imputation correction for measurement error in remotely sensed data proposed by Proctor et al. (2026) reduces the coefficient by 49% but leaves it statistically significant at the 5% level. Third, including day-of-week fixed effects yields similar results (Table I2), indicating that these patterns are not driven by scheduling high profile matches on

[38] Table I6 reports an alternate specification using bowler–batter pair fixed effects, with qualitatively similar results: the run-scoring probability increases by 0.40 percentage points per 10 $\mu g\,m^{-3}$. We use individual bowler and batter fixed effects in our main specification to align with the heterogeneity analyses in Section 5.4, which rely on bowler fixed effects.

[39] Sample mean $PM_{2.5}$ is 42.3 $\mu g\,m^{-3}$ (SD 20.0; range 14.2 to 159.9).

[40] We view this estimate as a lower bound on the total performance losses from pollution exposure. Pollution likely impairs both bowlers and batters; our estimates capture only the net disadvantage for the bowler, not the absolute effect on each.

[41] Productive output is defined, from the perspective of the bowler, as the probability that a delivery concedes no run, i.e., one minus the run-scoring probability. We obtain the 2.3% figure as follows. Our preferred linear specification implies that a 10 $\mu g\,m^{-3}$ increase in $PM_{2.5}$ raises the run-scoring probability by 0.42 percentage points. Median match-day $PM_{2.5}$ in the analytic sample is 36.7 $\mu g\,m^{-3}$, so moving to the WHO 24-hour air-quality guideline of 15 $\mu g\,m^{-3}$ represents a reduction of 21.7 $\mu g\,m^{-3}$. Applying the estimated effect to this gap yields a 0.91 percentage-point decline in the run-scoring probability, equivalently a 0.91 percentage-point gain in productive output. Expressed relative to mean productive output in the sample (0.40), this corresponds to a 2.3% increase.

Table 2: $PM_{2.5}$ exposure and run-scoring probability

	(1)	(2)	(3)	(4)	(5)
		Linear		Quintiles	Spline
	$\mathbb{1}$ (At least one run scored)				
Match-day $PM_{2.5}$	0.0041***	0.0027*	0.0042**		
	(0.0010)	(0.0014)	(0.0018)		
Q2 (Match-day $PM_{2.5}$)				0.0076	
				(0.0060)	
Q3 (Match-day $PM_{2.5}$)				0.010	
				(0.0070)	
Q4 (Match-day $PM_{2.5}$)				0.014	
				(0.0090)	
Q5 (Match-day $PM_{2.5}$)				0.028***	
				(0.011)	
Match-day $PM_{2.5}$ between 0 and 50 $\mu g\,m^{-3}$					0.0044
					(0.0037)
Match-day $PM_{2.5}$ between 50 and 160 $\mu g\,m^{-3}$					0.0041**
					(0.0020)
Weather controls		✓	✓	✓	✓
Bowler FE			✓	✓	✓
Batter FE			✓	✓	✓
Stadium-by-year FE			✓	✓	✓
Month FE			✓	✓	✓
Home stadium FE			✓	✓	✓
Innings FE			✓	✓	✓
Over phase FE			✓	✓	✓
N	183,572	183,572	183,556	183,556	183,556
R^2	0.00028	0.00039	0.047	0.047	0.047
p: quintiles jointly zero				0.08	
p: equal slopes across segments					0.95

Notes. The outcome variable is a binary indicator equal to 1 if at least one run is scored on a delivery and 0 otherwise (mean 0.599). All $PM_{2.5}$ variables are measured in 10 $\mu g\,m^{-3}$. Quintile 1 of match-day $PM_{2.5}$ is 14-27; Quintile 2 is 27-34, Quintile 3 is 34-41, Quintile 4 is 41-53, and Quintile 5 is 53-160 $\mu g\,m^{-3}$. Columns (1–3) enter match-day $PM_{2.5}$ linearly, adding weather controls in column (2) and fixed effects in column (3). Column (4) replaces the linear term with $PM_{2.5}$ quintile indicators, where the lowest quintile (Q1) is the omitted category. Column (5) instead enters match-day $PM_{2.5}$ as a continuous piecewise-linear spline with a single knot at 50 $\mu g\,m^{-3}$. Columns (2)–(5) include controls for weather conditions including temperature, temperature squared, relative humidity, wind speed, precipitation, atmospheric pressure, and solar radiation. Columns (3)–(5) additionally include fixed effects for individual bowlers, batters, stadium-by-year, month, innings, and over phase. Following Correia (2015), 16 singleton observations are dropped in columns (3)–(5). Standard errors (in parentheses) are two-way clustered at the match and bowler levels. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

days of the week when $PM_{2.5}$ tends to be lower. Fourth, we do not detect an association between strategies and $PM_{2.5}$ along the margins we test: bowler–batter pair arrangements (Table I6), more experienced bowlers playing in a match (Table I7), or the order in which bowlers bowl, conditional on bowling (Table I8). These tests support the idea that the change in run-scoring probability we observe is a degradation in performance, not a deliberate change in strategy.

The identification of this effect depends critically on controlling for weather conditions that correlate with pollution. A naive regression (column 1 of Table 2) yields a coefficient of 0.41 percentage points ($p < 0.01$), but adding weather controls in column 2 reduces this to 0.27 percentage points, reflecting omitted variable bias: $PM_{2.5}$ and humidity are negatively correlated ($\rho = -0.62$) while run-scoring decreases with humidity, and $PM_{2.5}$ and temperature are positively correlated ($\rho = 0.22$) while run-scoring increases with temperature.[42] The fixed effects in our preferred specification restore the coefficient magnitude, indicating that pollution's effect operates largely independently once we properly account for confounding weather factors.[43]

Importantly, the consistently positive relationship between pollution and run-scoring reveals an asymmetry: pollution disproportionately impairs bowlers relative to batters. Since each delivery represents direct competition, increased scoring probability necessarily implies bowlers' performance deteriorates more. This finding reflects differential physical demands—bowlers maintain a running approach requiring higher respiration rates while batters engage in intermittent activity—and aligns with the physiological evidence that establishes respiration as the primary channel through which people are exposed to particulate matter (Hamanaka and Mutlu, 2025).[44]

In Table 3, we test the hypothesis that bowlers are harmed by pollution by examining outcomes

[42] See Table I1 for raw correlations between run-scoring, $PM_{2.5}$, and all weather variables.

[43] This result is also confirmed using the post-double-selection (PDS) LASSO method (Belloni et al., 2014); see Appendix G.2 for details.

[44] An important distinction related to the length of the running approach (and thus respiration rates) among cricket bowlers is the bowling style, i.e., "pace" or "spin." Pace bowlers optimize the speed of the delivery, requiring a long running approach, whereas spin bowlers focus on spinning the ball, requiring a substantially shorter running approach. Our hypothesis that the higher respiration rates of bowlers relative to batters (as a result of their running approach) is responsible for the differential effect of $PM_{2.5}$ on bowlers vis-à-vis batters implies that pace bowlers should be more affected by $PM_{2.5}$ than spin bowlers. Table I5 presents results using the specification in column 3 of Table 2 with the addition of an interaction of $PM_{2.5}$ with a dummy for whether the bowler is a pace bowler. While we cannot reject the null hypothesis that the effect of $PM_{2.5}$ is the same across the two types of bowlers, the positive sign on the interaction points toward a larger harmful effect for pace bowlers, consistent with our hypothesized mechanism.

that are independently produced by the bowler and find evidence of bowler-specific impairment. We use data from the Hawk-Eye computer vision system that tracks the speed and location of the delivery and link these data at the delivery-level to 66% of matches in our analytic sample (Hawk-Eye, 2022).[45] We replicate the analyses in Table 2 for three bowler-independent outcomes: whether a delivery is ruled a "no-ball" (an illegal delivery attributable to a bowler fault), length (the distance between where the ball bounces and the batter, m), and speed (how fast the ball is bowled, $\mathrm{km\,h^{-1}}$) separately for pace and spin bowlers. For pace bowlers, pollution raises the probability of a no-ball by 0.045 percentage points per 10 $\mu\mathrm{g\,m^{-3}}$ ($p < 0.05$), relative to a base rate of 0.53% (column 1). Pace bowlers also systematically bowl the ball so that it bounces closer to the batter ("fuller" deliveries) as pollution levels rise: the linear estimate is negative but imprecise (column 3), while the quintile estimates are statistically significant from the third quintile onward (column 4). We cannot reject the null hypothesis that pollution does not affect the speed at which pace bowlers deliver the ball (column 5). In addition, neither the no-ball nor the length response is confined to the top quintile (columns 2 and 4): both are statistically significant by the third quintile and rise with match-day $PM_{2.5}$, and the fifth-quintile estimate is no larger than the fourth-quintile one for no-balls (0.29 versus 0.30 percentage points) and only modestly larger for length (0.29 versus 0.23 m).[46]

6.2 Non-linearity in the dose–response

We next ask whether the per-unit effect varies with the level of pollution. Column 4 of Table 2 replaces the continuous $PM_{2.5}$ measure in our preferred specification with quintile indicators (Equation 3). In this specification, only the fifth quintile shows a statistically significant effect: exposure to $PM_{2.5}$ above 53 $\mu\mathrm{g\,m^{-3}}$ increases run-scoring probability by 2.8 percentage points ($p < 0.01$) relative to the lowest quintile. The second through fourth quintiles show smaller, imprecisely estimated effects that rise steadily with exposure (0.8, 1.0, and 1.4 percentage points).

[45] Table I4 reports a within-spell test of speed fatigue.

[46] Table I3 replicates Table 3 with individual over fixed effects instead of over phase fixed effects. We find similar results overall, and stronger results for delivery length: the coefficient on match-day $PM_{2.5}$ in the linear specification (Column 3) is statistically significant at the 10% level.

Table 3: $PM_{2.5}$ and bowler-specific outcomes, pace and spin bowlers

	(1)	(2)	(3)	(4)	(5)	(6)
	No-ball (=1)		Length (m)		Speed (km/h)	
	Linear	Quintile	Linear	Quintile	Linear	Quintile
Panel A. Pace bowlers only						
Match-day $PM_{2.5}$	0.00045**		-0.02001		0.08726	
	(0.00020)		(0.01544)		(0.06763)	
Q2 (Match-day $PM_{2.5}$)		0.00089		-0.10681		0.03020
		(0.00089)		(0.06504)		(0.23543)
Q3 (Match-day $PM_{2.5}$)		0.00166*		-0.19424***		-0.07670
		(0.00096)		(0.07224)		(0.25313)
Q4 (Match-day $PM_{2.5}$)		0.00304***		-0.22727***		0.11211
		(0.00107)		(0.08335)		(0.29253)
Q5 (Match-day $PM_{2.5}$)		0.00286***		-0.28836***		0.33550
		(0.00110)		(0.09389)		(0.32634)
Outcome mean	0.0053	0.0053	6.8770	6.8770	127.7380	127.7380
N	118,508	118,508	80,913	80,913	79,346	79,346
R^2	0.0132	0.0132	0.1369	0.1370	0.7007	0.7007
Panel B. Spin bowlers only						
Match-day $PM_{2.5}$	-0.00011		-0.00130		-0.00511	
	(0.00016)		(0.01371)		(0.05780)	
Q2 (Match-day $PM_{2.5}$)		0.00106*		-0.07144		-0.31828
		(0.00054)		(0.04312)		(0.29196)
Q3 (Match-day $PM_{2.5}$)		0.00087*		-0.01871		-0.58126**
		(0.00051)		(0.04858)		(0.28258)
Q4 (Match-day $PM_{2.5}$)		0.00119		-0.04784		-0.47603
		(0.00100)		(0.06962)		(0.35483)
Q5 (Match-day $PM_{2.5}$)		0.00024		-0.01466		-0.74393**
		(0.00119)		(0.10183)		(0.36976)
Outcome mean	0.0014	0.0014	5.1843	5.1843	90.5525	90.5525
N	59,926	59,926	35,477	35,477	34,605	34,605
R^2	0.0140	0.0141	0.0810	0.0811	0.3334	0.3337
Weather controls	✓	✓	✓	✓	✓	✓
Bowler, batter, stadium-by-year, month, home stadium, innings, and over phase FE	✓	✓	✓	✓	✓	✓

Notes. Each pair of columns shares a dependent variable: no-ball (=1) in columns (1)-(2), delivery length in columns (3)-(4), and delivery speed in columns (5)-(6). The two panels share both the specification and the outcomes and differ only in which deliveries enter the sample. All three outcomes are produced by the bowler alone, independently of what the batter does, and so isolate bowler impairment. No-ball (=1) indicates a delivery ruled illegal on account of a bowler fault. Length is the distance in meters from the batter's stumps to the point where the delivery bounces, so a negative coefficient means fuller deliveries. Speed is the measured speed of the delivery in kilometers per hour, bottom-coded at the within-bowler-type (pace/spin) 5th percentile. In the first column of each pair, match-day $PM_{2.5}$ enters linearly and is measured in 10 $\mu g\,m^{-3}$; the second column instead enters indicators for the quintile of match-day $PM_{2.5}$, with the lowest quintile (Q1) omitted, so the reported coefficients are level shifts against Q1 in the outcome's own units. Quintiles are cut separately on each outcome's own estimation sample. Every regression includes controls for weather conditions including temperature, temperature squared, relative humidity, wind speed, precipitation, pressure, and solar radiation and fixed effects for individual bowlers, batters, stadium-by-year, month, innings, and over phase (powerplay/middle/death), as well as indicators for whether the bowling and batting teams are at their home stadium. Panel A restricts the sample to the 118,524 deliveries bowled by pace bowlers and Panel B to the 59,945 bowled by spin bowlers. (An additional 5,103 deliveries whose bowler could not be matched to bowling-style metadata enter neither panel, so the two panels do not sum to the 183,572 deliveries in the full sample.) Length and speed are ball-tracking measures observed only on the subset of 512 matches matched to Hawk-Eye data, so columns (3)-(6) are estimated on smaller samples than columns (1)-(2); for those columns bowler type is taken from the Hawk-Eye delivery-level bowling style rather than from the player metadata. Following Correia (2015), singleton observations are dropped, and sample sizes are reported for every column and panel. Standard errors (in parentheses) are two-way clustered at the match and bowler levels. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

This pattern is notable given that even our lowest quintile (14–27 $\mu g\,m^{-3}$) substantially exceeds the WHO's 15 $\mu g\,m^{-3}$ daily limit, and the second quintile (27–34 $\mu g\,m^{-3}$) is roughly double it.[47] The absence of significant effects in quintiles 2–4 relative to quintile 1, despite exposure levels considered hazardous by international standards, suggests that the differential impacts on bowlers vis-à-vis batters are detectable in our data only at the extreme pollution levels common in developing countries but rarely observed in settings typically studied.[48]

The 53 $\mu g\,m^{-3}$ threshold above which we detect significant effects lies well above the empirical support of the closest prior productivity studies: Archsmith et al. (2018) report a mean $PM_{2.5}$ of approximately 10 $\mu g\,m^{-3}$ and a maximum of approximately 40 $\mu g\,m^{-3}$ in their U.S. baseball umpire sample; Chang et al. (2016) report $PM_{2.5}$ ranging from 2 to 60 $\mu g\,m^{-3}$ in their U.S. pear-packing sample; and Adhvaryu et al. (2022) report a mean of 26 $\mu g\,m^{-3}$ in their Indian garment factory sample. Our quintile 5 thus lies outside the empirical range of these comparison settings, so prior work could not have observed the effects we detect at these levels.

Figure 4 plots the spline specification in Equation 4 with knots at the four WHO thresholds. The point estimates are relatively flat through moderate pollution levels, rise more steeply between 50 and 75 $\mu g\,m^{-3}$, and flatten thereafter. That said, the confidence band admits a straight line throughout, consistent with the equal slopes in column 5 of Table 2. Studies confined to low-pollution settings cannot observe this part of the dose–response, where billions of workers operate daily.

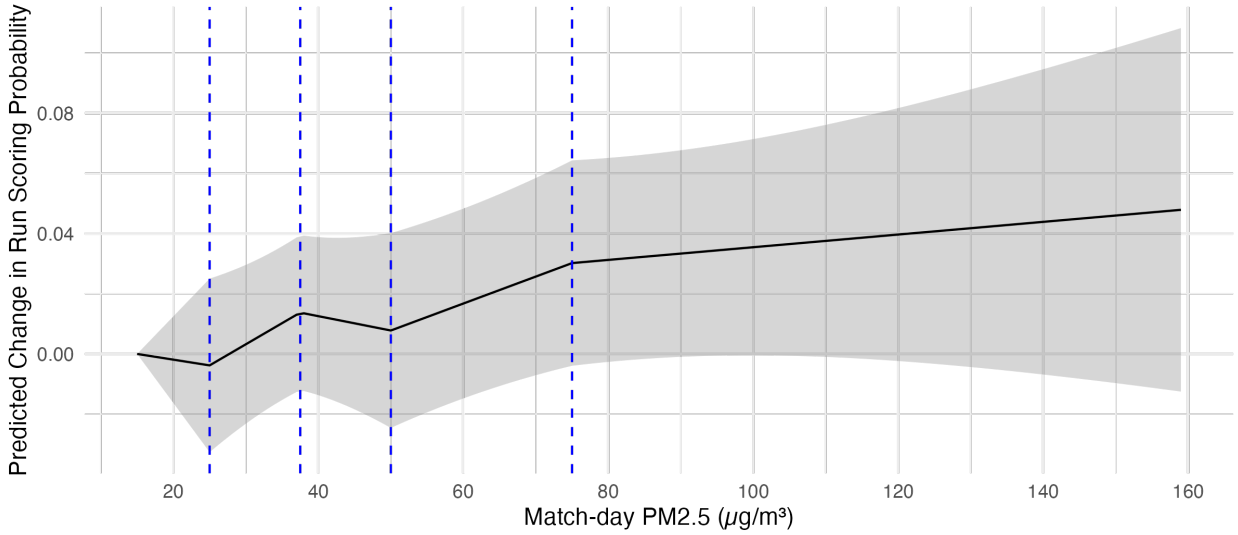
6.3 Evidence of adaptation to chronic exposure

We next examine whether workers adapt to chronic pollution exposure, potentially mitigating acute productivity losses. This analysis leverages variation in exposure histories that arise from how players are assigned to teams in cities with different baseline pollution through the IPL's salary cap

[47] Only 5 of 773 matches (0.60% of delivery observations) fall below WHO's 15 $\mu g\,m^{-3}$ daily limit, making comparing effects relative to that category unfeasible in this context.

[48] It is worth emphasizing that these results do not indicate that lower levels are harmless: the quintile estimates rise steadily across the distribution, and column 5, which enters match-day $PM_{2.5}$ as a piecewise-linear spline with a single knot at 50 $\mu g\,m^{-3}$, yields nearly identical slopes below and above the knot (0.44 and 0.41 percentage points per 10 $\mu g\,m^{-3}$), so we cannot reject a linear dose–response.

Figure 4: Effect of $PM_{2.5}$ on run-scoring probability (linear spline, WHO thresholds)



Notes. Figure displays predicted changes in run-scoring probability as a function of match-day $PM_{2.5}$ estimated using Equation 4 with $p = 1$ and knots at WHO targets: 25, 37.5, 50, and 75 $\mu g\, m^{-3}$. Dashed lines indicate knot locations. The shaded area represents 95% confidence intervals.

and auction system.[49]

6.3.1 Temporal dynamics: Short-term adaptation

We begin by examining how recent pollution exposure affects sensitivity to match-day pollution. Our primary measure is the bowler's mean $PM_{2.5}$ over the preceding T days ($\overline{P}_{J(j)d}$).

Table 4 reports the distributed-lag estimates from Equation 5 using a polynomial of degree 2.[50] Conditional on the preceding 30 days of exposure, the match-day effect is unchanged from

[49]The team assignment mechanism creates variation in short- and long-term exposure histories that appears to be unrelated to factors that affect player performance or the response to pollution shocks. While we cannot definitively rule out that exposure histories are correlated with other factors that may affect player performance or the marginal effect of contemporaneous pollution, we find no evidence in support of this claim. Variation in home stadium average $PM_{2.5}$ exposure is not correlated with team quality as indicated by the proportion of matches that teams win (Figure H5), nor is career $PM_{2.5}$ exposure correlated with the average career performance of bowlers as indicated by their career run-concession rate (Figure H6). This lack of systematic correlation is expected given that each cricket team is provided with an equal amount of funding with which to buy players in each season's auction for players. This rule in the tournament prevents teams from purchasing systematically better players on average due to wealth.

[50]We choose a polynomial of degree 2 since it is the lowest degree that allows the cumulative effect of $PM_{2.5}$ exposure to be non-monotone over the 30-day window. Allowing for additional flexibility by using polynomials of degree 3 and above does not improve the fit of the model while increasing standard errors at the 30-day lag. Despite this, the estimate of the effect of cumulative exposure over the 30 days preceding the match is quite similar using polynomials of degrees 0, 1, 2, 3, 4, remaining within the 0.34–0.36 percentage point range. Table I10 shows results for polynomials of degrees 0, 1, 2, 3, 4, as well as an unrestricted lag profile.

Table 4: Almon (polynomial) distributed lag over 30 days

	Almon polynomial, $Q = 2$ $\mathbb{1}$ (At least one run scored)					
Match-day $PM_{2.5}$	0.00418^{**} (0.00176)					
Cumulative effect of past exposure, lags 1 through k days ($k = 30$ is the whole window):						
$k =$	10	20	21	22	23	24
	0.00088	0.00266	0.00282	0.00296	0.00309^{*}	0.00320^{*}
	(0.00152)	(0.00187)	(0.00188)	(0.00187)	(0.00184)	(0.00180)
$k =$	25	26	27	28	29	30
	0.00329^{*}	0.00337^{**}	0.00342^{**}	0.00345^{**}	0.00346^{**}	0.00343^{**}
	(0.00175)	(0.00169)	(0.00163)	(0.00159)	(0.00158)	(0.00163)
N	183,556					
R^2	0.0472					
Weather controls	✓					
Bowler, batter, stadium-by-year, month, home stadium, innings, and over phase FE	✓					

Notes. The outcome variable is a binary indicator equal to 1 if at least one run is scored on a delivery and 0 otherwise (mean 0.599). All $PM_{2.5}$ variables are measured in 10 $\mu g\,m^{-3}$. The specification is the same as that in Table 2, column 3, plus the bowling team's $PM_{2.5}$ on each of the 30 days before the match. The profile is restricted to a polynomial of degree $Q = 2$ in the lag index s as described in Equation 6. The cumulative entries report $\sum_{s=1}^{k} \beta_s$, the effect of raising $PM_{2.5}$ by 10 $\mu g\,m^{-3}$ on each of the s days before the match. (This effect is calculated for each of days 1 through 30; we display it for a subset of these days for brevity.) Following Correia (2015), 16 singleton observations are dropped. Standard errors (in parentheses) are two-way clustered at the match and bowler levels. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 2 (0.42 percentage points per 10 $\mu g\,m^{-3}$, $p < 0.05$). The cumulative effect of raising $PM_{2.5}$ by 10 $\mu g\,m^{-3}$ on each of the preceding k days is small and imprecise over the first 10 days (0.09 percentage points) and 20 days (0.27 percentage points), becomes statistically significant at 23 days (0.31 percentage points, $p < 0.10$) and at the 5% level from 26 days (0.34 percentage points), and is largest at 29 days (0.35 percentage points) before flattening; over the full 30-day window it is 0.34 percentage points ($p < 0.05$). Recent exposure therefore has a level effect on current performance that accumulates over roughly three to four weeks in this context.[51]

Because the cumulative effect of past exposure becomes detectable at three to four weeks and

[51]We note that the horizon at which the cumulative effect becomes statistically significantly different from zero is not a definitive measure of "over what range past exposure matters." This is because cumulative dose rises mechanically with the window length while precision improves mechanically as well since there is more variation for identification over longer exposure windows. We therefore supplement this conclusion with Table I11. This table shows that point estimates suggest most of the effect is in the 6–25 day range (Panel B), with days 21–30 contributing approximately 22% of the total cumulative effect (0.08 of 0.34 percentage points, Panel A). We choose 30 days because it is approximately one month and therefore a convenient benchmark. Our main adaptation results are robust to alternate choices of a 20-day (Table I26) or 24-day (Table I27) window.

stabilizes over the 26–30-day range (Table 4), we use a 30-day window for our main specifications. Columns 1 and 2 of Table 5 present our main short-term adaptation results. Column 1 estimates the baseline continuous specification, with both match-day $PM_{2.5}$ and past 30-day mean entering linearly alongside their interaction. The main effects are nearly identical: 0.67 percentage points for match-day $PM_{2.5}$ and 0.60 percentage points for the 30-day mean—statistically indistinguishable from each other ($p = 0.74$, Table 5). However, a 10 $\mu g\,m^{-3}$ increase in the 30-day mean implies 30 times the cumulative dose of a single day's exposure. That past and present exposure have similar coefficients despite this vast difference indicates contemporaneous exposure is far more harmful per unit exposure.

The marginal effect of match-day pollution varies substantially with bowlers' recent exposure. Figure 5 plots the marginal effect of match-day $PM_{2.5}$ within each segment of the spline in column 2 of Table 5.[52] Below 50 $\mu g\,m^{-3}$, at low recent exposure (10th percentile, 32 $\mu g\,m^{-3}$), a 10 $\mu g\,m^{-3}$ increase in match-day pollution increases run-scoring by approximately 0.8 percentage points ($p = 0.06$), while at high recent exposure (90th percentile, 64 $\mu g\,m^{-3}$), the same match-day increase has no positive effect (approximately −0.3 percentage points, not statistically different from zero). The difference between the two, −1.1 percentage points, is statistically significant at the 5% level. Above 50 $\mu g\,m^{-3}$, the marginal effects at both exposure levels are small (approximately 0.3 and 0.5 percentage points) and not statistically different from one another. The below-50 differential indicates that bowlers with higher recent pollution exposure are substantially less sensitive to acute pollution shocks. The continuous interaction term in column 1 of Table 5 is consistent with this heterogeneity: it is negative but not statistically significant ($p = 0.42$), reflecting that the adaptation effect is concentrated below 50 $\mu g\,m^{-3}$. This finding is not an artifact of the 30-day window: the interaction term is negative and statistically significant at the 5% level for every window from 20 to 30 days (Figure H12).[53]

[52] Figure H10 shows the same estimation results presented in levels.

[53] Column 2 of Table I14 shows the same pattern with quintiles of match-day $PM_{2.5}$ in place of the spline, while keeping past 30-day mean $PM_{2.5}$ continuous, as in column 4 of Table 2. Every quintile's interaction with past exposure is negative, but only the fifth quintile's is statistically significant (−0.010, $p < 0.05$). This implies that moving from the first to the fifth quintile of match-day $PM_{2.5}$ raises the run-scoring probability by 4.0 percentage points for bowlers at the 10th percentile of past exposure but by only 0.8 percentage points at the 90th ($p < 0.05$ for the difference), results that

Table 5: Short-term adaptation to $PM_{2.5}$

	(1) Linear	(2) Spline, knot at 50 μg m^{-3}
	𝟙 (At least one run scored)	
Match-day $PM_{2.5}$	0.00674*	
	(0.00357)	
Past 30-day $PM_{2.5}$	0.00595*	0.01651**
	(0.00344)	(0.00644)
Match-day $PM_{2.5}$ × Past 30-day $PM_{2.5}$	-0.00053	
	(0.00065)	
Match-day $PM_{2.5}$ between 0 and 50 μg m^{-3}		0.01901**
		(0.00823)
Match-day $PM_{2.5}$ between 50 and 160 μg m^{-3}		0.00071
		(0.00516)
0–50 μg m^{-3} × Past 30-day $PM_{2.5}$		-0.00342**
		(0.00168)
50–160 μg m^{-3} × Past 30-day $PM_{2.5}$		0.00072
		(0.00107)
Weather controls	✓	✓
Bowler, batter, stadium-by-year, month, home stadium, innings, and over phase FE	✓	✓
N	183,556	183,556
R^2	0.05	0.05
p: match-day = past 30-day	0.74	
p: dose–response linear		0.11
p: no moderation	0.42	0.12
p: moderation constant across segments		0.09

Notes. The dependent variable is an indicator equal to 1 if at least one run is scored on the delivery and 0 otherwise (mean 0.599). All $PM_{2.5}$ variables are measured in 10 μg m^{-3}. Past 30-day $PM_{2.5}$ is the bowling team's mean $PM_{2.5}$ over the 30 days before the match. In column (1), match-day $PM_{2.5}$ and past 30-day are in levels and are interacted. Column (2) replaces the linear match-day dose with a continuous piecewise-linear spline with one knot at 50 μg m^{-3} and interacts each segment slope with past 30-day $PM_{2.5}$. The reported p-values are for tests that: the two segment slopes are equal, so the dose–response is linear; there is no moderation, which in column (1) is the single interaction and in column (2) is both segment × past 30-day interactions jointly; and the two interactions of column (2) are equal to one another, which is the restriction that reduces its moderation to the single linear-by-linear parameter of column (1). Both columns include controls for temperature, temperature squared, relative humidity, wind speed, precipitation, atmospheric pressure, and solar radiation, and indicators for whether the bowling and batting teams are at their home stadium, together with fixed effects for individual bowlers, batters, stadium-by-year, month, innings, and over phase. Following Correia (2015), 16 singleton observations are dropped. Standard errors (in parentheses) are two-way clustered at the match and bowler levels. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6.3.2 Long-term adaptation across career exposure

We now examine whether the adaptation patterns observed over 30-day windows extend to career-average exposure. Columns 1 and 2 of Table 6 present results using each bowler's mean $PM_{2.5}$

are consistent with the spline. Specifically, each quintile coefficient is the effect of moving from the lowest quintile to that one, and its interaction measures how much adaptation shrinks that effect as past exposure rises. In the spline, adaptation operates only below 50 μg m^{-3}. Because the top quintile begins at 53 μg m^{-3}, the move from the first quintile to the fifth spans that entire below-50 range and so picks up the adaptation effect in full; the moves to the second, third, and fourth quintiles span progressively less of it and carry progressively smaller interactions (−0.002, −0.007, and −0.008).

Table 6: Long-term adaptation to $PM_{2.5}$

	(1) Linear	(2) Spline, knot at 50 μg m^{-3}
	𝟙 (At least one run scored)	
Match-day $PM_{2.5}$	0.01289**	
	(0.00537)	
Match-day $PM_{2.5}$ × Career $PM_{2.5}$	-0.00193*	
	(0.00116)	
Match-day $PM_{2.5}$ between 0 and 50 μg m^{-3}		0.03170**
		(0.01395)
Match-day $PM_{2.5}$ between 50 and 160 μg m^{-3}		0.00272
		(0.00775)
0–50 μg m^{-3} × Career $PM_{2.5}$		-0.00657**
		(0.00320)
50–160 μg m^{-3} × Career $PM_{2.5}$		0.00039
		(0.00180)
Weather controls	✓	✓
Bowler, batter, stadium-by-year, month, home stadium, innings, and over phase FE	✓	✓
N	183,556	183,556
R^2	0.05	0.05
p: dose–response linear		0.12
p: no moderation	0.10	0.08
p: moderation constant across segments		0.11

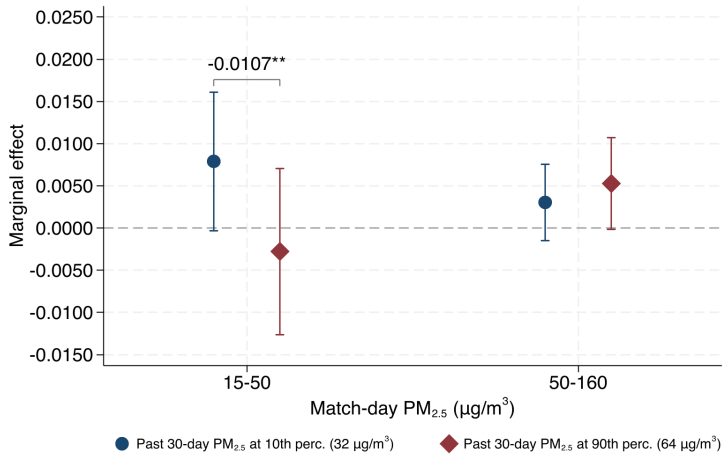
Notes. The dependent variable is an indicator equal to 1 if at least one run is scored on the delivery and 0 otherwise (mean 0.599). All $PM_{2.5}$ variables are measured in 10 μg m^{-3}. Career $PM_{2.5}$ is the bowler's mean $PM_{2.5}$ exposure across all matches played in the cricket league. In column (1), match-day $PM_{2.5}$ and career $PM_{2.5}$ are in levels and are interacted. Column (2) replaces the linear match-day dose with a continuous piecewise-linear spline with one knot at 50 μg m^{-3} and interacts each segment slope with career $PM_{2.5}$. The reported p-values are for tests that: the two segment slopes are equal, so the dose–response is linear; there is no moderation, which in column (1) is the single interaction and in column (2) is both segment × career interactions jointly; and the two interactions of column (2) are equal to one another, which is the restriction that reduces its moderation to the single linear-by-linear parameter of column (1). Both columns include controls for temperature, temperature squared, relative humidity, wind speed, precipitation, atmospheric pressure, and solar radiation, and indicators for whether the bowling and batting teams are at their home stadium, together with fixed effects for individual bowlers, batters, stadium-by-year, month, innings, and over phase. Following Correia (2015), 16 singleton observations are dropped. Standard errors (in parentheses) are two-way clustered at the match and bowler levels. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

across all IPL matches as the measure of long-term exposure. Column 1 of Table 6 estimates the linear interaction specification parallel to our short-term analysis.[54] The interaction coefficient is −0.0019, negative and statistically significant ($p < 0.10$), pointing toward smaller responses to match-day pollution among bowlers with higher career-average exposure.[55] The economic

[54]We also estimate a mirrored response for batters as a robustness check in Table I16. In this specification, the interaction for bowlers is negative and statistically significant while the interaction term for batters is not statistically different from zero. This robustness check strengthens our main finding that bowlers adapt over their careers, while finding no evidence of batter adaptation. We do not estimate this mirrored specification for batters in for short-term (past 30-day exposure) due to correlation between recent bowler and batter $PM_{2.5}$ exposure ($\rho = 0.16$).

[55]These results are robust to using place-based measures rather than person-specific career averages. When we replace individual career exposure with team home stadium pollution, interaction coefficients remain consistently negative but become smaller and statistically insignificant, indicating that measurement precision matters: person-specific measures

Figure 5: Match-day $PM_{2.5}$ marginal effects by spline segment and past 30-day $PM_{2.5}$



Notes. This figure shows how the marginal effects of match-day $PM_{2.5}$ differ along two dimensions: (i) the level of match-day $PM_{2.5}$ exposure and (ii) the level of past 30-day $PM_{2.5}$ exposure. Coefficient estimates and 95% confidence intervals come from column (2) of Table 5. Each coefficient is the implied marginal effect of a 10 $\mu g\,m^{-3}$ increase in match-day $PM_{2.5}$ on the run-scoring probability within one segment of the spline. The horizontal axis is the match-day $PM_{2.5}$ range each segment covers, and within each range the two markers set past 30-day $PM_{2.5}$ to the 10th percentile of past 30-day $PM_{2.5}$ (32 $\mu g\,m^{-3}$) and to the 90th (64 $\mu g\,m^{-3}$). (They are dodged left and right so their intervals do not overlap.) * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

magnitude is substantial: at zero career exposure, the marginal effect is 1.3 percentage points; at mean career exposure (42.6 $\mu g\,m^{-3}$), it falls to 0.47 percentage points; at high career exposure (60 $\mu g\,m^{-3}$), it attenuates close to zero, though remaining weakly positive.[56]

The marginal effects implied by column 1 of Table 6 at the 10th and 90th percentiles of career exposure (32.7 and 53.8 $\mu g\,m^{-3}$; reported in Table I15) are 0.66 and 0.25 percentage points, respectively—a 62% reduction in the acute sensitivity ($p < 0.10$ for the difference). To benchmark this against prior work, the reduction is larger than the ozone-adaptation effect of approximately

capturing actual cumulative exposure based on players' movements over time yield larger adaptation effects than cruder place-based proxies (Table I18).

[56]An alternative—and coarser—way to conceptualize long-term exposure to air pollution is simply to assign players exposure based on their country. We observe the national cricket team for which each bowler in the IPL plays, which we use as a proxy for their primary country of residence. We categorize bowlers into those from South Asian countries and those not. We assume those from South Asian countries are used to higher pollution levels. In Table I19 we present the same specification as column 1 of Table 6, replacing the Career $PM_{2.5}$ measure with a dummy for whether a bowler is from a South Asian country. We see qualitatively similar results: while we cannot reject the null hypothesis that the effect of $PM_{2.5}$ is the same across the two types of bowlers, the negative sign on the interaction points toward bowlers from South Asia experiencing smaller marginal effects of $PM_{2.5}$, consistent with our hypothesis that people who experience chronically high levels of $PM_{2.5}$ partially adapt to it.

43% documented by Mullins (2018)[57] for U.S. collegiate track and field athletes, extending the finding that workers chronically exposed to higher pollution levels develop reduced sensitivity to acute shocks.

Column 2 of Table 6 interacts career exposure with the linear spline in match-day $PM_{2.5}$.[58] As with recent exposure, adaptation operates below 50 $\mu g\,m^{-3}$: the below-50 slope falls by 0.66 percentage points for each 10 $\mu g\,m^{-3}$ of career $PM_{2.5}$ ($p < 0.05$), whereas the above-50 interaction is small, positive, and not statistically significant (0.04 percentage points). The two interactions are jointly significant ($p = 0.08$), although the test that they equal one another does not reject at conventional levels ($p = 0.11$).[59]

We thoroughly investigate the concern of endogenous exit and find that it is not responsible for these results. Exit from the IPL is common: the average number of exiting bowlers per season is 27 (against an average of 34 new bowlers per season, Table I23). Bowlers who play more often, have been in the IPL for longer, and are from a South Asian country and are younger are more likely to appear in the next IPL season; those who perform worse (as measured by runs conceded per delivery) are less likely to appear in the next IPL season (Table I24).[60] When we weight the sample by inverse probability of attrition, we find that the sensitivity to match-day $PM_{2.5}$ is higher in the weighted sample than the unweighted one (1.0 vs. 0.41 percentage points, $p < 0.05$, Table I25). That the weighted estimate is larger is consistent with the theory that bowlers who are more sensitive to $PM_{2.5}$ may be more likely to exit, all else equal. However, this dynamic does not explain away the adaptation results we observe across the unweighted sample. When we estimate Column 2 of Table 6 with the weighted sample, we can still reject the null hypothesis of no moderation at the 10% level.

[57] This is the percent decrease in the marginal effect of ozone for those in the top bin of prior exposure to the middle bin using the estimates in Table 2 of Mullins (2018).

[58] Figure H11 presents these estimation results in predicted levels (analogous to Figure H10 for recent exposure).

[59] Column 2 of Table I15 examines heterogeneity across the match-day pollution distribution using quintile indicators. Consistent with both our baseline findings and the short-term adaptation results, only the fifth quintile of match-day $PM_{2.5}$ exhibits a statistically significant negative interaction with career exposure (−0.016, $p < 0.10$); the fourth-quintile interaction is similar in size but imprecise (−0.015). As in the short-term analysis, results line up with the spline: each interaction measures how much adaptation shrinks the effect of moving from the lowest quintile to a higher one, and the interactions are largest for the fourth and fifth quintiles (−0.015 and −0.016), whose moves span most or all of the below-50 range in which the spline in column 2 of Table 6 finds the adaptation effect.

[60] Notably, we do not find evidence that sensitivity to $PM_{2.5}$ is correlated with age, though we interpret these results with caution given the limited age range available in this sample.

As an additional robustness check, we estimate Equation 8 on short-career subsamples in Table I20. In every subsample, the estimate on the interaction of match-day $PM_{2.5}$ with career $PM_{2.5}$ is negative. While it is not statistically significant in all samples, this is plausibly due to a much smaller sample size of deliveries and bowlers in the more restrictive samples.[61] We also estimate the spline specification in Column 2 of Table 5 for the sample of bowlers who appear in at least three seasons (the mean), restricting to their first three seasons. Here, we find that the negative interaction term on the below 50 $\mu g\, m^{-3}$ portion of the spline is even larger in magnitude and more statistically precise than when estimated in the full sample (−0.91 percentage points, $p < 0.01$, Table I22). In summary, we find no evidence that this estimate of adaptation over a career spanning horizon is driven by players who are sensitive to $PM_{2.5}$ exiting the IPL.

6.4 The tradeoff between adaptation and cumulative harm

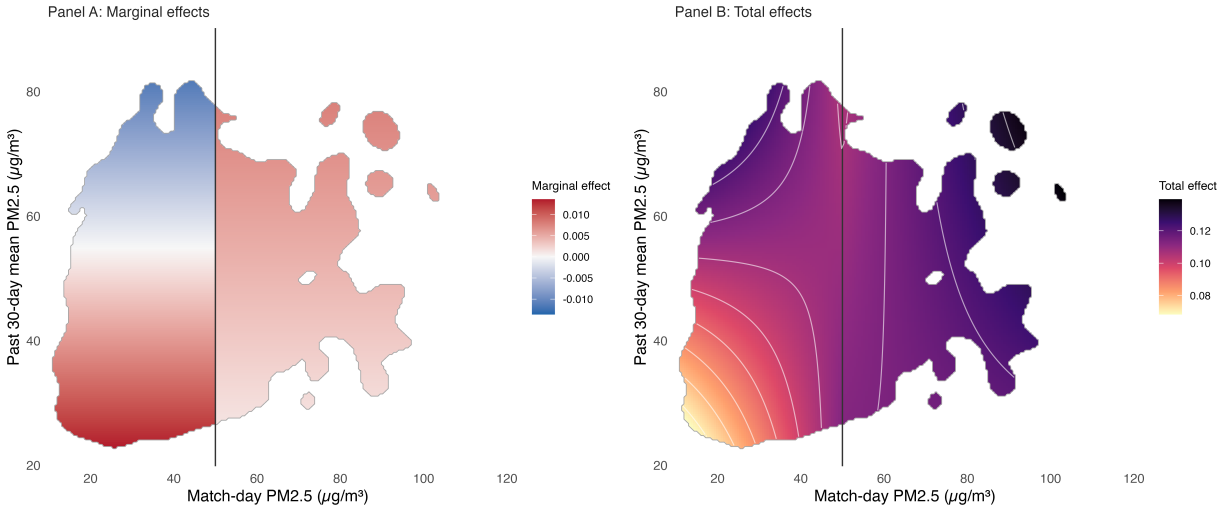
Our findings establish that past exposure generates two opposing effects: it reduces sensitivity to acute pollution shocks (adaptation) while simultaneously degrading baseline performance (cumulative harm). This raises a critical question: under what conditions, if any, does the protective adaptation effect of raising past exposure outweigh the cumulative damage?

We address this question by examining both marginal effects and total effects of match-day and past 30-day $PM_{2.5}$ exposure. Column 2 of Table 5 also identifies the marginal effect of past 30-day exposure on current performance as a function of match-day pollution. This reverses the perspective from our previous analyses: rather than examining how past exposure moderates the effect of present pollution, we examine how present pollution moderates the effect of past exposure.

As shown in column 2 of Table 5, the level effect of past exposure is positive and substantial (1.7 percentage points per 10 $\mu g\, m^{-3}$, $p < 0.05$), indicating that higher past exposure harms current performance through cumulative physiological damage. This marginal effect declines as match-day

[61] We also estimate an analogous version of this table with long-career subsamples (i.e., keeping bowlers who appear in at least X seasons, using only their first X seasons of data) in Table I21. Results are consistent: every estimate of the interaction term is negative, and across the samples we choose (1, 2, 3, 5, and 8 seasons), 3 of the 5 interaction estimates are statistically significant.

Figure 6: Marginal and total effects



Notes. This figure compares the marginal and total effects of contemporaneous and short-run past $PM_{2.5}$ on run-scoring. Effects are calculated using the estimated coefficients in column (2) of Table 5, a linear spline in match-day $PM_{2.5}$ with one knot at 50 μg m^{-3} and each segment slope interacted with the bowling team's past 30-day mean $PM_{2.5}$. The vertical line at 50 μg m^{-3} in each panel indicates the location of the spline's knot. The plane is the set of (match-day $PM_{2.5}$, past 30-day $PM_{2.5}$) pairs, and color indicates the marginal effects (Panel A) and total effects relative to $PM_{2.5}$=0 for both measures (Panel B). White color indicates regions outside the joint support of the two exposure measures.

pollution rises toward 50 μg m^{-3} (by 0.34 percentage points for every 10 μg m^{-3} of match-day $PM_{2.5}$, $p < 0.05$); above 50 μg m^{-3} it no longer declines (the corresponding interaction is 0.07 percentage points and not statistically significant). Panel A of Figure 6 shows the complementary perspective: below 50 μg m^{-3}, the marginal effect of match-day $PM_{2.5}$ is positive at low past exposure and turns negative at high past exposure, while above 50 μg m^{-3} it is small and positive at all levels of past exposure.

Two caveats apply to this observation. First, the estimates above 50 μg m^{-3} are imprecise: the slope of the above-50 segment and its interaction with past exposure are both statistically indistinguishable from zero (Table 5, column 2). Over the observed range of exposures, raising both acute and chronic exposure impairs productivity, even as adaptation moderates the acute response.

Second, this is a statement about the marginal effect, not the total effect: raising past exposure can reduce harm at the margin without making any exposure level beneficial relative to a low-exposure benchmark. Considering the total effect of $PM_{2.5}$ exposure makes it clear that there is no level of

exposure—either on the day of the match or over the previous 30 days—that is beneficial relative to a scenario with low pollution. Figure 6 plots the total effect of $PM_{2.5}$ exposure on the run-scoring probability, calculated as the predicted run-scoring probability at given levels of match-day and past 30-day $PM_{2.5}$ minus the predicted run-scoring probability in a benchmark scenario where both match-day and 30-day $PM_{2.5}$ are zero. When viewed in this way, the effect on run-scoring probability is above zero—indicating harm for bowlers—at any level of match-day or past 30-day exposure.[62]

The visualization of the total effect in Figure 6 reveals two additional patterns. First, increasing match-day exposure by one unit results in a steeper gradient of harm (i.e., larger marginal effects) than increasing past 30-day exposure by one unit. This is indicated by the fact that moving from left to right results in crossing more contour lines than moving from bottom to top and is a visual presentation of the result in column 1 of Table 5 which shows that match-day $PM_{2.5}$ is more harmful per unit than past 30-day $PM_{2.5}$. Second, the effect of increasing match-day $PM_{2.5}$ is dependent on the level of exposure to $PM_{2.5}$ in the past 30-days. This is evident by comparing the result of moving along the match-day $PM_{2.5}$ axis while holding past 30-day $PM_{2.5}$ constant at a high-level (say, 80 $\mu g\, m^{-3}$) versus moving along the same axis while holding past 30-day $PM_{2.5}$ constant at a low-level (say, 20 $\mu g\, m^{-3}$). When past 30-day $PM_{2.5}$ is low, the effect of raising match-day $PM_{2.5}$ is larger than when it is high. This is the visual analog of the negative coefficient on the interaction between the below-50 segment and past 30-day $PM_{2.5}$ in column 2 of Table 5 and is consistent with adaptation.

[62] Intuitively, the joint relationship between present exposure, past exposure, and harm can be pictured as the elevation surface in Figure 6, with the two exposure measures defining the horizontal plane and harm corresponding to height. The protective (marginal) result says that, holding present exposure fixed, raising past exposure moves downhill to a lower elevation. That point nonetheless remains well above the base of the surface—the benchmark of low exposure on both margins—so the total effect is still harmful.

7 Discussion and conclusion

This paper documents how air pollution affects worker productivity in a high-pollution developing-country context, yielding three key insights. First, this effect is statistically detectable only at the extreme levels above 53 $\mu g\,m^{-3}$ that billions of workers experience daily but that are rarely studied. Second, workers partially adapt to chronic exposure over both short-term (30-day) and long-term (career-spanning) horizons. However, this adaptation comes at a substantial cost: our third insight is that the cumulative harm from sustained exposure far outweighs any protective benefit of adaptation.

The asymmetric effects across worker roles provide insight into the mechanisms through which pollution impairs productivity. Bowlers, who maintain high respiration rates during their running approach, experience greater performance degradation than batters, who engage in more intermittent physical activity. This pattern aligns with physiological evidence establishing respiration as the primary channel for particulate matter exposure. For workplace policy, this suggests that pollution control measures should prioritize protection for workers in physically intensive roles, and that task assignments might strategically account for pollution exposure during high-pollution episodes.

Our evidence indicates that workers accustomed to higher pollution levels are substantially less sensitive to acute pollution shocks over both 30-day and career-spanning horizons. Below 50 $\mu g\,m^{-3}$, a match-day increase that raises the run-scoring probability by 0.8 percentage points for a bowler with low recent exposure has no detectable effect for a bowler with high recent exposure, and the acute response is 62% smaller for bowlers with high career exposure than for those with low career exposure. In both cases the attenuation operates along the lower part of the dose–response curve; above 50 $\mu g\,m^{-3}$ the acute effect is small and does not vary with exposure history. Adaptation thus blunts the response to moderate pollution rather than to extreme episodes, consistent with physiological and behavioral adjustments that develop through sustained exposure.

However, adaptation does not resolve the productivity costs of pollution. The protective effect never outweighs cumulative harm relative to a low-exposure benchmark: the total effect of exposure is harmful at every observed combination of match-day and past exposure. For virtually all policy-relevant exposure scenarios, both acute and chronic pollution unambiguously impair

productivity.

These findings carry three important policy implications. First, because the effects we detect lie at levels far above those typically studied, interventions targeting high-pollution episodes—such as temporary factory closures, construction bans, or vehicle restrictions during pollution spikes—may generate substantial productivity benefits in heavily polluted regions.

Second, populations typically exposed to clean air are especially vulnerable to pollution spikes. Wildfire smoke in the United States and Canada provides a salient contemporary example: populations accustomed to clean air experience intermittent exposure to hazardous pollution without the physiological adaptation that might partially buffer impacts. Our findings suggest such episodic exposures may be particularly harmful precisely because affected populations lack adaptive capacity developed through sustained exposure.

Third, and most importantly, our results therefore reinforce rather than undermine the case for aggressive pollution control in chronically polluted settings. Adaptation is not a solution to pollution; it is evidence of ongoing harm—the continuing costs that workers bear in persistently polluted environments. For the billions of workers operating in such environments across the developing world, reducing pollution exposure remains the main path to protecting both productivity and long-term health.

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Supplemental Appendix to "The Pollution–Productivity Curve: Non-Linear Effects and Adaptation in High-Pollution Environments"

Matthew S. Brooks[†]
University of California, Davis

Faraz Usmani[‡]
Global Energy Alliance for People and Planet

[†]University of California, Davis. msbrooks@ucdavis.edu.

[‡]Global Energy Alliance for People and Planet. faraz.usmani@energyalliance.org.

Appendix A Bowler and batter physical demands

The distinct physical demands across playing positions in Twenty20 cricket create variation in how air quality might affect different types of players. Fast bowlers face the most intense physiological requirements (Noakes and Durandt, 2000). During delivery, a bowler accelerates through their run-up reaching speeds of approximately 22 kilometers per hour, plants their front foot with ground reaction forces between 2.4 and 5.8 times their body weight, and decelerates rapidly after ball release (Bartlett, 2003). In Twenty20 cricket, fast bowlers perform approximately 23 sprints per hour of play, with significantly less recovery time between high-intensity efforts compared to other positions (Petersen et al., 2010). During bowling spells, heart rates can reach 180–190 beats per minute, and in hot conditions, bowlers have recorded sweat rates up to 1.5 liters per hour, comparable to marathon runners (Noakes and Durandt, 2000).

For batters, the physical requirements combine intermittent high-intensity running with periods of technical performance (Noakes and Durandt, 2000), surrounded by periods of rest. In Twenty20 cricket, batters perform approximately 15 sprints per hour (Petersen et al., 2010). The activity pattern, however, is highly intermittent: mean heart rates during a day's cricket rarely exceed 128 beats per minute for batters. Over a complete Twenty20 innings, batters cover approximately 3.5 kilometers in total distance—a relatively small amount considering that innings last about 1.5 hours. Only about 20% of this distance is covered in high-intensity running (Petersen et al., 2010). Unlike fast bowlers who run before delivering each ball, a batter's main strenuous task is not running, but the technical demands of facing deliveries from bowlers that can exceed 140 kilometers per hour (Bartlett, 2003). These factors indicate that batters' requirements are less physically demanding than bowlers' requirements, suggesting that bowlers may be more affected by air pollution than are batters. The discrete nature of cricket's bowler–batter interactions generates rich data on individual performance, enabling precise measurement of productivity effects.

Appendix B Glutathione's role in cellular adaptation to repeated pollution exposure

B.1 Physiological mechanism: increased glutathione production

When air pollution enters the body through respiration, it brings with it a variety of harmful chemicals that the body must process in order to expel. The first step of this process, known as Phase I metabolism, involves converting the molecule into a more polar version of itself—often a reactive electrophile—which can result in it becoming more toxic. This is a preliminary step which is necessary for Phase II, in which the body makes the molecule more water soluble, and thus easier to excrete. Glutathione in the lungs (in both its substrate and enzyme form), performs this second step by attaching itself to the molecules produced from Phase I (a process known as conjugation), neutralizing the reactive electrophiles, thereby completing Phase II metabolism.

At low levels of pollution, there is enough glutathione present at baseline to neutralize the harmful effects of air pollution. As air pollution increases, however, the body becomes unable to produce glutathione at a sufficient rate to conjugate incoming electrophiles from air pollution. The rate limiting step in the chemical reaction for the body to produce glutathione is the enzyme γ-glutamylcysteine synthetase (γ-GCS), which itself must be produced by cells in the lungs. The

key adaptation that the environmental toxicology literature has identified is that, when exposed to air pollution repeatedly, cells augment their ability to produce γ-GCS, enabling them to produce glutathione more quickly in response to incoming air pollution.

While the body does not become completely immune to air pollution at all levels, augmenting the ability to produce γ-GCS enables it to endure higher levels of air pollution than it would otherwise while suffering minimal damage from an acute episode of exposure. To visualize the effects of this adaptation response, Figure H4 shows magnified images of cells from the lungs of mice after a single day of air pollution exposure (Panel B) and seven days of exposure (Panel C), relative to a control group of mice who were exposed to clean air during the same period (Panel A). The key insight is that the mice who were exposed over seven days have an intact cellular structure that is more similar to the control cells than to the ones that experienced only a single episode of exposure.

B.2 Empirical tests of glutathione-driven tolerance

In randomized controlled trials exposing mice to varying levels of air pollution over time, cells in the respiratory tract and lungs—i.e., the cells hit hardest by air pollution—can mitigate the damage from air pollution using an antioxidant called glutathione. Glutathione is present in both humans and mice and is understood to perform a similar function in humans as in mice (Ketterer et al., 1983), an insight that serves as the basis for an extensive literature investigating the effects of air pollution on mice to learn about implications for humans.

While the body does not become completely immune to air pollution at all levels, this physiological adaptation allows it to endure higher levels of air pollution than it would otherwise while suffering minimal damage from an acute episode of exposure. West et al. (2002) identifies glutathione as a mechanism for tolerance. Consistent with this adaptation mechanism, West et al. (2003) shows that mice who were exposed over seven days to air pollution have an intact cellular structure that is more similar to mice who were never exposed to air pollution than to mice that experienced only a single episode of exposure. (To visualize the effects of this adaptation response, Figure H4 shows magnified images of cells from the lungs of mice after a single day of air pollution exposure (Panel B) and seven days of exposure (Panel C), relative to a control group of mice who were exposed to clean air during the same period (Panel A).)

Appendix C Conceptual framework

First, the marginal effect of pollution on performance may vary with the level of pollution. As depicted in the left panel of Figure H1, a linear damage function implies constant marginal effects across all pollution levels. However, the relationship could be concave (middle panel), where the drop in productivity from an increase in pollution is larger at higher pollution levels.[63]

Second, the effect of pollution may vary systematically across individuals based on their typical exposure levels, consistent with adaptation.[64] The right panel of Figure H1 illustrates how

[63] Alternatively, the relationship could be convex, though the key insight remains: there is little theoretical justification for assuming constant marginal effects across all pollution levels.

[64] Conceptually, adaptation can be viewed as another form of non-linearity. Whereas a non-linear dose–response means that marginal effects vary across the level of contemporaneous exposure, adaptation means that marginal effects vary across the level of typical exposure.

performance might differ between individuals accustomed to high versus low pollution levels.[65] The figure suggests a potential crossing of damage functions: individuals adapted to high pollution might perform better than their low-pollution counterparts when pollution is high, but worse when pollution is low.

Third, the impact of pollution on performance may vary by the type of task performed, with tasks that require certain types of effort or prolonged exposure being more vulnerable to pollution's effects. In addition, as discussed in Section 2.2, accumulated exposure may generate both negative effects (cumulative physiological damage) and positive effects (behavioral and physiological acclimatization), with the net impact depending on their relative magnitudes.

Appendix D Validating Wang et al. (2024) $PM_{2.5}$ estimates

We choose the U.S. AirNow network to perform validation since it is a high quality ground monitor network that was not included in the training data for the calibration of the model in Wang et al. (2024). (Wang et al. (2024) use ground monitor readings from India's Central Pollution Control Board to train their model.) Figure H7 demonstrates a strong correlation between the Wang et al. (2024) dataset and ground readings. By contrast, Figure H8 reports the results of an analogous exercise with aerosol optical depth (AOD) from the Moderate Resolution Imaging Spectroradiometer (MODIS), and shows a much weaker correlation. We therefore choose Wang et al. (2024) as our source of data for $PM_{2.5}$ concentrations. In both validation exercises, we compare estimates from the data product in the cell containing the U.S. AirNow ground monitor. For Wang et al. (2024), these are 10 km $\times$ 10 km grid cells. For MODIS AOD, these are 3 km $\times$ 3 km grid cells.

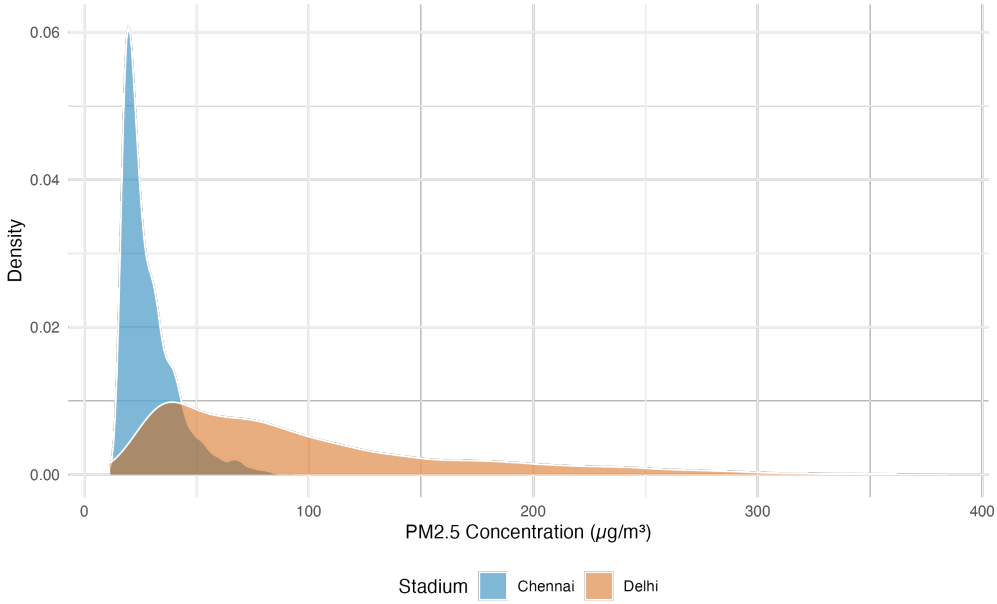
Appendix E Measuring past exposure to $PM_{2.5}$

Our approach to exploring heterogeneity in the short-run effect of pollution by exposure to pollution in the long-run builds on the person-based approach to measuring pollution exposure histories in environmental toxicology (Dodson et al., 2007; Garcia et al., 2013). In this approach, we assign a measure of accumulated pollution exposure to each person, taking into account their travel itinerary over a fixed window of time. Following the climate econometrics literature (Dell et al., 2014), we conceptualize each team's home stadium as having its own pollution "climate," where daily pollution levels are realizations from this distribution. When teams play a match at a stadium in a rival's city, they experience a new pollution climate—daily pollution levels there are realizations from a different distribution rather than from the same one. This framework is consistent with the observed data indicating diverging average pollution levels across stadiums (Figure 2). For example, in Figure E1 we see that although there is a common support of the probability mass at low pollution levels, the Delhi stadium regularly experiences pollution levels that the stadium in Chennai never experiences.

This approach, to our knowledge, is new to the economics literature and offers several advantages, along with a few challenges. The advantage of measuring person-specific adaptation is that we can track exposure to pollution for the same individual over time. This is a more precise way to

[65] Similarly, we could categorize damage functions into different tasks, with different damage functions for each type of task.

Figure E1: Probability density functions of $PM_{2.5}$ in two example stadiums



Notes. This figure displays the probability density function (PDF) of $PM_{2.5}$ concentration in two example stadiums with divergent distributions of $PM_{2.5}$.

measure pollution exposure than assuming, for example, that all individuals with an address in a district live and work in that district and therefore are exposed continuously to the pollution levels there. People move fairly regularly, especially lower-income migratory workers, who may work in one place during one season of the year and reside in another during other parts of the year. This movement may also occur within a day: for example, sleeping in a high-pollution exposed area but working in a low-exposure area, or vice versa.

The challenges of the person-based approach is developing a conceptually sound measure of long-term exposure to pollution. Typically, the long-term exposure level for a unit is considered to be fixed. However, in the IPL context, players may switch teams across seasons. Among players who played in more than one season in our sample, the mean number of times each player switched teams is 0.97,[66] with a maximum of 7. The traditional space-based approach to long-term exposure measures would suggest assigning a long-term $PM_{2.5}$ level to a player based on the location of their home-stadium, but this is problematic when their home stadium changes across years. We therefore explore alternative definitions of long-term exposure, discuss the advantages and drawbacks of each, and make a recommendation for how to measure this consistently in similar settings.

[66]The average number of team switches is over twice as high—2.04—when weighting players according to the number of deliveries in which they appear in our sample.

Appendix F Alternative definitions of short-term past exposure to $PM_{2.5}$

In addition to the measure of exposure defined in Section 4.3.1,

$$\overline{\mathrm{P}}_{J(j)d} \equiv \frac{1}{T}\sum_{\tau=1}^{T}\mathrm{P}_{J(j)\tau}, \tag{F.1}$$

we define two alternative measures of short-term past exposure:

1. Number of days in past T days where $PM_{2.5}$ was above Z threshold
   e.g., WHO thresholds, i.e., $Z \in \{25, 37.5, 50, 75\}$

$$P_{J(j)d} = \sum_{\tau=1}^{T}\mathbf{1}(P_{J(j)\tau} > Z) \tag{F.2}$$

2. Degree day analogous measure (number of units above Z threshold for each day above Z threshold in past T days)

$$P_{J(j)d} = \sum_{d=1}^{T}[\mathbf{1}(P_{J(j)\tau} > Z)](P_{j\tau} - Z) \tag{F.3}$$

We note that the measures in Equations F.2, F.3 are piece-wise linear functions of the measure in Equation F.1. We therefore expect them to yield somewhat similar results. However, each measure presumes a conceptually distinct hypothesis as to how past exposure to $PM_{2.5}$ may affect the body. Use of the measure in Equation F.1 would suggest that the mean level of exposure in the past, say, 30 days is meaningful. Measuring the mean, however, may obscure substantial variation in intensity of exposure over time. For example, in the hypothetical extreme case of someone who is exposed alternately to a day with perfectly clean air (i.e., 0 $\mu g\,m^{-3}$) and then a day with some of the most polluted air, say, 80 $\mu g\,m^{-3}$, the measure in Equation F.1 would be 40 $\mu g\,m^{-3}$. This would be the same as that measure for someone who is exposed to days of exactly 40 $\mu g\,m^{-3}$ each day, even though the experience of being exposed to extreme pollution and then clean air alternately may be quite different than being exposed to moderate pollution consistently.

The measure in Equation F.2 corrects for this coarseness to some extent by counting how many "bad" days of pollution there were for a team in a given time window, where "bad" is a threshold that can be adjusted. The underlying idea is that there is a certain threshold beyond which exposure to $PM_{2.5}$ is harmful and below which it is not. We use the thresholds that WHO sets (World Health Organization, 2021) for the Air Quality Guideline (25 $\mu g\,m^{-3}$, Interim Target 3 (37.5 $\mu g\,m^{-3}$), Interim Target 2 (50 $\mu g\,m^{-3}$) and Interim Target 1 (75 $\mu g\,m^{-3}$), and explore robustness of the measure to each threshold. Returning to the example above, using the threshold of 50 $\mu g\,m^{-3}$ would mean that the measure in Equation F.2 would be 15 in the alternating case and 0 in the consistent case.

The measure in Equation F.3 is a further refinement of that in Equation F.2—it quantifies not only whether a day has "bad" pollution, but also "how bad" it is; in other words, by how much it exceeds a given threshold. This method is analogous to a cooling degree day in the climate

literature. The underlying idea that it matters not only whether $PM_{2.5}$ exceeds a threshold, but by how much it exceeds it. Several studies in the epidemiological literature have adopted a similar approach (Lin et al., 2018; Chen et al., 2020; Xiao et al., 2022) but, to our knowledge, such an approach to measurement has not yet been explored in the economics literature. Returning to the first example and using a threshold of 50 $\mu g\,m^{-3}$, the team exposed to alternate 0 and 80 $\mu g\,m^{-3}$ days for 30 days would be assigned $30 \times 15 = 450$ for this measure, while the team exposed to 40 $\mu g\,m^{-3}$ consistently would be assigned 0. The advantage of Equation F.3 relative to Equation F.2 is that Equation F.3 makes it possible to distinguish between a scenario where on the "high" days in the alternating scenario the high is, say, 55 $\mu g\,m^{-3}$ versus 80 $\mu g\,m^{-3}$. The former measure would be the same in both cases (since the "high" day is above 50 in both cases) while the latter would reflect this variation.

For matches early in each season, the lookback window extends into the pre-season period, during which we assume teams are present at their home stadiums; we extend back 30 days before the season start, so that a 30-day window is always complete.[67] This assumption reflects the fact that teams tend to start training in their home stadiums several weeks before the season starts.

Note that we report results for the mean measure in Equation F.1 only, with its horizon determined by the distributed-lag specification of Section 5.3 (Table 4) rather than by a search over windows. The count and degree-day measures are defined here to make explicit the alternative hypotheses about how past exposure may operate.

Appendix G Additional robustness checks

G.1 Robustness: incorporating variation in number of runs scored

In our main specifications, we define the outcome as a binary indicator for whether a bowler conceded a run. This definition abstracts away from variation in the number of runs scored, conditional on scoring any run. As a robustness check, we estimate two models that incorporate variation in the number of runs scored. The first is an OLS regression where the outcome is the number of runs scored per delivery. Second, we estimate an ordered logit model which treats each number of runs scored as a discrete category with a higher number indicating poor performance for the bowler and bowling team:

$$\log\left(\frac{P(R_{ij\ell t} \leq r)}{P(R_{ij\ell t} > r)}\right) = \beta \mathrm{P}_{\ell d} + \mathbf{X}'_{\ell d}\gamma + \psi_j + \phi_i + \delta_{\ell y} + \mu_m + \theta_{n(t)} + \eta_{o(t)} + \Lambda_i + \Delta_j + \varepsilon_{ij\ell t} \quad \text{(G.1)}$$

where $R_{ij\ell t}$ is the number of runs scored ($R_{ij\ell t} \in \{0, 1, 2, 3, 4, 5, 6\}$). The OLS model results are reported in Table I12. The results estimating Equation G.1 are reported in Table I13. Both sets of results are qualitatively similar to those estimating analogous regressions in Tables 2, 5 and 6.

[67] For windows longer than 30 days, this would truncate the lookback for matches within the first weeks of a season and mechanically lower the count variables in Equations F.2 and F.3; with $T = 30$ the truncation does not bind.

G.2 Robustness: weather interactions

Our identification relies on match-day pollution being as-good-as-randomly assigned with respect to performance potential. The IPL's scheduling rules—matches scheduled months in advance based on stadium availability and broadcast considerations—strongly support this assumption. However, we address potential confounding from weather conditions that both influence $PM_{2.5}$ and directly affect performance.

We implement Post-Double Selection (PDS) Lasso (Belloni et al., 2014) with $PM_{2.5}$ as the treatment variable, regressing run-scoring on 29 potential controls: home stadium dummies, linear and quadratic weather terms, and pairwise weather interactions. The procedure selects no controls as outcome predictors but identifies one $PM_{2.5}$ predictor: humidity-wind interactions. The final regression yields a coefficient of 0.41 percentage points ($p < 0.01$), similar in magnitude to our baseline result of 0.42 percentage points, confirming our results are not driven by complex weather patterns.

As a benchmark, we repeat this procedure with temperature as the treatment variable. We cannot reject the null hypothesis that temperature's coefficient (-0.06 percentage points, $p = 0.595$) is zero, even though matches occur in warm conditions (21.7 °C to 37.3 °C with a mean of 29.5 °C) where heat stress might substantially impair performance.

G.3 Robustness: alternative measures of long-term past exposure to $PM_{2.5}$

Table I17 presents seven alternative definitions of long-term $PM_{2.5}$. Our preferred definition for long-term exposure to $PM_{2.5}$ is the mean exposure to $PM_{2.5}$ across all IPL matches (in all seasons) for a given bowler. This is reported in the first row and has as many distinct values as there are bowlers. The advantage of this measure is that it is bowler-specific, reflecting the fact that the bowler is the unit of observation that could potentially adapt over time. The downside of this measure is that it incorporates future data on $PM_{2.5}$ exposure into past matches. We find that pollution levels are serially correlated within bowler, which supports the use of this metric.

In the second row, we calculate a similar measure, but instead of taking the mean across all matches in the IPL (2008-2022) we split the panel in two (2008-2014 and 2015-2022) and take the mean exposure to $PM_{2.5}$ across matches in the first half of the panel. Since there are fewer distinct players in the first half of the IPL, the number of distinct values is substantially smaller. This measure addresses the temporal problem of the first measure since we can restrict estimating to the second half of the panel, using the first half as a baseline. A major drawback in doing so, however, is loss of statistical power from loss of sample size.

The remaining five rows of the table report means of $PM_{2.5}$ in the location of the home-stadium for a given IPL team. There are 15 teams that have been in the IPL over 2008–2022.[68] In row 3, we calculate this mean over the 10 years prior to the IPL (1998-2007) to get a baseline exposure level. In row 4, we calculate the mean over the period of the IPL (2008-2022). In the next two alternative definitions, we account for the fact that IPL matches occur only in three months of the year—March, April, and May—and construct measures analogous to those in rows 3 and 4. The final row uses the mean $PM_{2.5}$ at the team's home stadium over the 21 days before each season's start, which varies by team-season rather than by team. These measures all assign long-term pollution to a bowler on the

[68] Two pairs (Deccan Chargers and Sunrisers Hyderabad; Pune Warriors and Rising Pune Supergiant) share a home stadium.

basis of the team they play on, but since they can switch teams between seasons, these measures are not bowler-specific, but bowler-team specific. We include all alternative definitions of long-term exposure as robustness checks.

We report estimates of Equation 8 in Table I18 for the long-term definitions of $PM_{2.5}$ in row 2 of Table I17 (column 1); the definition in row 1 is our main specification (Table 6). For the definitions in rows 3–7, we alter the specification to include bowling team-by-year fixed effects instead of bowler fixed effects, since the assignment of long-term $PM_{2.5}$ to each bowler changes depending on what team they are on in each year.[69] We estimate the slightly modified specification:

$$R_{ij\ell t} = \beta_1 \mathrm{P}_{\ell d} + \beta_2 \mathrm{P}_{\ell d} \times \overline{\mathrm{P}}_{J(j,y)} + \mathbf{X}'_{\ell d}\gamma + \psi_{yJ(j,y)} + \phi_i + \delta_{\ell y} + \mu_m + \theta_{n(t)} + \eta_{o(t)} + \varepsilon_{ij\ell t} \qquad \text{(G.2)}$$

where $\psi_{yJ(j,y)}$ represent bowling team-by-year fixed effects. Note that $J(j, y)$ is a function that maps bowler j in year y to team J.

[69] Note that we include both bowling team-by-year fixed effects and stadium-by-year fixed effects since these represent two different things: the stadium is the stadium at which the match is played (which may or may not be a home stadium for either team), while the bowling team-by-year fixed effects control for the team's home stadium.

Appendix H Additional figures

Figure H1: Dose response of performance to pollution

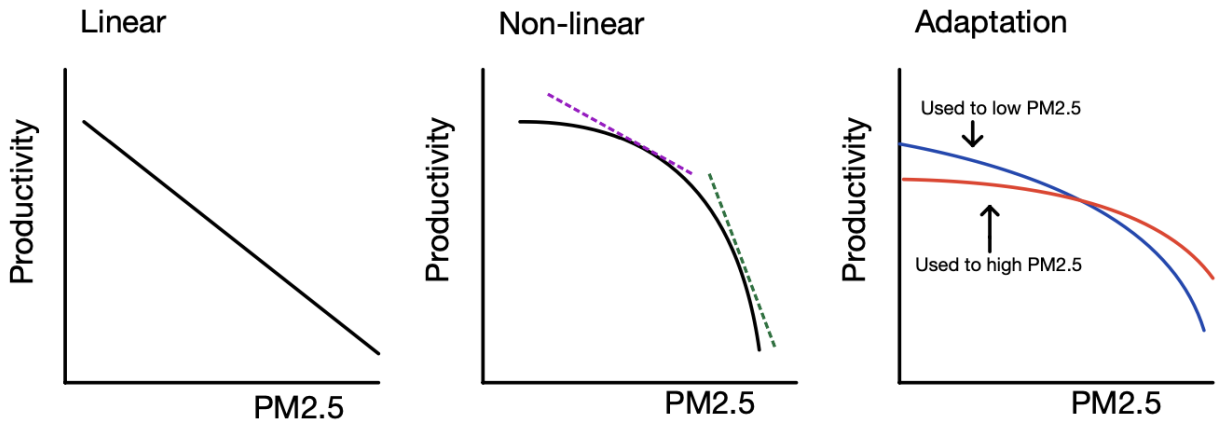
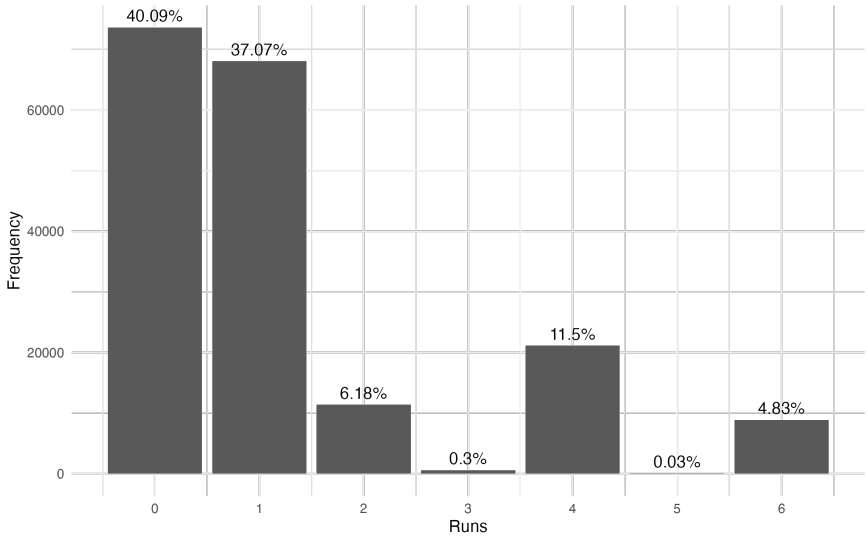
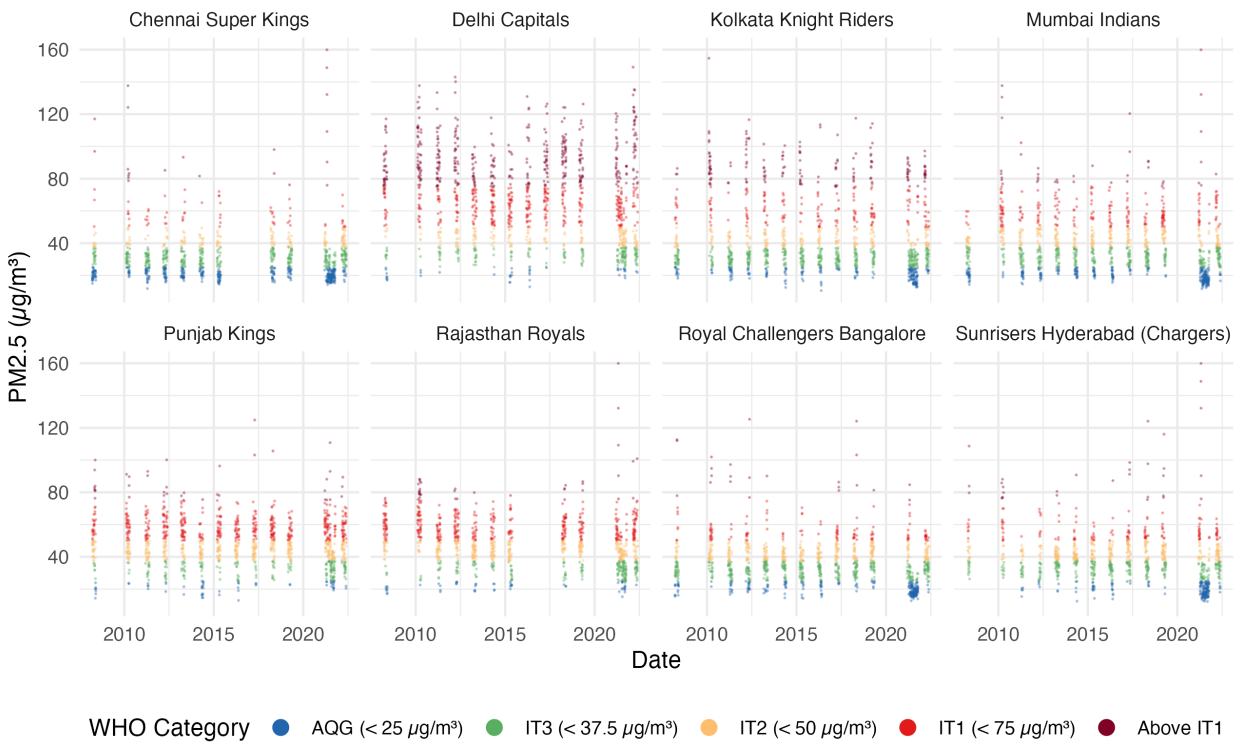


Figure H2: Histogram of runs



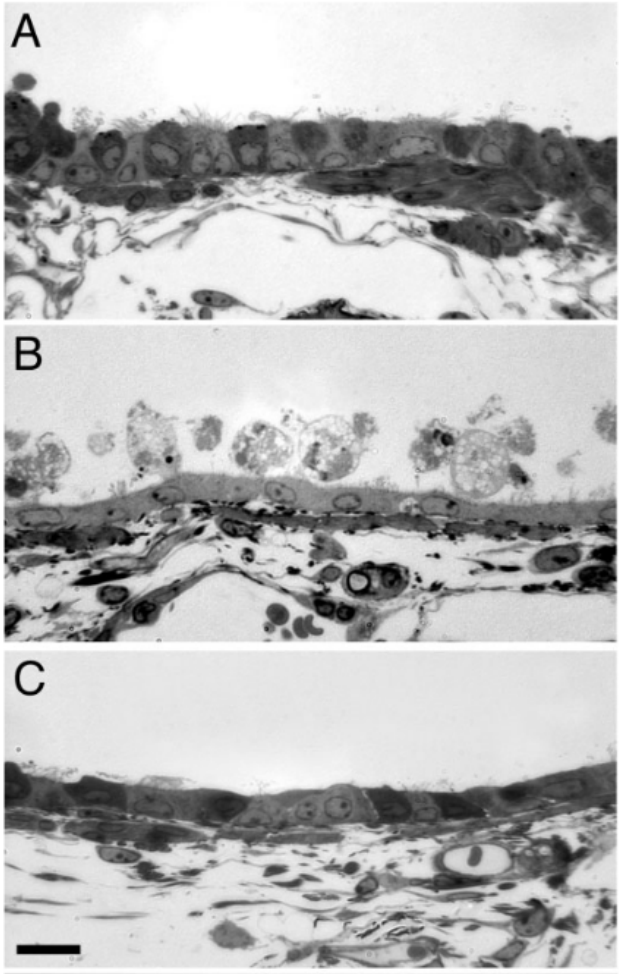
Notes. This figure shows the distribution of the outcome variable (runs scored) in our analysis. The unit of observation is a delivery (N=183,572).

Figure H3: Team $PM_{2.5}$ exposure histories in IPL, 2008–2022



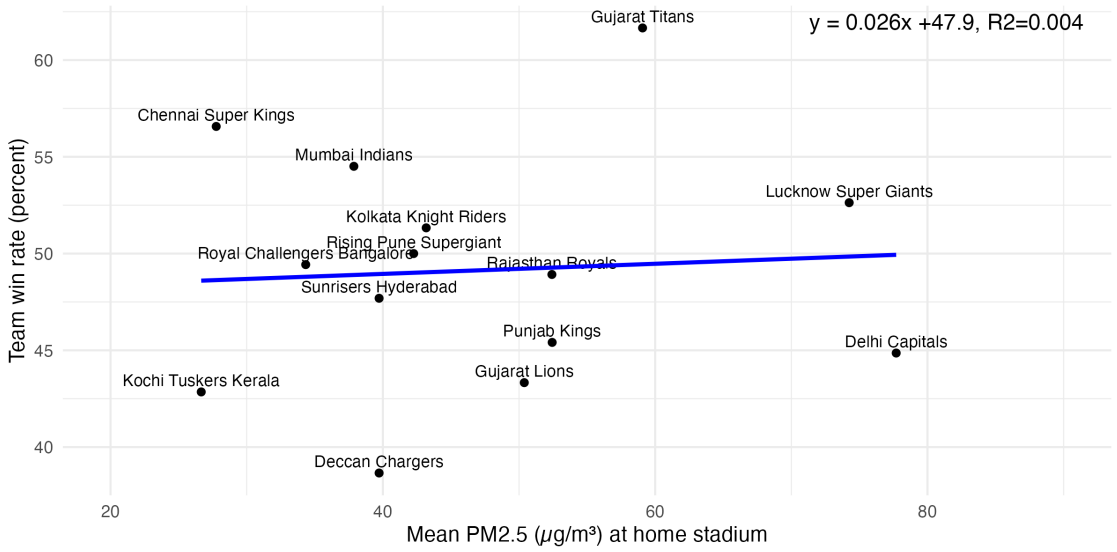
Notes. This figure displays the $PM_{2.5}$ exposure history of each IPL team in the years 2008–2022. Teams are assumed to be exposed to the level of $PM_{2.5}$ in their home stadium, unless they are playing a match at an away stadium, in which case they are exposed to the $PM_{2.5}$ at that location. The IPL takes place in three months each year and we assume teams are present and training in their home stadiums 30 days before the start date of each season. We do not interpolate $PM_{2.5}$ exposure between seasons, leading to blank spaces between the clusters of dots for each season.

Figure H4: Cellular adaptation to repeated exposure to air pollution



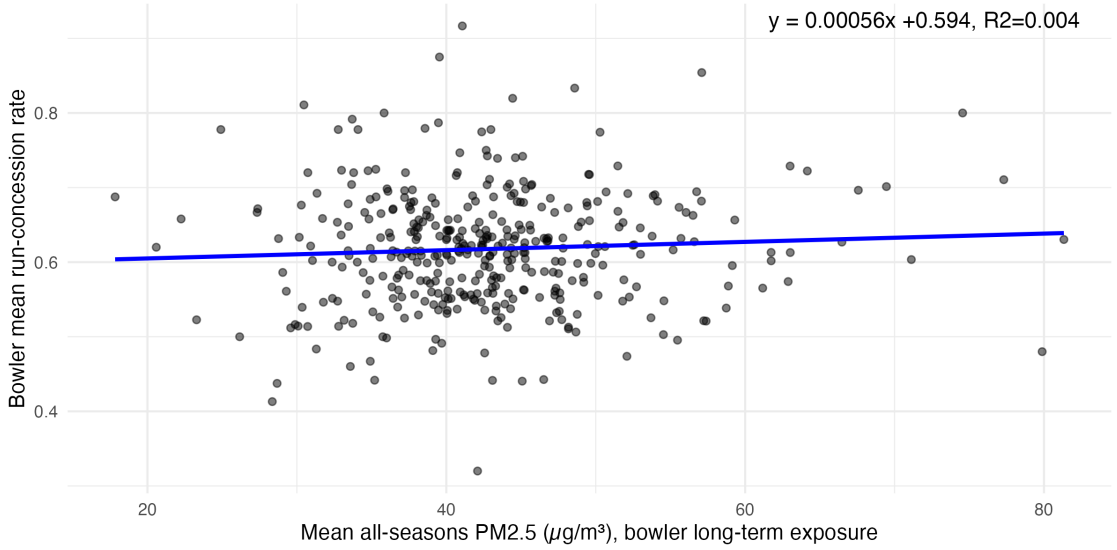
Notes. This figure, reproduced from West et al. (2003), displays three panels of microscopic images of cells in the lungs of mice. Panel A shows cells from mice that were in the control group and breathed clean air throughout the experiment. Panel B shows cells from mice from the treatment group that was exposed to polluted air for one day, and had their cells imaged 24 hours after. Panel C shows cells from mice that were exposed to the same level of air pollution as those in Panel B, except they were exposed for 7 days instead of 1. As for the cells in Panel B, their cells are imaged 24 hours after their last exposure.

Figure H5: Team quality is not correlated with long-term $PM_{2.5}$ exposure



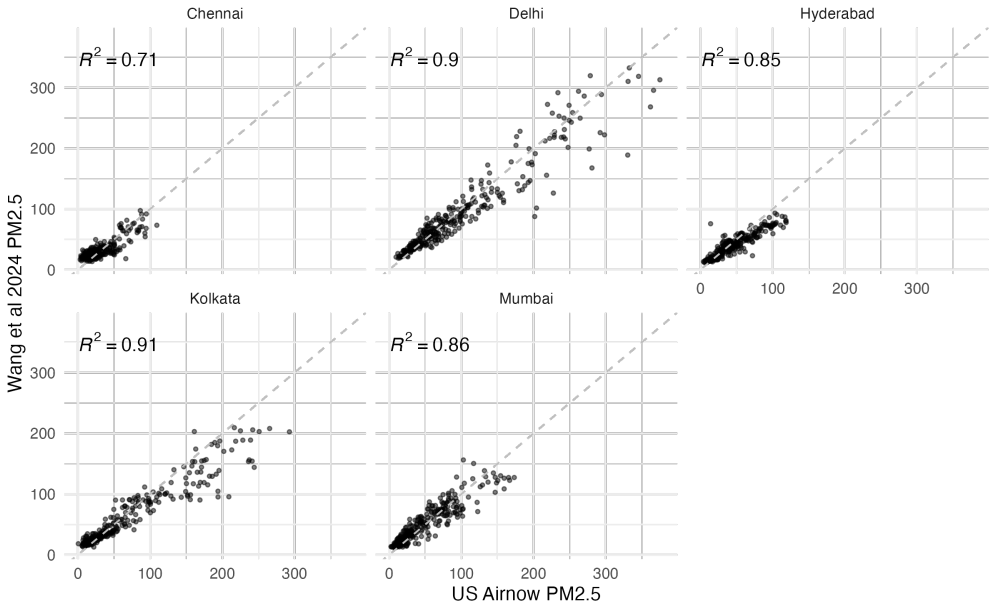
Notes. This figure shows the win rate (number of matches won out of all matches played) for IPL teams as plotted against the average $PM_{2.5}$ at the team's home stadium. Average $PM_{2.5}$ is defined as the mean level of $PM_{2.5}$ at the team's home stadium during March, April, and May (the months when IPL matches typically occur) for the study period, 2008-2022. The lack of significant relationship between win rate and $PM_{2.5}$ is robust to alternative definitions of long-term $PM_{2.5}$, such as including non-IPL months, or including $PM_{2.5}$ levels before the IPL franchise began. Note that the Pune Warriors, who played only 2011-2013 and had the lowest win-rate (26.7%, more than 2 standard deviations below the mean) are excluded from this graph. The lack of significant relationship also holds when including them.

Figure H6: Bowler quality is not correlated with long-term $PM_{2.5}$ exposure



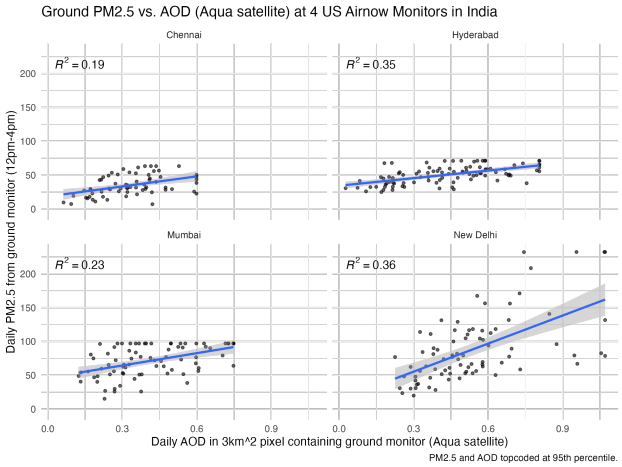
Notes. This figure shows the run-concession rate (number of deliveries that result in a run divided by the total number deliveries) for bowlers as plotted against the average $PM_{2.5}$ the bowler is exposed to across all matches in the IPL (i.e., career $PM_{2.5}$ in Equation 8). Bowlers who deliver fewer than 24 deliveries (the theoretical maximum for one match) are excluded from this plot; this results in excluding 56 bowlers. The non-significant relationship holds when including these bowlers.

Figure H7: U.S. AirNow vs. Wang et al. (2024) daily $PM_{2.5}$



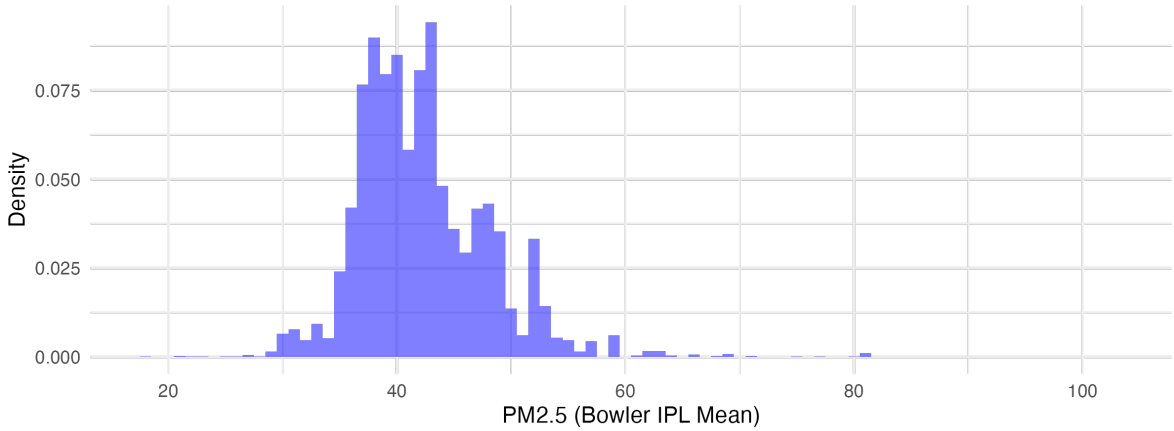
Notes. This figure displays the correlation between Wang et al. (2024) estimates of $PM_{2.5}$ and ground observations of $PM_{2.5}$ at five cities with a U.S. AirNow ground monitor, a network of high-quality monitors that are calibrated to EPA standards. Each dot represents a daily mean value of $PM_{2.5}$. The 45 degree line is shown for reference. Values of Wang et al. (2024) $PM_{2.5}$ are taken from the 10 km $\times$ 10 km grid cell containing the latitude and longitude coordinates of the U.S. AirNow ground monitor. Importantly, the U.S. AirNow network is not included as training data for the Wang et al. (2024) model.

Figure H8: U.S. AirNow vs. MODIS AOD



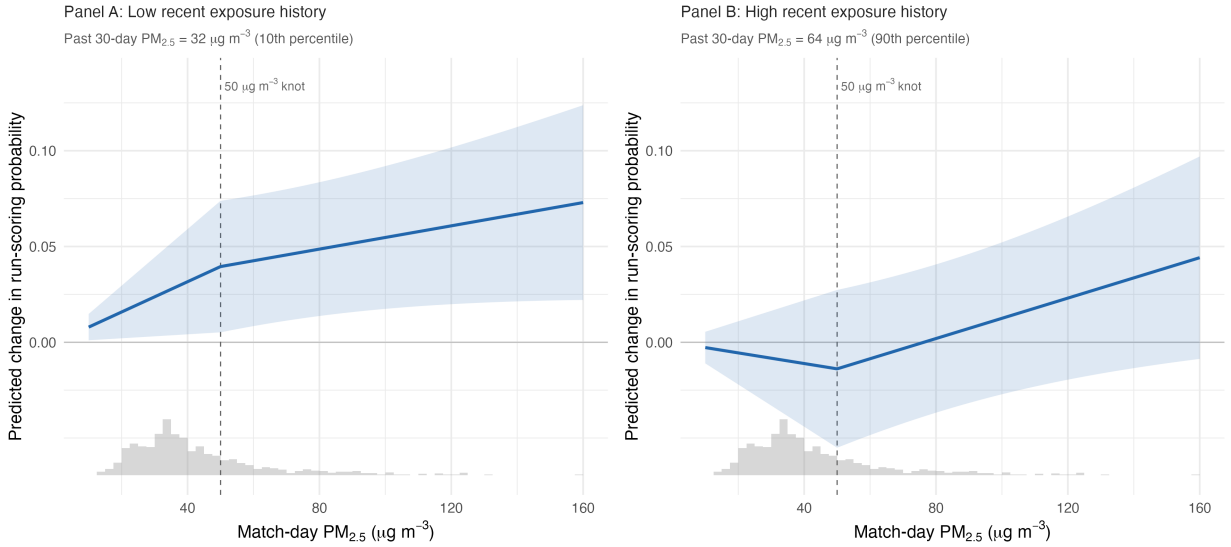
Notes. This figure displays the correlation between Aerosol Optical Depth from the MODIS Aqua satellite (which passes over India approximately once per day in the afternoon) with ground observations of $PM_{2.5}$ at five cities with a U.S. AirNow ground monitor. Each dot represents a daily mean value of $PM_{2.5}$. AOD is calculated as the mean AOD in the 3 km $\times$ 3 km pixel containing the ground monitor. Both $PM_{2.5}$ and AOD are topcoded at the 95th percentile.

Figure H9: Histogram of bowler career $PM_{2.5}$ mean in IPL matches 2008–2022



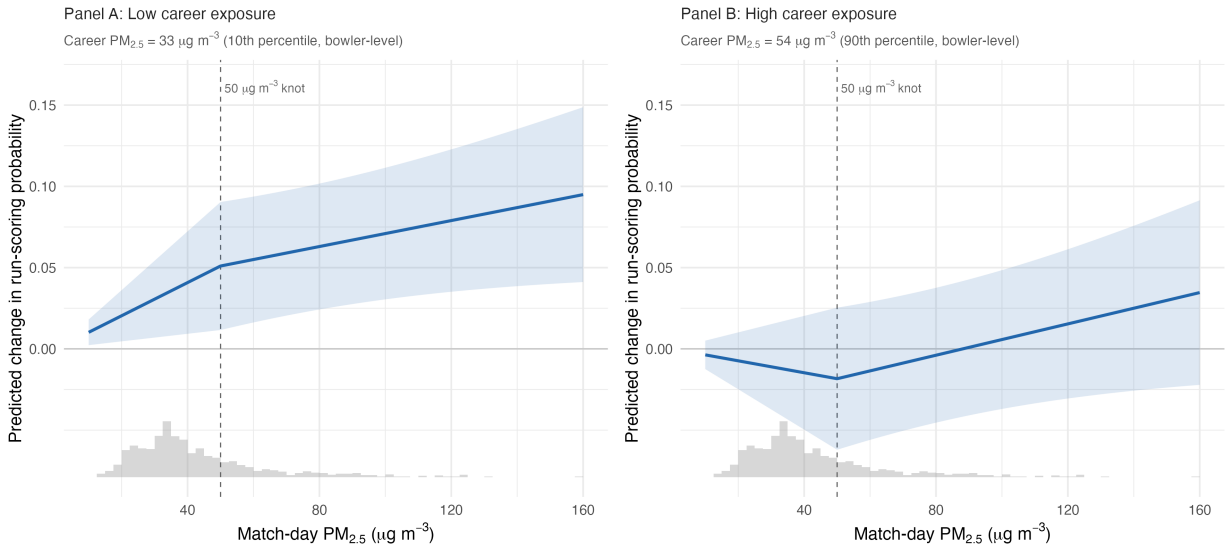
Notes. This figure displays the distribution of bowler mean $PM_{2.5}$ defined as the mean $PM_{2.5}$ exposure across all matches for a given bowler in the IPL 2008-2022 (i.e., $\overline{P}_{j0}$ in Equation 8).

Figure H10: Adaptation to $PM_{2.5}$ by short-term exposure history



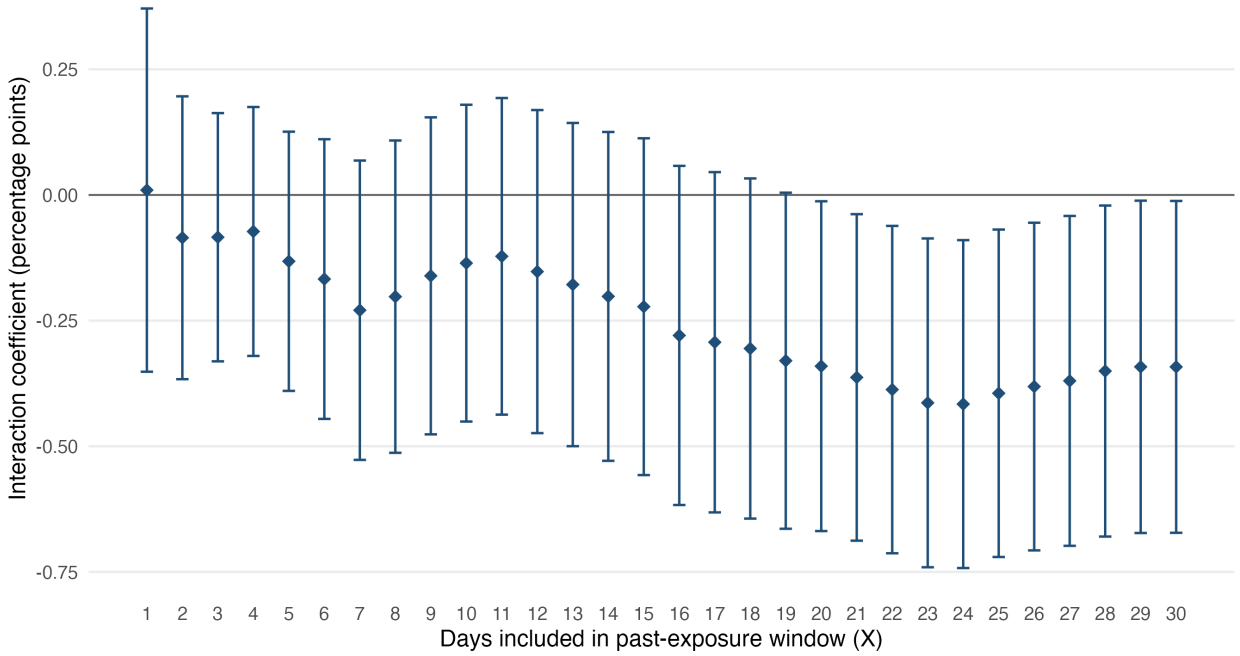
Notes. This figure plots the model-implied effect of match-day $PM_{2.5}$ on run-scoring probability, relative to $PM_{2.5} = 0$, holding past 30-day $PM_{2.5}$ exposure fixed at its 10th percentile (Panel A) or 90th percentile (Panel B) across deliveries. Effects come from a linear spline in match-day $PM_{2.5}$ with a single knot at 50 $\mu g\,m^{-3}$, interacted with past 30-day $PM_{2.5}$, with weather controls and fixed effects for stadium-by-year, bowler, batter, innings, month, and over phase. Shaded bands are 90% confidence intervals from standard errors two-way clustered by match and bowler. Tick marks along the x-axis show the distribution of match-day $PM_{2.5}$ in the estimation sample.

Figure H11: Adaptation to $PM_{2.5}$ by career exposure history



Notes. This figure plots the model-implied effect of match-day $PM_{2.5}$ on run-scoring probability, relative to $PM_{2.5} = 0$, holding a bowler's career $PM_{2.5}$ exposure (mean across all IPL matches) fixed at its 10th percentile (Panel A) or 90th percentile (Panel B) across bowlers. Effects come from a linear spline in match-day $PM_{2.5}$ with a single knot at 50 $\mu g\,m^{-3}$, interacted with career $PM_{2.5}$ (which enters the interaction only, since it is bowler-constant and otherwise absorbed by the bowler fixed effect), with weather controls and fixed effects for stadium-by-year, bowler, batter, innings, month, and over phase. Shaded bands are 90% confidence intervals from standard errors two-way clustered by match and bowler. Tick marks along the x-axis show the distribution of match-day $PM_{2.5}$ in the estimation sample.

Figure H12: Robustness of short-term adaptation across varying past-exposure windows



Notes. Figure plots point estimates and 95% confidence intervals for the coefficient on the interaction term "0–50 $\mu g\, m^{-3} \times$ Past X-day $PM_{2.5}$" as in Table 5 for X ranging from 1 to 30.

Appendix I Additional tables

Table I1: $PM_{2.5}$ and weather correlation matrix

	Run	$PM_{2.5}$	Temperature	Precipitation	Radiation	Wind	Humidity
Run	1	0.017	0.007	-0.002	0.003	0.001	-0.014
$PM_{2.5}$	0.017	1	0.218	-0.153	0.230	-0.182	-0.619
Temperature	0.007	0.218	1	-0.254	0.232	0.234	-0.366
Precipitation	-0.002	-0.153	-0.254	1	-0.520	-0.070	0.253
Radiation	0.003	0.230	0.232	-0.520	1	0.108	-0.385
Wind	0.001	-0.182	0.234	-0.070	0.108	1	0.155
Humidity	-0.014	-0.619	-0.366	0.253	-0.385	0.155	1

Notes. Run is a binary indicator equal to 1 if at least one run is scored on a delivery and 0 otherwise. Correlations computed from delivery-level data. $PM_{2.5}$ and weather variables are measured at the match-level.

Table I2: $PM_{2.5}$ exposure and run-scoring probability: adding day-of-week fixed effects

	(1)	(2)	(3)	(4)	(5)
		Linear		Quintiles	Spline
	$\mathbb{1}$ (At least one run scored)				
Match-day $PM_{2.5}$	0.0041***	0.0027*	0.0040**		
	(0.0010)	(0.0014)	(0.0017)		
Q2 (Match-day $PM_{2.5}$)				0.0083	
				(0.0060)	
Q3 (Match-day $PM_{2.5}$)				0.010	
				(0.0070)	
Q4 (Match-day $PM_{2.5}$)				0.014	
				(0.0090)	
Q5 (Match-day $PM_{2.5}$)				0.028***	
				(0.010)	
Match-day $PM_{2.5}$ between 0 and 50 $\mu g\,m^{-3}$					0.0043
					(0.0037)
Match-day $PM_{2.5}$ between 50 and 160 $\mu g\,m^{-3}$					0.0038*
					(0.0020)
Weather controls		✓	✓	✓	✓
Bowler FE			✓	✓	✓
Batter FE			✓	✓	✓
Stadium-by-year FE			✓	✓	✓
Month FE			✓	✓	✓
Home stadium FE			✓	✓	✓
Innings FE			✓	✓	✓
Over phase FE			✓	✓	✓
Day-of-week FE			✓	✓	✓
N	183,572	183,572	183,556	183,556	183,556
R^2	0.00028	0.00039	0.047	0.047	0.047

Notes. The outcome variable is a binary indicator equal to 1 if at least one run is scored on a delivery and 0 otherwise (mean 0.599). All $PM_{2.5}$ variables are measured in 10 $\mu g\,m^{-3}$. Quintile 1 of match-day $PM_{2.5}$ is 14-27; Quintile 2 is 27-34, Quintile 3 is 34-41, Quintile 4 is 41-53, and Quintile 5 is 53-160 $\mu g\,m^{-3}$. Columns (1–3) enter match-day $PM_{2.5}$ linearly, adding weather controls in column (2) and fixed effects in column (3). Column (4) replaces the linear term with $PM_{2.5}$ quintile indicators, where the lowest quintile (Q1) is the omitted category. Column (5) instead enters match-day $PM_{2.5}$ as a continuous piecewise-linear spline with a single knot at 50 $\mu g\,m^{-3}$. Columns (1)-(2) are identical to the corresponding columns of Table 2, since neither carries any fixed effects yet. Columns (2)–(5) include controls for weather conditions including temperature, temperature squared, relative humidity, wind speed, precipitation, atmospheric pressure, and solar radiation. Columns (3)–(5) additionally include fixed effects for individual bowlers, batters, stadium-by-year, month, innings, and over phase and, unlike Table 2, for the day of week of the match. Match-day $PM_{2.5}$ is orthogonal to day of week ($p = 0.986$, a joint test conditional on stadium-by-season fixed effects), so this row checks whether Table 2's estimates are sensitive to scheduling patterns rather than testing for a substantively new confound. Following Correia (2015), 16 singleton observations are dropped in columns (3)–(5). Standard errors (in parentheses) are two-way clustered at the match and bowler levels. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table I3: $PM_{2.5}$ and bowler-specific outcomes, pace and spin bowlers: per-over fixed effects

	(1)	(2)	(3)	(4)	(5)	(6)
	No-ball (=1)		Length (m)		Speed (km/h)	
	Linear	Quintile	Linear	Quintile	Linear	Quintile
Panel A. Pace bowlers only						
Match-day $PM_{2.5}$	0.00046**		-0.02534*		0.09643	
	(0.00020)		(0.01523)		(0.06601)	
Q2 (Match-day $PM_{2.5}$)		0.00093		-0.12151*		0.02253
		(0.00090)		(0.06480)		(0.23399)
Q3 (Match-day $PM_{2.5}$)		0.00170*		-0.20463***		-0.06866
		(0.00097)		(0.06997)		(0.25666)
Q4 (Match-day $PM_{2.5}$)		0.00304***		-0.23251***		0.12297
		(0.00106)		(0.07962)		(0.28551)
Q5 (Match-day $PM_{2.5}$)		0.00287**		-0.28731***		0.34882
		(0.00111)		(0.08762)		(0.31745)
Outcome mean	0.0053	0.0053	6.8770	6.8770	127.7380	127.7380
N	118,508	118,508	80,913	80,913	79,346	79,346
R^2	0.0137	0.0137	0.1492	0.1494	0.7027	0.7027
Panel B. Spin bowlers only						
Match-day $PM_{2.5}$	-0.00011		-0.00144		-0.00886	
	(0.00016)		(0.01375)		(0.05865)	
Q2 (Match-day $PM_{2.5}$)		0.00101*		-0.07242*		-0.33202
		(0.00053)		(0.04338)		(0.29127)
Q3 (Match-day $PM_{2.5}$)		0.00084		-0.01967		-0.57915**
		(0.00051)		(0.05008)		(0.28985)
Q4 (Match-day $PM_{2.5}$)		0.00116		-0.04706		-0.50099
		(0.00103)		(0.07042)		(0.36734)
Q5 (Match-day $PM_{2.5}$)		0.00025		-0.01496		-0.74104*
		(0.00118)		(0.10247)		(0.38370)
Outcome mean	0.0014	0.0014	5.1843	5.1843	90.5525	90.5525
N	59,926	59,926	35,477	35,477	34,605	34,605
R^2	0.0145	0.0145	0.0818	0.0819	0.3381	0.3385
Weather controls	✓	✓	✓	✓	✓	✓
Bowler, batter, stadium-by-year, month, home stadium, innings, and over FE	✓	✓	✓	✓	✓	✓

Notes. This table repeats Table 3 with full per-over fixed effects in place of over-phase fixed effects. Each pair of columns shares a dependent variable: no-ball (=1) in columns (1)-(2), delivery length in columns (3)-(4), and delivery speed in columns (5)-(6). The two panels share both the specification and the outcomes and differ only in which deliveries enter the sample. All three outcomes are produced by the bowler alone, independently of what the batter does, and so isolate bowler impairment. No-ball (=1) indicates a delivery ruled illegal on account of a bowler fault. Length is the distance in meters from the batter's stumps to the point where the delivery bounces, so a negative coefficient means fuller deliveries. Speed is the measured speed of the delivery in kilometers per hour, bottom-coded at the within-bowler-type (pace/spin) 5th percentile. In the first column of each pair, match-day $PM_{2.5}$ enters linearly and is measured in 10 $\mu g\,m^{-3}$; the second column instead enters indicators for the quintile of match-day $PM_{2.5}$, with the lowest quintile (Q1) omitted, so the reported coefficients are level shifts against Q1 in the outcome's own units. Quintiles are cut separately on each outcome's own estimation sample. Every regression includes controls for weather conditions including temperature, temperature squared, relative humidity, wind speed, precipitation, pressure, and solar radiation and fixed effects for individual bowlers, batters, stadium-by-year, month, innings, and over, as well as indicators for whether the bowling and batting teams are at their home stadium. Panel A restricts the sample to the 118,524 deliveries bowled by pace bowlers and Panel B to the 59,945 bowled by spin bowlers. (An additional 5,103 deliveries whose bowler could not be matched to bowling-style metadata enter neither panel, so the two panels do not sum to the 183,572 deliveries in the full sample.) Length and speed are ball-tracking measures observed only on the subset of 512 matches matched to Hawk-Eye data, so columns (3)-(6) are estimated on smaller samples than columns (1)-(2); for those columns bowler type is taken from the Hawk-Eye delivery-level bowling style rather than from the player metadata. Following Correia (2015), singleton observations are dropped, and sample sizes are reported for every column and panel. Standard errors (in parentheses) are two-way clustered at the match and bowler levels. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table I4: $PM_{2.5}$ and speed decline within a bowling spell

	All	Pace	Spin
	Delivery speed (km/h)		
Delivery number in spell	-0.06948***	-0.14037***	0.08913***
	(0.01987)	(0.01999)	(0.02551)
$PM_{2.5}$ × delivery number in spell	-0.00255	-0.00037	-0.00520
	(0.00251)	(0.00297)	(0.00369)
N	113,975	79,360	34,615

Notes. The unit of observation is a delivery. The outcome is delivery speed in km/h. Each column represents a separate regression. A bowling spell is defined as a set of deliveries that a bowler delivers consecutively, without rest. The delivery number is defined as the sequential position within the bowler's spell (i.e. 1 for the 1st delivery within the spell, 2 for the second, etc.). Each column shows delivery speed regressed on delivery number and delivery number interacted with match-day $PM_{2.5}$. Each regression includes a fixed effect for each spell. Standard errors in parentheses are two-way clustered at the match and bowler levels. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table I5: Fast bowlers: $PM_{2.5}$ exposure and run-scoring probability

	(1) 𝟙 (At least one run scored)
Match-day $PM_{2.5}$	0.0039*
	(0.0020)
Match-day $PM_{2.5}$ X Fast bowler	0.00029
	(0.0014)
Weather controls	✓
Bowler FE	✓
Batter FE	✓
Stadium-by-year FE	✓
Month FE	✓
Home stadium FE	✓
Innings FE	✓
Over phase FE	✓
N	178,452
R^2	0.047

Notes. The outcome variable is a binary indicator equal to 1 if at least one run is scored on a delivery and 0 otherwise (mean 0.599). $PM_{2.5}$ is measured in 10 $\mu g\, m^{-3}$. Fast is a dummy variable equal to 1 if the bowler is a fast bowler; 0 if otherwise. 61.6% of bowlers are fast bowlers. The categorization is unobserved for 12 bowlers; 5,103 observations associated with these bowlers are dropped. Columns (1) presents results from a regression of this run indicator on match-day $PM_{2.5}$ levels (measured in 10 $\mu g\, m^{-3}$). It includes fixed effects for individual batters, bowlers, stadium-by-year, home stadium, month, innings, and over phase. It includes controls for weather conditions including temperature, temperature squared, relative humidity, precipitation, atmospheric pressure, solar radiation, and wind speed. Following Correia (2015), 17 singleton observations are dropped. Standard errors (in parentheses) are two-way clustered at the match and bowler levels. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table I6: $PM_{2.5}$ exposure and run-scoring probability: alternate specification with bowler-batter pair fixed effects

	(1)	(2)	(3)	(4)	(5)
		Linear		Quintiles	Spline
		$\mathbb{1}$ (At least one run scored)			
Match-day $PM_{2.5}$	0.0041***	0.0027*	0.0040*		
	(0.0010)	(0.0014)	(0.0023)		
Q2 (Match-day $PM_{2.5}$)				0.012*	
				(0.0064)	
Q3 (Match-day $PM_{2.5}$)				0.0078	
				(0.0083)	
Q4 (Match-day $PM_{2.5}$)				0.022**	
				(0.011)	
Q5 (Match-day $PM_{2.5}$)				0.036***	
				(0.013)	
Match-day $PM_{2.5}$ between 0 and 50 $\mu g\,m^{-3}$					0.0061
					(0.0045)
Match-day $PM_{2.5}$ between 50 and 160 $\mu g\,m^{-3}$					0.0031
					(0.0026)
Weather controls		✓	✓	✓	✓
Bowler-batter pair FE			✓	✓	✓
Stadium-by-year FE			✓	✓	✓
Month FE			✓	✓	✓
Home stadium FE			✓	✓	✓
Innings FE			✓	✓	✓
Over phase FE			✓	✓	✓
N	183,572	183,572	181,556	181,556	181,556
R^2	0.00028	0.00039	0.15	0.15	0.15

Notes. The outcome variable is a binary indicator equal to 1 if at least one run is scored on a delivery and 0 otherwise (mean 0.599). All $PM_{2.5}$ variables are measured in 10 $\mu g\,m^{-3}$. Quintile 1 of match-day $PM_{2.5}$ is 14-27; Quintile 2 is 27-34, Quintile 3 is 34-41, Quintile 4 is 41-53, and Quintile 5 is 53-160 $\mu g\,m^{-3}$. Columns (1–3) enter match-day $PM_{2.5}$ linearly, adding weather controls in column (2) and fixed effects in column (3). Column (4) replaces the linear term with $PM_{2.5}$ quintile indicators, where the lowest quintile (Q1) is the omitted category. Column (5) instead enters match-day $PM_{2.5}$ as a continuous piecewise-linear spline with a single knot at 50 $\mu g\,m^{-3}$. Columns (1)-(2) are identical to the corresponding columns of Table 2, since neither carries any fixed effects yet. Columns (3)–(5) replace the individual bowler and batter fixed effects with a single fixed effect for each bowler-batter pair, and additionally include fixed effects for stadium-by-year, month, home stadium, innings, and over phase. Columns (2)–(5) include controls for weather conditions including temperature, temperature squared, relative humidity, wind speed, precipitation, atmospheric pressure, and solar radiation. Following Correia (2015), 2,016 singleton observations are dropped in columns (3)–(5). Standard errors (in parentheses) are two-way clustered at the match and bowler levels. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table I7: Bowler experience and match-day $PM_{2.5}$

	(1) Share of bowlers with ≥ 3 prior IPL seasons
Match-day $PM_{2.5}$	0.00101 (0.00323)
Weather controls	✓
Stadium-by-year FE	✓
Team FE	✓
N	1,544
R^2	0.78166

Notes. The unit of observation is team-match (i.e. one row per team per match). The dependent variable is the share of a team's bowlers used in a match who have already completed at least 3 prior IPL seasons as of the match's season (mean 0.43). $PM_{2.5}$ is measured in 10 $\mu g\, m^{-3}$. The regression includes fixed effects for stadium-by-year and team, and controls for temperature, temperature squared, relative humidity, precipitation, atmospheric pressure, solar radiation, and wind speed. Standard errors (in parentheses) are clustered at the match level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table I8: Career $PM_{2.5}$ exposure and order in the bowling sequence

	(1) Bowling order	(2) First in bowling order (0/1)
Match-day $PM_{2.5}$	-0.02932	0.01287
	(0.04008)	(0.01004)
Match-day $PM_{2.5}$ × Career $PM_{2.5}$	0.00584	-0.00300
	(0.00878)	(0.00222)
Weather controls	✓	✓
Bowler FE	✓	✓
Stadium-by-year FE	✓	✓
Innings FE	✓	✓
N	9,097	9,097
R^2	0.53848	0.38284

Notes. The unit of observation is a bowler's spell. The outcome in column 1 is bowling order. Bowling order is defined as the rank of introduction (1 = first bowler used that innings, 2, second bowler used that innings, etc.; mean 3.49). The outcome in column 2 is an indicator for that takes the value 1 if bowling order is equal to 1; 0 otherwise (mean 0.17). $PM_{2.5}$ variables are measured in 10 $\mu g\, m^{-3}$. Career $PM_{2.5}$ is the bowler's mean match-day $PM_{2.5}$ across all matches played in the IPL. All columns include fixed effects for individual bowlers, stadium-by-year, and innings, and controls for temperature, temperature squared, relative humidity, precipitation, atmospheric pressure, solar radiation, and wind speed. Standard errors (in parentheses) are clustered at the match level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table I9: $PM_{2.5}$ exposure and run-scoring probability (ground-monitor $PM_{2.5}$)

	(1)	(2)	(3)	(4)	(5)
		Linear		Quintiles	Spline
		$\mathbb{1}$ (At least one run scored)			
Match-day $PM_{2.5}$	0.0030***	0.0021	0.0018		
	(0.00090)	(0.0015)	(0.0015)		
Q2 (Match-day $PM_{2.5}$)				-0.0033	
				(0.011)	
Q3 (Match-day $PM_{2.5}$)				-0.0070	
				(0.011)	
Q4 (Match-day $PM_{2.5}$)				-0.013	
				(0.014)	
Q5 (Match-day $PM_{2.5}$)				-0.0048	
				(0.017)	
Match-day $PM_{2.5}$ between 0 and 50 $\mu g\,m^{-3}$					-0.0029
					(0.0045)
Match-day $PM_{2.5}$ over 50 $\mu g\,m^{-3}$					0.0026*
					(0.0015)
Weather controls		✓	✓	✓	✓
Bowler FE			✓	✓	✓
Batter FE			✓	✓	✓
Stadium-by-year FE			✓	✓	✓
Month FE			✓	✓	✓
Home stadium FE			✓	✓	✓
Innings FE			✓	✓	✓
Over phase FE			✓	✓	✓
N	50,355	50,355	50,342	50,342	50,342
R^2	0.00043	0.00067	0.048	0.048	0.048

Notes. This table restricts observations to those from stadiums colocated with a ground $PM_{2.5}$ monitor (N=50,355 vs. N=183,572 deliveries in the full sample). The outcome variable is a binary indicator equal to 1 if at least one run is scored on a delivery and 0 otherwise (mean 0.602 in this subsample). $PM_{2.5}$, measured in 10 $\mu g\,m^{-3}$, are 24-hour averages reported from the US AirNow air quality monitoring network at 5 U.S. consulates in Chennai, Kolkata, New Delhi (the embassy), Mumbai, and Hyderabad. Distances between the ground monitor at the consulate and the cricket stadium in each of these 5 respective cities ranges from 1.63 to 23.4 km with a mean of 10.9 km (median 11.0 km). Quintile 1 of ground-monitor match-day $PM_{2.5}$ is 7-24; Quintile 2 is 24-32, Quintile 3 is 33-43, Quintile 4 is 43-62, and Quintile 5 is 62-296 $\mu g\,m^{-3}$. Columns (1–3) enter match-day $PM_{2.5}$ linearly, adding weather controls in column (2) and fixed effects in column (3). Column (4) replaces the linear term with $PM_{2.5}$ quintile indicators, where the lowest quintile (Q1) is the omitted category. Column (5) instead enters match-day $PM_{2.5}$ as a continuous piecewise-linear spline with a single knot at 50 $\mu g\,m^{-3}$, the same knot as Table 2's column (5). Columns (2)–(5) include controls for weather conditions including temperature, temperature squared, relative humidity, precipitation, atmospheric pressure, solar radiation, and wind speed. Columns (3)–(5) additionally include fixed effects for individual batters, bowlers, stadium-by-year, month, home stadium, innings, and over phase. Following Correia (2015), 13 singleton observations are dropped in columns (3)–(5). Standard errors (in parentheses) are two-way clustered at the match and bowler levels. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table I10: Almon distributed lag over 30 days: unrestricted lags and polynomial degrees 0 to 4

	(1)	(2)	(3)	(4)	(5)	(6)
		Almon polynomial in lag order				
	Unrestricted lags	$Q = 0$	$Q = 1$	$Q = 2$	$Q = 3$	$Q = 4$
		$\mathbb{1}$ (At least one run scored)				
Match-day $PM_{2.5}$	0.00672***	0.00411**	0.00410**	0.00418**	0.00410**	0.00417**
	(0.00228)	(0.00177)	(0.00177)	(0.00176)	(0.00182)	(0.00183)
Cumulative effect of past exposure, lags 1 through:						
10 days	-0.00136	0.00119**	0.00128	0.00088	0.00097	0.00097
	(0.00275)	(0.00053)	(0.00148)	(0.00152)	(0.00153)	(0.00153)
20 days	0.00148	0.00239**	0.00248	0.00266	0.00277	0.00259
	(0.00326)	(0.00107)	(0.00179)	(0.00187)	(0.00188)	(0.00208)
30 days (whole window)	-0.00018	0.00358**	0.00360**	0.00343**	0.00353**	0.00344*
	(0.00300)	(0.00160)	(0.00160)	(0.00163)	(0.00165)	(0.00175)
p: profile is degree Q	—	0.0567	0.0479	0.0525	0.0516	0.0395
R^2 given up vs col. (1)	—	0.00024	0.00024	0.00024	0.00024	0.00024
p: profile is flat	—	—	0.9519	0.8670	0.9474	0.9813
N	183,556	183,556	183,556	183,556	183,556	183,556
R^2	0.0474	0.0472	0.0472	0.0472	0.0472	0.0472
Weather controls	✓	✓	✓	✓	✓	✓
Bowler, batter, stadium-by-year, month, home stadium, innings, and over phase FE	✓	✓	✓	✓	✓	✓

Notes. The outcome variable is a binary indicator equal to 1 if at least one run is scored on a delivery and 0 otherwise (mean 0.599). All $PM_{2.5}$ variables are measured in 10 $\mu g\,m^{-3}$. Every column is the specification of Table 2, column 3, plus the bowling team's $PM_{2.5}$ on each of the 30 days before the match; the columns differ only in what is assumed about the shape of the lag profile. Column (1) leaves the 30 lag coefficients β_s unrestricted. Columns (2) through (6) restrict the profile to a polynomial of degree Q in the lag index s, as in Equation 6; column (4), $Q = 2$, is the model reported in Table 4. Degree 0 is a flat profile, which is a regression on the simple 30-day mean of past exposure used elsewhere in the paper: its whole-window total, 0.00358, reproduces the coefficient on that mean estimated directly, 0.00358, to every reported digit. The cumulative rows report $\sum_{s=1}^{k} \beta_s$, the effect of raising $PM_{2.5}$ by 10 $\mu g\,m^{-3}$ on each of the k days before the match; each is a linear combination of the estimated coefficients with a delta-method standard error. The orthonormal basis $g_0, \ldots, g_Q$ of Equation 6 can be completed to a basis $g_0, \ldots, g_{30-1}$ spanning all 30 lags, so column (1) is exactly reparameterised as a regression on 30 such terms and the degree-Q restriction is $\alpha_{Q+1} = \cdots = \alpha_{30-1} = 0$. The first p-value tests that restriction against column (1) with a cluster-robust Wald F-test on $30 - 1 - Q$ degrees of freedom; the row beneath reports the R^2 of column (1) minus the column's own, the fit the restriction gives up. The second p-value is the joint test within each column that the shape terms $\alpha_1, \ldots, \alpha_Q$ are zero, i.e. that no day in the window matters more than any other; it is undefined at degree 0. The whole-window total of the degree-2 model is statistically indistinguishable from the unrestricted sum in column (1) (Hausman-style $p = 0.151$). Following Correia (2015), 16 singleton observations are dropped. Standard errors (in parentheses) are two-way clustered at the match and bowler levels. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table I11: Contribution of days beyond k to the 30-day cumulative effect

	(1) Almon polynomial, $Q = 2$	(2) Unrestricted lags
	$\mathbb{1}$ (At least one run scored)	
Cumulative effect of past exposure, lags 1–30	0.00343**	-0.00018
	(0.00163)	(0.00300)
Panel A. Contribution of days $k+1$ through 30, for k equal to:		
5 days	0.00322**	0.00266
	(0.00150)	(0.00188)
10 days	0.00255	0.00118
	(0.00157)	(0.00186)
15 days	0.00165	0.00322
	(0.00162)	(0.00202)
20 days	0.00077	-0.00167
	(0.00163)	(0.00192)
25 days	0.00014	0.00045
	(0.00125)	(0.00169)
Joint test of the 5 rows (p)	0.1397	0.0442
degrees of freedom	3	5
Panel B. Contribution of non-overlapping five-day blocks, days:		
1–5	0.00021	-0.00284
	(0.00109)	(0.00261)
6–10	0.00067	0.00147
	(0.00061)	(0.00134)
11–15	0.00090	-0.00204
	(0.00066)	(0.00179)
16–20	0.00088	0.00489***
	(0.00059)	(0.00169)
21–25	0.00063	-0.00212
	(0.00052)	(0.00162)
26–30	0.00014	0.00045
	(0.00125)	(0.00169)
N	183,556	183,556
R^2	0.0472	0.0474
Weather controls	✓	✓
Bowler, batter, stadium-by-year, month, home stadium, innings, and over phase FE	✓	✓

Notes. The outcome variable is a binary indicator equal to 1 if at least one run is scored on a delivery and 0 otherwise (mean 0.599). All $PM_{2.5}$ variables are measured in 10 $\mu g\,m^{-3}$. Both columns are the specification of Table 2, column 3, plus the bowling team's $PM_{2.5}$ on each of the 30 days before the match, with match-day $PM_{2.5}$ entering unrestricted. Column (1) restricts the lag profile to a polynomial of degree $Q = 2$ in the lag index s, as in Equation 6, and is the model of Table 4; column (2) leaves the 30 lag coefficients β_s unrestricted. The first row is the whole-window cumulative effect $\sum_{s=1}^{30} \beta_s$, the effect of raising $PM_{2.5}$ by 10 $\mu g\,m^{-3}$ on each of the 30 days before the match. Panel A reports the part of that total contributed by the days beyond day k, so each row asks whether a window of k days would miss anything a 30-day window captures. Panel B reports the sum of the lag coefficients over non-overlapping five-day blocks and asks which days carry the effect. Every entry is a separate linear combination of the estimated coefficients with a delta-method standard error, not a difference of cumulative rows, whose standard errors carry no information about the covariance between two overlapping sums. The joint test in Panel A is a cluster-robust Wald test that all 5 contrasts are zero; in column (1) it has 3 rather than 5 degrees of freedom, because a degree-2 profile has only 3 free parameters and the contrasts cannot span more dimensions than that. With standard errors of the same order as the point estimates, a row that is not distinguishable from zero is a statement about power, not evidence that those days do not matter. A quadratic is symmetric about its vertex, so in column (1) the two end blocks of Panel B (days 1–5 and 26–30) are constrained by the functional form to be similar; column (2) imposes no shape and is reported for that reason, at the cost of precision: adjacent daily exposures are highly collinear, so its block sums alternate in sign from one block to the next and are better read through the joint test than one at a time. Following Correia (2015), 16 singleton observations are dropped. Standard errors (in parentheses) are two-way clustered at the match and bowler levels. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table I12: Adaptation to $PM_{2.5}$ exposure, number of runs scored outcome

	(1)	(2)	(3)
	Number of runs scored		
Match-day $PM_{2.5}$	0.00095	0.0019	0.0051***
	(0.00058)	(0.0012)	(0.0017)
Past 30-day $PM_{2.5}$		0.0019*	
		(0.0010)	
Match-day $PM_{2.5}$ X Past 30-day $PM_{2.5}$		-0.000019	
		(0.000018)	
Match-day $PM_{2.5}$ X Career $PM_{2.5}$			-0.000092**
			(0.000036)
Weather controls	✓	✓	✓
Bowler FE	✓	✓	✓
Batter FE	✓	✓	✓
Stadium-by-year FE	✓	✓	✓
Month FE	✓	✓	✓
Home stadium FE	✓	✓	✓
Innings FE	✓	✓	✓
Over phase FE	✓	✓	✓
N	183,556	183,556	183,556
R^2	0.028	0.028	0.028

Notes. The outcome variable is the number of runs scored (0, 1, 2, 3, 4, 5, or 6; mean 1.25). $PM_{2.5}$ is measured in $\mu g\, m^{-3}$. Past 30-day $PM_{2.5}$ is the bowler's mean $PM_{2.5}$ exposure over the preceding 30 days. Career $PM_{2.5}$ is defined as the mean $PM_{2.5}$ in all of a bowler's games in the IPL. Regressions include fixed effects for individual bowlers, batters, stadium-by-year, month, innings, and over phase. Regressions also include controls for weather conditions including temperature, temperature squared, relative humidity, wind speed, precipitation, atmospheric pressure, and solar radiation. Standard errors (in parentheses) are two-way clustered at the match and bowler levels. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table I13: Adaptation to $PM_{2.5}$ exposure, number of runs scored (ordered logit)

	(1)	(2)	(3)
	Number of runs scored (ordered logit)		
Match-day $PM_{2.5}$	0.016*	0.027	0.056**
	(0.0085)	(0.018)	(0.023)
Past 30-day $PM_{2.5}$		0.025	
		(0.017)	
Match-day $PM_{2.5}$ X Past 30-day $PM_{2.5}$		-0.0024	
		(0.0032)	
Match-day $PM_{2.5}$ X Career $PM_{2.5}$			-0.0090*
			(0.0047)
Weather controls	✓	✓	✓
Bowler FE	✓	✓	✓
Batter FE	✓	✓	✓
Stadium-by-year FE	✓	✓	✓
Month FE	✓	✓	✓
Home stadium FE	✓	✓	✓
Innings FE	✓	✓	✓
Over phase FE	✓	✓	✓
N	183,572	183,572	183,572

Notes. The outcome variable is the number of runs scored (0, 1, 2, 3, 4, 5, or 6; mean 1.25). Regressions are estimated using ordered logit as in Equation G.1. $PM_{2.5}$ is measured in $\mu g\, m^{-3}$. Past 30-day $PM_{2.5}$ is the bowler's mean $PM_{2.5}$ exposure over the preceding 30 days. Career $PM_{2.5}$ is defined as the mean $PM_{2.5}$ in all of a bowler's games in the IPL. Regressions include fixed effects for individual bowlers, batters, stadium-by-year, month, innings, and over phase. Regressions also include controls for weather conditions including temperature, temperature squared, relative humidity, wind speed, precipitation, atmospheric pressure, and solar radiation. Standard errors (in parentheses) are two-way clustered at the match and bowler levels. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table I14: Adaptation to $PM_{2.5}$ exposure: short-term

	(1)	(2)
	𝟙 (At least one run scored)	
Match-day $PM_{2.5}$	0.00674*	
	(0.00357)	
Past 30-day $PM_{2.5}$	0.00595*	0.00965**
	(0.00344)	(0.00430)
Match-day $PM_{2.5}$ × Past 30-day $PM_{2.5}$	-0.00053	
	(0.00065)	
Q2 (Match-day $PM_{2.5}$)		0.01453
		(0.01929)
Q3 (Match-day $PM_{2.5}$)		0.03831*
		(0.02034)
Q4 (Match-day $PM_{2.5}$)		0.04514*
		(0.02307)
Q5 (Match-day $PM_{2.5}$)		0.07368***
		(0.02550)
Q2 (Match-day $PM_{2.5}$) × Past 30-day $PM_{2.5}$		-0.00206
		(0.00466)
Q3 (Match-day $PM_{2.5}$) × Past 30-day $PM_{2.5}$		-0.00693
		(0.00473)
Q4 (Match-day $PM_{2.5}$) × Past 30-day $PM_{2.5}$		-0.00756
		(0.00505)
Q5 (Match-day $PM_{2.5}$) × Past 30-day $PM_{2.5}$		-0.01032**
		(0.00518)
Implied effect of match-day $PM_{2.5}$: linear slope in (1), Q5 vs. Q1 in (2)		
Low exposure (p10)	0.00502**	0.04018***
	(0.00202)	(0.01248)
High exposure (p90)	0.00337	0.00796
	(0.00205)	(0.01461)
p: high = low	0.419	0.047
N	183,556	183,556
R^2	0.0472	0.0472
Weather controls	✓	✓
Bowler, batter, stadium-by-year, month, home stadium, innings, and over phase FE	✓	✓

Notes. The dependent variable is an indicator equal to 1 if at least one run is scored on the delivery and 0 otherwise (mean 0.599). All $PM_{2.5}$ variables are measured in 10 $\mu g\,m^{-3}$. Past 30-day $PM_{2.5}$ is the bowler's mean exposure over the preceding 30 days. Column (1) enters both linearly with their interaction; column (2) replaces match-day $PM_{2.5}$ with quintile indicators (lowest quintile omitted), interacted with the 30-day mean. The panel below the coefficients reports the implied effect of match-day $PM_{2.5}$ at the 10th and 90th percentile of past 30-day $PM_{2.5}$ (32.5 and 63.7 $\mu g\,m^{-3}$). p: high = low tests the interaction. Standard errors (in parentheses) are two-way clustered at the match and bowler levels. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table I15: Adaptation to $PM_{2.5}$ exposure: long-term

	(1)	(2)
	𝟙 (At least one run scored)	
Match-day $PM_{2.5}$	0.01289**	
	(0.00537)	
Match-day $PM_{2.5}$ × Career $PM_{2.5}$	-0.00193*	
	(0.00116)	
Q2 (Match-day $PM_{2.5}$)		-0.00557
		(0.03456)
Q3 (Match-day $PM_{2.5}$)		0.03546
		(0.03412)
Q4 (Match-day $PM_{2.5}$)		0.07731*
		(0.04048)
Q5 (Match-day $PM_{2.5}$)		0.09761**
		(0.03884)
Q2 (Match-day $PM_{2.5}$) × Career $PM_{2.5}$		0.00307
		(0.00834)
Q3 (Match-day $PM_{2.5}$) × Career $PM_{2.5}$		-0.00625
		(0.00816)
Q4 (Match-day $PM_{2.5}$) × Career $PM_{2.5}$		-0.01517
		(0.00932)
Q5 (Match-day $PM_{2.5}$) × Career $PM_{2.5}$		-0.01630*
		(0.00890)
Implied effect of match-day $PM_{2.5}$: linear slope in (1), Q5 vs. Q1 in (2)		
Low exposure (p10)	0.00656***	0.04430***
	(0.00218)	(0.01341)
High exposure (p90)	0.00247	0.00985
	(0.00212)	(0.01493)
p: high = low	0.096	0.068
N	183,556	183,556
R^2	0.0472	0.0472
Weather controls	✓	✓
Bowler, batter, stadium-by-year, month, home stadium, innings, and over phase FE	✓	✓

Notes. The dependent variable is an indicator equal to 1 if at least one run is scored on the delivery and 0 otherwise (mean 0.599). All $PM_{2.5}$ variables are measured in 10 $\mu g\,m^{-3}$. Career $PM_{2.5}$ is a bowler's mean exposure across all IPL matches, entered continuously. Column (1) enters match-day $PM_{2.5}$ linearly, interacted with career $PM_{2.5}$; column (2) replaces it with quintile indicators (lowest quintile omitted). The panel below the coefficients reports the implied effect of match-day $PM_{2.5}$ at the 10th and 90th percentile of career $PM_{2.5}$ (32.7 and 53.8 $\mu g\,m^{-3}$). p: high = low tests the interaction. Standard errors (in parentheses) are two-way clustered at the match and bowler levels. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table I16: Long-term adaptation to $PM_{2.5}$ exposure: alternate specification with batter exposure

	(1) $\mathbb{1}$ (At least one run scored)
Match-day $PM_{2.5}$	0.015* (0.0081)
Match-day $PM_{2.5}$ X Career $PM_{2.5}$ (bowler)	-0.0020* (0.0012)
Match-day $PM_{2.5}$ X Career $PM_{2.5}$ (batter)	-0.00041 (0.0011)
Weather controls	✓
Home stadium FE	✓
Stadium-by-year FE	✓
Over phase FE	✓
Month FE	✓
Innings FE	✓
Bowler FE	✓
Batter FE	✓
N	183,556
R^2	0.047

Notes. The dependent variable is an indicator equal to 1 if at least one run is scored on the delivery and 0 otherwise (mean 0.599). Match-day $PM_{2.5}$ is measured in 10 $\mu g\,m^{-3}$. Career $PM_{2.5}$ is the bowler's (batter's) mean $PM_{2.5}$ exposure over all matches played in the cricket league. Column (1) includes the interaction of both with $PM_{2.5}$ on the day of the match. Column (1) includes controls for weather conditions including temperature, temperature squared, relative humidity, wind speed, precipitation, atmospheric pressure, and solar radiation, and indicators for whether the bowling and batting teams are at their home stadium, together with fixed effects for individual bowlers, batters, stadium-by-year, month, innings, and over phase. Standard errors (in parentheses) are two-way clustered at the match and bowler levels. Following Correia (2015), 16 singleton observations are dropped. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table I17: Summary statistics of long-run $PM_{2.5}$ variables

Definition	Distinct values	Mean	SD	Min	Max
Bowler $PM_{2.5}$					
All IPL, 2008–2022	445	42.62	9.66	17.83	103.16
IPL, 2008–2014	136	40.50	7.48	22.76	69.13
Team stadium $PM_{2.5}$					
1998–2007	15	45.47	17.42	25.11	83.59
2008–2022	15	50.12	19.61	28.27	96.23
2008–2022, IPL months	15	46.67	14.86	26.66	77.70
1998–2007, IPL months	15	42.54	14.27	22.28	70.49
Preseason	110	51.08	15.14	23.44	100.13

Notes. This table reports summary statistics for seven alternative definitions of long-run $PM_{2.5}$ exposure, computed at the level given by Distinct values. Bowler $PM_{2.5}$ is a bowler's mean $PM_{2.5}$ exposure across all IPL matches played, either over the full 2008–2022 sample or restricted to matches in 2008–2014 (for bowlers who also appear in 2015–2022). Team stadium $PM_{2.5}$ is the mean $PM_{2.5}$ at a team's home stadium over the indicated window; IPL months restricts to March–May. Preseason $PM_{2.5}$ is the mean $PM_{2.5}$ at a team's home stadium over the 21 days preceding that season's start.

Table I18: Robustness of long-term $PM_{2.5}$ measures

	(1)	(2)	(3)	(4)	(5)	(6)
	$\mathbb{1}$ (At least one run scored)					
Match-day $PM_{2.5}$	0.0056	0.0068**	0.0066**	0.0073**	0.0061**	0.0078***
	(0.0060)	(0.0030)	(0.0026)	(0.0029)	(0.0025)	(0.0028)
$PM_{2.5}$ X Bowler $PM_{2.5}$ (IPL Seasons 2008-2014)	-0.0014					
	(0.0013)					
$PM_{2.5}$ X Bowler Team Stadium $PM_{2.5}$ (Preseason)		-0.00044				
		(0.00046)				
$PM_{2.5}$ X Bowler Team Stadium $PM_{2.5}$ (1998-2007)			-0.000041			
			(0.000039)			
$PM_{2.5}$ X Bowler Team Stadium $PM_{2.5}$ (1998-2007, IPL Months)				-0.000058		
				(0.000048)		
$PM_{2.5}$ X Bowler Team Stadium $PM_{2.5}$ (2008-2022)					-0.000029	
					(0.000033)	
$PM_{2.5}$ X Bowler Team Stadium $PM_{2.5}$ (2008-2022, IPL Months)						-0.000074
						(0.000051)
Weather controls	✓	✓	✓	✓	✓	✓
Bowler FE	✓					
Bowling Team-by-year FE		✓	✓	✓	✓	✓
Batter FE	✓	✓	✓	✓	✓	✓
Stadium-by-year FE	✓	✓	✓	✓	✓	✓
Month FE	✓	✓	✓	✓	✓	✓
Innings FE	✓	✓	✓	✓	✓	✓
Over phase FE	✓	✓	✓	✓	✓	✓
N	53,607	183,558	183,558	183,558	183,558	183,558
R^2	0.048	0.042	0.042	0.042	0.042	0.042

Notes. The outcome variable is a binary indicator equal to 1 if at least one run is scored on a delivery and 0 otherwise. All columns include controls for weather conditions including temperature, temperature squared, relative humidity, wind speed, precipitation, atmospheric pressure, and solar radiation, and fixed effects for batter, stadium-by-year, month, innings, and over phase; column (1) additionally includes fixed effects for individual bowlers, while columns (2)–(6) include fixed effects for bowling teams-by-year in place of individual bowlers. Standard errors (in parentheses) are two-way clustered at the match and bowler levels. Column (1) restricts the sample to bowlers who appear in both the first and second half of the panel (first half: 2008-2014, second half: 2015-2022), analyzing their performance in the second half using their mean $PM_{2.5}$ exposure during IPL games in the first half. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table I19: South Asian bowlers: $PM_{2.5}$ exposure and run-scoring probability

	(1) $\mathbb{1}$ (At least one run scored)
Match-day $PM_{2.5}$	0.0049***
	(0.0018)
Match-day $PM_{2.5}$ X South Asian bowler	-0.0012
	(0.0012)
Weather controls	✓
Bowler FE	✓
Batter FE	✓
Stadium-by-year FE	✓
Month FE	✓
Home stadium FE	✓
Innings FE	✓
Over phase FE	✓
N	178,459
R^2	0.047

Notes. The outcome variable is a binary indicator equal to 1 if at least one run is scored on a delivery and 0 otherwise (mean 0.599). $PM_{2.5}$ is measured in 10 $\mu g\, m^{-3}$. South Asian bowler is a dummy variable equal to 1 if the bowler plays on a national cricket team in a South Asian country; 0 if otherwise. 60.45% of bowlers are South Asian by this categorization. The categorization is unobserved for 11 bowlers; 5,096 observations associated with these bowlers are dropped. Columns (1) presents results from a regression of this run indicator on match-day $PM_{2.5}$ levels (measured in 10 $\mu g\, m^{-3}$). It includes fixed effects for individual batters, bowlers, stadium-by-year, home stadium, month, innings, and over phase. It includes controls for weather conditions including temperature, temperature squared, relative humidity, precipitation, atmospheric pressure, solar radiation, and wind speed. Following Correia (2015), 17 singleton observations are dropped. Standard errors (in parentheses) are two-way clustered at the match and bowler levels. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table I20: Short-career ladder for the long-term adaptation interaction

	(1) Bowlers in ≤ 1 season (168 bowlers)	(2) Bowlers in ≤ 2 seasons (248 bowlers)	(3) Bowlers in ≤ 3 seasons (292 bowlers)	(4) Bowlers in ≤ 5 seasons (364 bowlers)	(5) Bowlers in ≤ 8 seasons (413 bowlers)
	𝟙 (At least one run scored)				
Match-day $PM_{2.5}$	0.01561	0.01152	0.02715***	0.01864**	0.01465**
	(0.02254)	(0.01143)	(0.01029)	(0.00724)	(0.00574)
Match-day $PM_{2.5}$ × Career $PM_{2.5}$ (entire career)	-0.00395	-0.00190	-0.00485**	-0.00231	-0.00225*
	(0.00382)	(0.00205)	(0.00200)	(0.00154)	(0.00123)
Weather controls	✓	✓	✓	✓	✓
Bowler FE	✓	✓	✓	✓	✓
Batter FE	✓	✓	✓	✓	✓
Stadium-by-year FE	✓	✓	✓	✓	✓
Month FE	✓	✓	✓	✓	✓
Innings FE	✓	✓	✓	✓	✓
Over phase FE	✓	✓	✓	✓	✓
N	11,859	25,983	38,915	73,980	123,010
R^2	0.10239	0.07957	0.06758	0.05356	0.04939

Notes. The dependent variable is an indicator equal to 1 if at least one run is scored on the delivery and 0 otherwise. $PM_{2.5}$ is measured in 10 μg m^{-3}. Each column restricts the sample to bowlers whose entire IPL career is at most N seasons (N = 1, 2, 3, 5, or 8, per column), keeping all of their seasons, and recomputes career $PM_{2.5}$ on that restricted sample. All columns include controls for temperature, temperature squared, relative humidity, wind speed, precipitation, atmospheric pressure, and solar radiation, and indicators for whether the bowling and batting teams are at their home stadium, together with fixed effects for individual bowlers, batters, stadium-by-year, month, innings, and over phase. Standard errors (in parentheses) are two-way clustered at the match and bowler levels. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table I21: Career-length ladder for the long-term adaptation interaction

	(1) Bowlers in ≥ 1 season (445 bowlers)	(2) Bowlers in ≥ 2 seasons (277 bowlers)	(3) Bowlers in ≥ 3 seasons (197 bowlers)	(4) Bowlers in ≥ 5 seasons (111 bowlers)	(5) Bowlers in ≥ 8 seasons (44 bowlers)
	𝟙 (At least one run scored)				
Match-day $PM_{2.5}$	0.01045	0.02524***	0.01552**	0.01563**	0.02622***
	(0.00873)	(0.00679)	(0.00640)	(0.00763)	(0.00819)
Match-day $PM_{2.5}$ × Career $PM_{2.5}$ (first N seasons)	-0.00086	-0.00398***	-0.00258*	-0.00249	-0.00445**
	(0.00154)	(0.00136)	(0.00136)	(0.00162)	(0.00180)
Weather controls	✓	✓	✓	✓	✓
Bowler FE	✓	✓	✓	✓	✓
Batter FE	✓	✓	✓	✓	✓
Stadium-by-year FE	✓	✓	✓	✓	✓
Month FE	✓	✓	✓	✓	✓
Innings FE	✓	✓	✓	✓	✓
Over phase FE	✓	✓	✓	✓	✓
N	45,278	66,553	78,351	91,283	64,815
R^2	0.06712	0.05426	0.05055	0.04927	0.05337

Notes. The dependent variable is an indicator equal to 1 if at least one run is scored on the delivery and 0 otherwise. $PM_{2.5}$ is measured in 10 μg m^{-3}. Each column restricts the sample to bowlers with at least N IPL seasons (N = 1, 2, 3, 5, or 8, per column), keeping only their first N seasons of data, and recomputes career $PM_{2.5}$ on that restricted sample. All columns include controls for temperature, temperature squared, relative humidity, wind speed, precipitation, atmospheric pressure, and solar radiation, and indicators for whether the bowling and batting teams are at their home stadium, together with fixed effects for individual bowlers, batters, stadium-by-year, month, innings, and over phase. Standard errors (in parentheses) are two-way clustered at the match and bowler levels. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table I22: Long-term adaptation to $PM_{2.5}$, bowlers with at least 3 seasons, first 3 seasons only

	(1) Linear	(2) Spline, knot at 50 µg m^{-3}
	$\mathbb{1}$ (At least one run scored)	
Match-day $PM_{2.5}$	0.01552**	
	(0.00640)	
Match-day $PM_{2.5}$ × Career $PM_{2.5}$ (first 3 seasons)	-0.00258*	
	(0.00136)	
Match-day $PM_{2.5}$ between 0 and 50 µg m^{-3}		0.04096***
		(0.01369)
Match-day $PM_{2.5}$ between 50 and 160 µg m^{-3}		0.00264
		(0.00937)
0–50 µg m^{-3} × Career $PM_{2.5}$ (first 3 seasons)		-0.00909***
		(0.00317)
50–160 µg m^{-3} × Career $PM_{2.5}$ (first 3 seasons)		0.00038
		(0.00197)
Weather controls	✓	✓
Bowler FE	✓	✓
Batter FE	✓	✓
Stadium-by-year FE	✓	✓
Month FE	✓	✓
Home stadium controls	✓	✓
Innings FE	✓	✓
Over phase FE	✓	✓
N	78,351	78,351
R^2	0.05	0.05
p: dose–response linear		0.05
p: no moderation	0.06	0.01
p: moderation constant across segments		0.03

Notes. The dependent variable is an indicator equal to 1 if at least one run is scored on the delivery and 0 otherwise. All $PM_{2.5}$ variables are measured in 10 µg m^{-3}. The sample is restricted to the 197 bowlers who appear in at least 3 IPL seasons, keeping only each bowler's first 3 seasons of data. Across all 445 bowlers, the mean career length is 3.2 seasons (median 2). Career $PM_{2.5}$ is recomputed on this restricted sample, so it is each bowler's mean $PM_{2.5}$ over his first 3 seasons only. In column (1), match-day $PM_{2.5}$ and career $PM_{2.5}$ are in levels and are interacted. Column (2) replaces the linear match-day dose with a continuous piecewise-linear spline with one knot at 50 µg m^{-3} and interacts each segment slope with career $PM_{2.5}$. The reported p-values are for tests that: the two segment slopes are equal, so the dose–response is linear; there is no moderation; and the two interactions of column (2) are equal to one another. Both columns include controls for temperature, temperature squared, relative humidity, wind speed, precipitation, atmospheric pressure, and solar radiation, and indicators for whether the bowling and batting teams are at their home stadium, together with fixed effects for individual bowlers, batters, stadium-by-year, month, innings, and over phase. Following Correia (2015), 21 singleton observations are dropped. Standard errors (in parentheses) are two-way clustered at the match and bowler levels. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table I23: IPL bowler entry and exit, by season

Season	First-time bowlers	Exiting bowlers
2008	98	32
2010	63	23
2011	44	30
2012	29	31
2013	29	38
2014	17	19
2015	17	18
2016	20	23
2017	24	28
2018	22	18
2019	26	34
2021	14	26
2022	42	n/a
Mean	34.23	26.67
S.d.	23.51	6.60
Min	14	18
Median	26.0	27.0
Max	98	38

Notes. A player is any bowler observed in an IPL delivery (match-player pairs, collapsed to one observation per player-season; team switches within a season are not treated as separate observations). A player is first-time in season s if s is the earliest season in which the player is observed. A player exits in season s if s is the player's last observed season and s is not the final season in the sample (the final season, 2022, is right-censored – exit status is unknown for players still active then – and is marked n/a and excluded from the summary panel below). Summary statistics are computed across the 13 seasons (first-time players) and 12 seasons excluding the final one (exits).

Table I24: Season-to-season retention probit (bowler-season panel)

	(1) $\mathbb{1}$(bowls in the IPL next season)
Log deliveries this season	0.238***
	(0.077)
Matches this season	0.071***
	(0.019)
Run-concession rate this season	0.549
	(0.555)
Runs conceded per delivery this season	-0.348**
	(0.144)
Wickets per 100 deliveries this season	0.008
	(0.012)
Wides + no-balls per 100 deliveries this season	-0.003
	(0.012)
Share of deliveries in powerplay	-0.178
	(0.190)
Share of deliveries at the death	-0.381
	(0.244)
Mean match-day $PM_{2.5}$ this season	-0.019
	(0.029)
Age	-0.051***
	(0.009)
IPL seasons played so far (incl. this one)	0.059***
	(0.021)
Pace bowler	-0.103
	(0.094)
South Asia	0.170*
	(0.088)
Season dummies	✓
N (bowler-seasons)	1,403
Pseudo R^2	0.175
AUC (ROC)	0.773
Correctly classified (%)	72.3
Sensitivity (%)	85.9
Specificity (%)	46.5

Notes. One observation per bowler-season in the estimation panel (IPL seasons played in India, 2008–2022). The dependent variable is 1 if the bowler bowled at least one delivery in the following IPL season, measured in the full ball-by-ball record through 2024 and including the two seasons played outside India (2009, 2020), so it is defined for every panel season. All regressors are measured in the current season. AUC is the area under the receiver operating characteristic curve; correctly classified, sensitivity, and specificity are computed at the $\hat{p} \geq 0.5$ classification cutoff. Standard errors (in parentheses) are conventional probit standard errors. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table I25: Contemporaneous and career-adaptation estimates under sequential attrition weights

	(1)	(2)	(3)	(4)	(5)	(6)
	Contemporaneous		Linear × career		Spline × career	
	Unweighted	Seq. IPW	Unweighted	Seq. IPW	Unweighted	Seq. IPW
Match-day $PM_{2.5}$	0.00413**	0.01015***	0.01002*	0.02246**		
	(0.00178)	(0.00364)	(0.00511)	(0.00887)		
Match-day $PM_{2.5}$ between 0 and 50 µg m^{-3}					0.03034**	0.02991
					(0.01394)	(0.01855)
Match-day $PM_{2.5}$ between 50 and 160 µg m^{-3}					-0.00092	0.02206*
					(0.00756)	(0.01220)
Match-day $PM_{2.5}$ × Career $PM_{2.5}$			-0.00131	-0.00278		
			(0.00111)	(0.00175)		
0–50 µg m^{-3} × Career $PM_{2.5}$					-0.00635**	-0.00654
					(0.00321)	(0.00442)
50–160 µg m^{-3} × Career $PM_{2.5}$					0.00120	-0.00191
					(0.00176)	(0.00272)
Weather and home-stadium controls	✓	✓	✓	✓	✓	✓
Bowler, batter, stadium-by-year, month, innings, over-phase FE	✓	✓	✓	✓	✓	✓
N	178,459	178,459	178,459	178,459	178,459	178,459
R^2	0.0472	0.0578	0.0473	0.0578	0.0473	0.0579
p: no moderation			0.24	0.11	0.13	0.09
p: cols (1)=(2)		0.04				
Weight: 99th pct.		25.7		25.7		25.7
Weight: max		150.8		150.8		150.8

Notes. The dependent variable is an indicator equal to 1 if at least one run is scored on the delivery. $PM_{2.5}$ variables are in 10 µg m^{-3}. Columns (1)–(2) estimate the Table 2 column (3) specification; columns (3)–(6) the linear and spline career-adaptation specifications of Table 6, columns (1) and (2). Even-numbered columns weight each delivery by the sequential inverse-probability-of-retention weight $w_{is} = \prod_{t<s} 1/\hat{p}_{it}$, the product over the bowler's earlier observed seasons of the inverse predicted probability of returning the following season from the probit of Table I24 (weight 1 in the bowler's first season). Bowlers with a missing retention probability in any season are excluded from all columns. p: no moderation tests the interaction term(s); p: cols (1)=(2) tests whether the match-day $PM_{2.5}$ coefficient differs between columns (1) and (2); because both columns use the same deliveries under different weights, it is computed from a single stacked model in which every regressor and fixed effect is interacted with a column indicator, with standard errors two-way clustered at the match and bowler levels. Following Correia (2015), 17 singleton observations are dropped. Standard errors (in parentheses) are two-way clustered at the match and bowler levels and treat the weights as known. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table I26: Short-term adaptation to $PM_{2.5}$

	(1) Linear	(2) Spline, knot at 50 μg m^{-3}
	𝟙 (At least one run scored)	
Match-day $PM_{2.5}$	0.00638*	
	(0.00357)	
Past 20-day $PM_{2.5}$	0.00600*	0.01657**
	(0.00350)	(0.00642)
Match-day $PM_{2.5}$ × Past 20-day $PM_{2.5}$	-0.00049	
	(0.00068)	
Match-day $PM_{2.5}$ between 0 and 50 μg m^{-3}		0.01836**
		(0.00789)
Match-day $PM_{2.5}$ between 50 and 160 μg m^{-3}		0.00006
		(0.00515)
0–50 μg m^{-3} × Past 20-day $PM_{2.5}$		-0.00341**
		(0.00167)
50–160 μg m^{-3} × Past 20-day $PM_{2.5}$		0.00085
		(0.00110)
Weather controls	✓	✓
Bowler FE	✓	✓
Batter FE	✓	✓
Stadium-by-year FE	✓	✓
Month FE	✓	✓
Home stadium controls	✓	✓
Innings FE	✓	✓
Over phase FE	✓	✓
N	183,556	183,556
R^2	0.05	0.05
p: dose–response linear		0.10
p: no moderation	0.47	0.12
p: moderation constant across segments		0.08

Notes. The dependent variable is an indicator equal to 1 if at least one run is scored on the delivery and 0 otherwise (mean 0.599). All $PM_{2.5}$ variables are measured in 10 μg m^{-3}. Past 30-day $PM_{2.5}$ is the bowling team's mean $PM_{2.5}$ over the 30 days before the match. In column (1), match-day $PM_{2.5}$ and past 30-day are in levels and are interacted. Column (2) replaces the linear match-day dose with a continuous piecewise-linear spline with one knot at 50 μg m^{-3} and interacts each segment slope with past 30-day $PM_{2.5}$. The reported p-values are for tests that: the two segment slopes are equal, so the dose–response is linear; there is no moderation, which in column (1) is the single interaction and in column (2) is both segment × past 30-day interactions jointly; and the two interactions of column (2) are equal to one another, which is the restriction that reduces its moderation to the single linear-by-linear parameter of column (1). Both columns include controls for temperature, temperature squared, relative humidity, wind speed, precipitation, atmospheric pressure, and solar radiation, and indicators for whether the bowling and batting teams are at their home stadium, together with fixed effects for individual bowlers, batters, stadium-by-year, month, innings, and over phase. Following Correia (2015), 16 singleton observations are dropped. Standard errors (in parentheses) are two-way clustered at the match and bowler levels. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table I27: Short-term adaptation to $PM_{2.5}$

	(1) Linear	(2) Spline, knot at 50 µg m^{-3}
	𝟙 (At least one run scored)	
Match-day $PM_{2.5}$	0.00726**	
	(0.00366)	
Past 24-day $PM_{2.5}$	0.00678*	0.01947***
	(0.00351)	(0.00631)
Match-day $PM_{2.5}$ × Past 24-day $PM_{2.5}$	-0.00066	
	(0.00068)	
Match-day $PM_{2.5}$ between 0 and 50 µg m^{-3}		0.02186***
		(0.00794)
Match-day $PM_{2.5}$ between 50 and 160 µg m^{-3}		-0.00021
		(0.00525)
0–50 µg m^{-3} × Past 24-day $PM_{2.5}$		-0.00416**
		(0.00166)
50–160 µg m^{-3} × Past 24-day $PM_{2.5}$		0.00091
		(0.00111)
Weather controls	✓	✓
Bowler FE	✓	✓
Batter FE	✓	✓
Stadium-by-year FE	✓	✓
Month FE	✓	✓
Home stadium controls	✓	✓
Innings FE	✓	✓
Over phase FE	✓	✓
N	183,556	183,556
R^2	0.05	0.05
p: dose–response linear		0.05
p: no moderation	0.34	0.04
p: moderation constant across segments		0.04

Notes. The dependent variable is an indicator equal to 1 if at least one run is scored on the delivery and 0 otherwise (mean 0.599). All $PM_{2.5}$ variables are measured in 10 µg m^{-3}. Past 30-day $PM_{2.5}$ is the bowling team's mean $PM_{2.5}$ over the 30 days before the match. In column (1), match-day $PM_{2.5}$ and past 30-day are in levels and are interacted. Column (2) replaces the linear match-day dose with a continuous piecewise-linear spline with one knot at 50 µg m^{-3} and interacts each segment slope with past 30-day $PM_{2.5}$. The reported p-values are for tests that: the two segment slopes are equal, so the dose–response is linear; there is no moderation, which in column (1) is the single interaction and in column (2) is both segment × past 30-day interactions jointly; and the two interactions of column (2) are equal to one another, which is the restriction that reduces its moderation to the single linear-by-linear parameter of column (1). Both columns include controls for temperature, temperature squared, relative humidity, wind speed, precipitation, atmospheric pressure, and solar radiation, and indicators for whether the bowling and batting teams are at their home stadium, together with fixed effects for individual bowlers, batters, stadium-by-year, month, innings, and over phase. Following Correia (2015), 16 singleton observations are dropped. Standard errors (in parentheses) are two-way clustered at the match and bowler levels. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

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